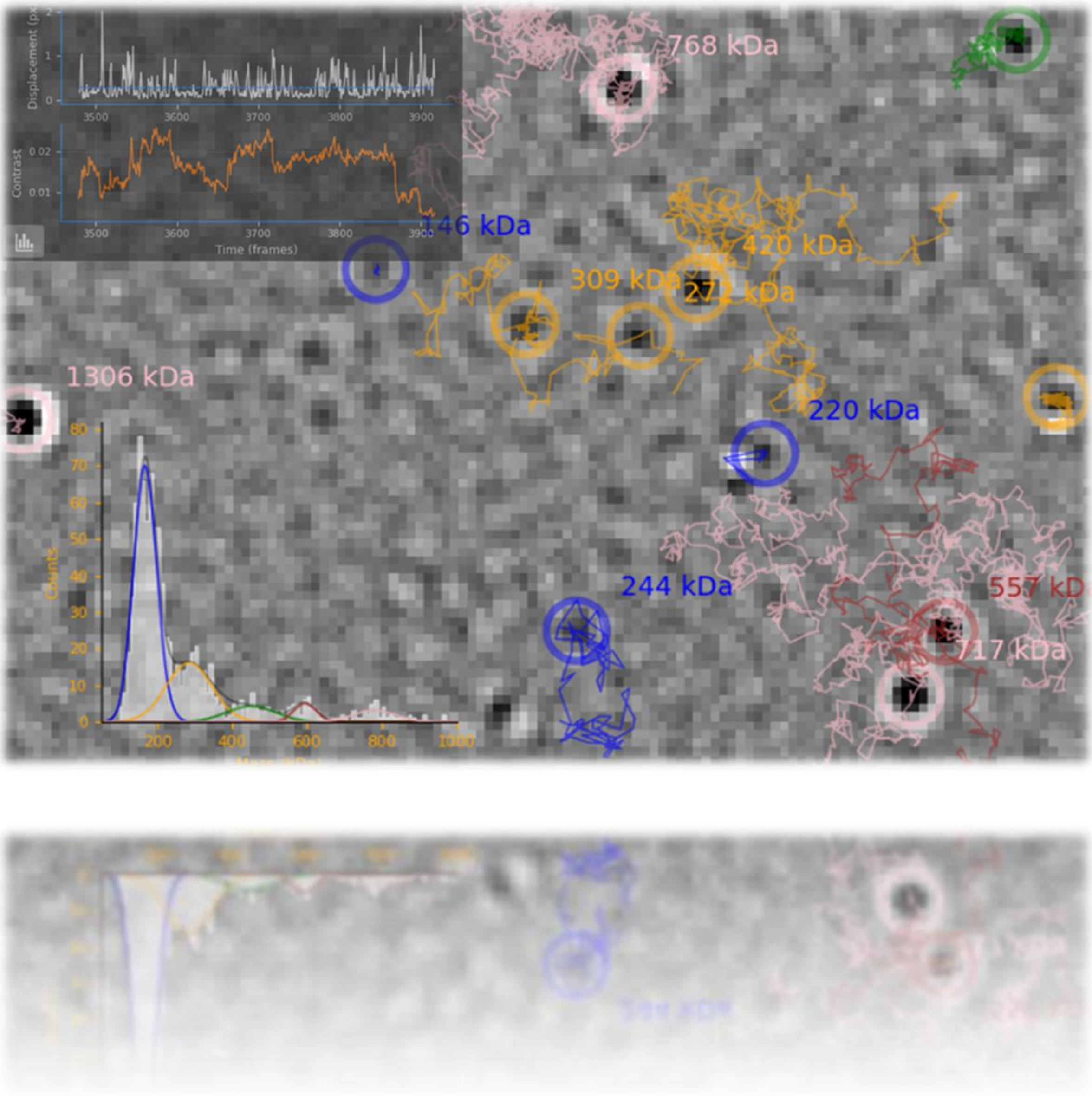
OpenMASS
v2025r

User Manual



Contents

Chapter 1 – Getting Started

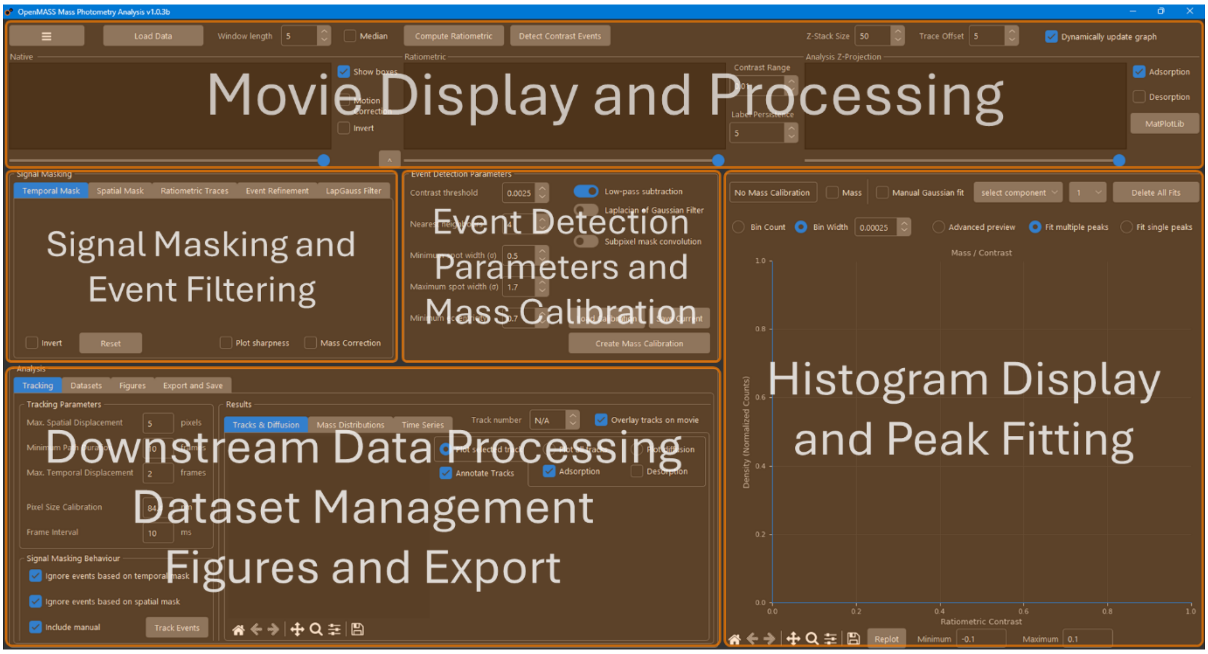
1.1 Overview

OpenMASS (Open mass analysis software suite) is an open source and free software package for analysing mass photometry data from Refeyn® mass photometer instruments. The software is written in python (version 3.11.0) and can be launched directly from source after running the following command after cloning the git repo into a new python 3.11.0 virtual environment:

```
pip install -r requirements.txt
```

Alternatively, a standalone windows exe file is also available. This version does not require installation and can run directly from the folder containing the exe. This is convenient if you need to run it from a USB or network drive on any PC quickly. The software is designed to analyse '.mp' files, '.h5' files, or '.tiff' files. If the MP format is used, Movies saved from Refeyn AcquireMP® should be saved uncompressed (This can be changed in AcquireMP® advanced settings). If the MP format used by your instrument doesn't work, you may need to convert to '.h5' or '.tiff' in Refeyn DiscoverMP® first.

The main user interface of the software is currently limited to 1920x1080 screen resolution, suitable for most modern monitors, but may be an issue on older hardware. **Make sure display scaling is 100%.** The layout of the UI is split into sections as follows:



The ‘Movie Display and Processing’ section has views for the native and ratiometric movie as well as the filtered z-projection that is analysed. The buttons to load data, set window size, z-stack size, motion correction, and background subtraction mode are also located in this ribbon and adjacent to the image views.

The ‘Signal Masking and Event Filtering’ section contains controls for applying temporal or spatial masks to exclude regions in time or position. Contrast vs time traces and event filtering based on trace fitting for landing assay type experiments are accessible here as well. Finally, the difference of Gaussians approximation for the Laplacian of Gaussian PSF filter can be tuned here.

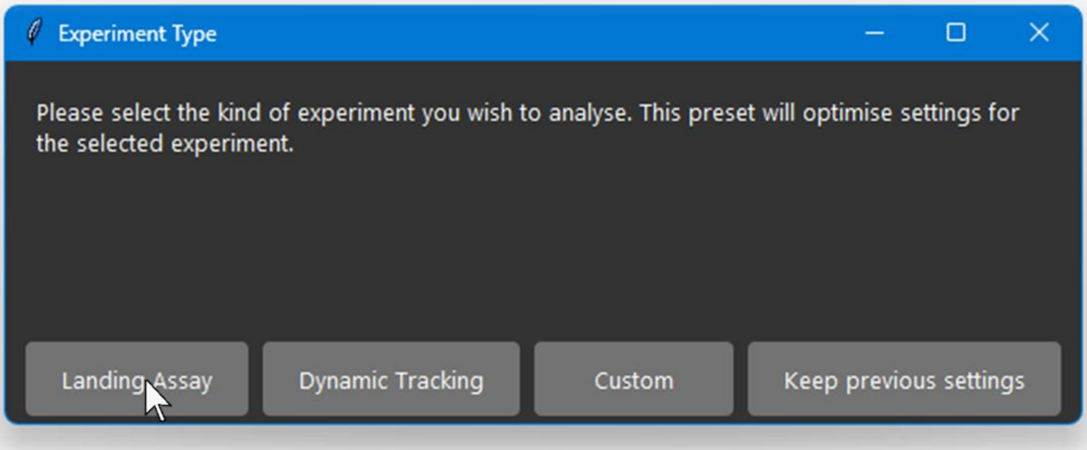
The ’Event Detection Parameters and Mass Calibration’ section is where settings for fitting contrast events live. The Mass calibration window can be opened from here and mass calibrations can be loaded and saved.

The ‘Histogram Display and Peak Fitting’ section is where histograms of contrast events are plotted and fitted. Track mass distributions can be displayed and fitted here too. Users can choose between fitting single peaks, or multiple peaks at once using a mixture model. It is also possible to open an advanced fitting window to finetune fitting and solver parameters and choose between symmetric and asymmetric Gaussian mixture models.

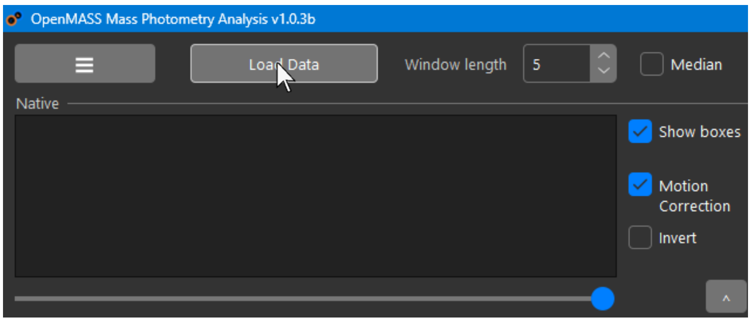
The ‘Downstream Data Processing, Data Management, Figures and Export’ section Contains all the controls for further processing of data including tracking and time series analysis. All kinds of data can be imported into the dataset manager from which figures can be made. Movies and event, contrast, mass, and tracking data can also be exported from here.

1.2 Quick-Start Guide

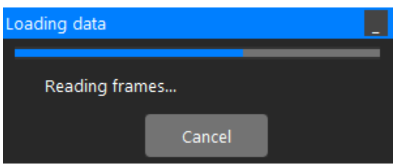
When the program first opens you will be prompted to choose an experiment type. Here, we will begin with a landing assay of the Invitrogen® NativeMark™ unstained protein ladder which we will use to create a mass calibration. Click 'Landing Assay' in the dialogue box:

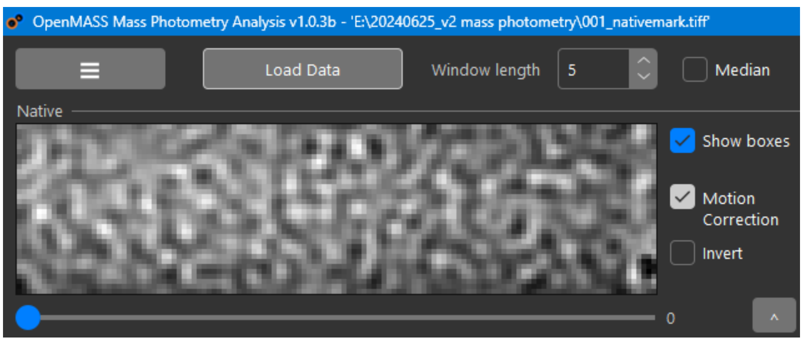


This will load and apply the default landing assay settings profile which is optimised for general use. Next Click 'Load Data' in the top left of the UI above the native image view and open the MP file:

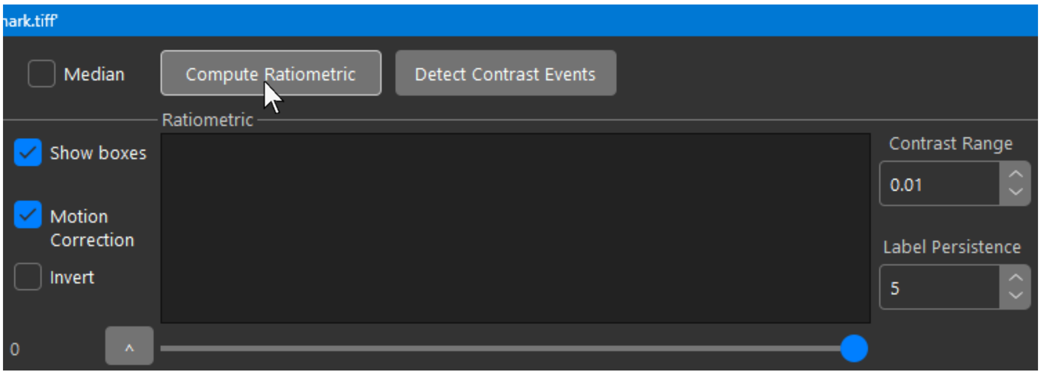


The native movie will then be loaded and displayed in the native image view:

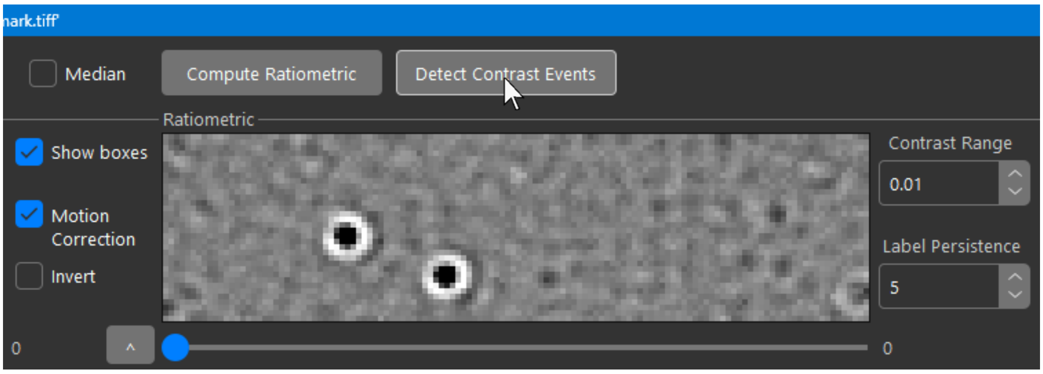




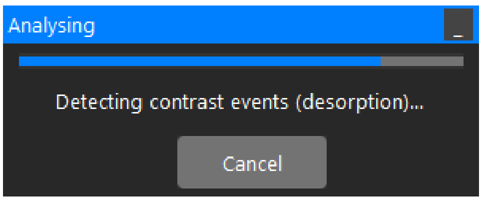
Next, Click 'Compute Ratiometric' above the ratiometric image view:

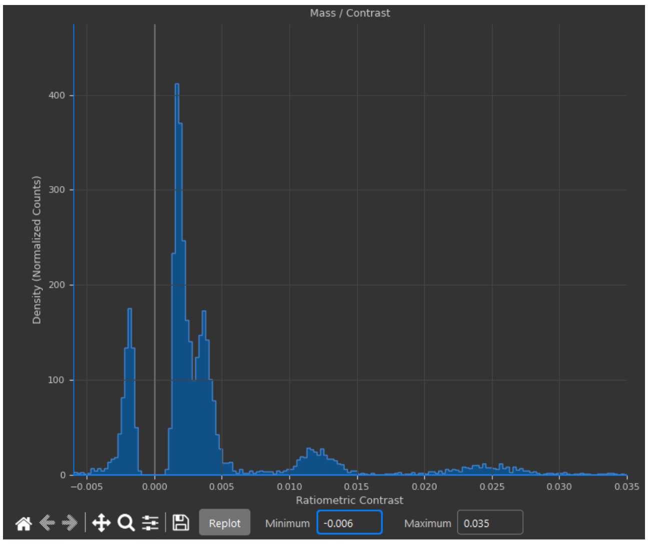


The ratiometric movie will be calculated. Wait for motion correction to finish processing and the frames will be displayed in the ratiometric image view. You can scroll through frames with the horizontal scrollbar below. Next Click 'Detect Contrast Events' above the ratiometric image view:

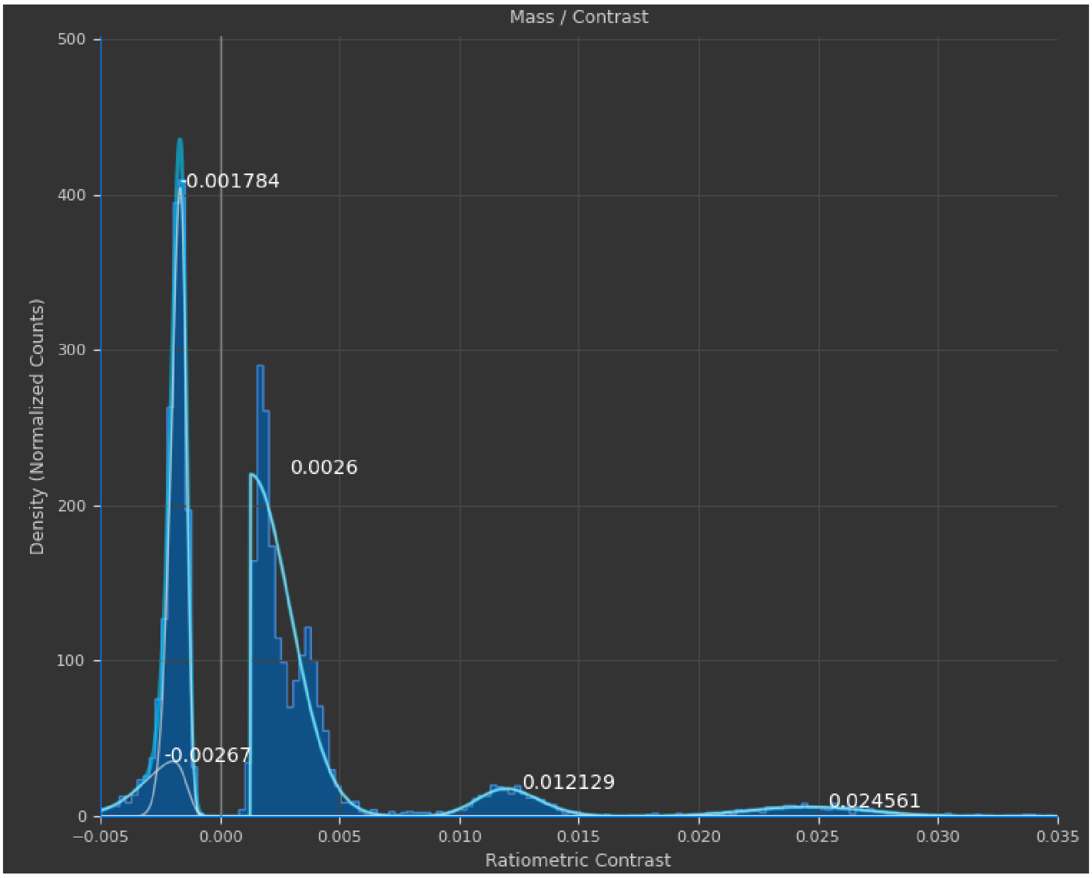


Events will be calculated and the histogram in the bottom right will be periodically updated with events detected until the process finishes:

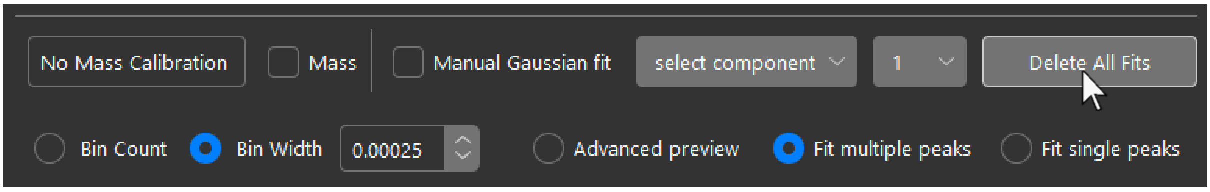




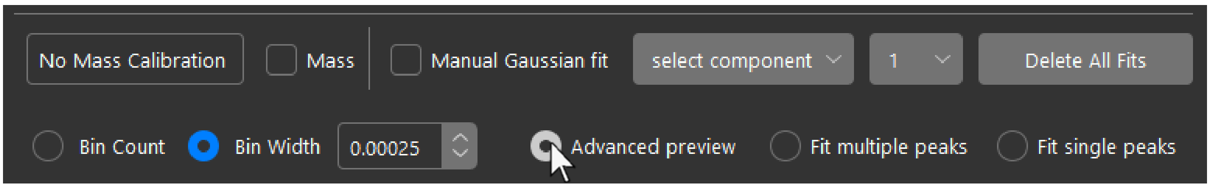
OpenMASS will try to automatically fit the peaks with a skewed Gaussian mixture model. The result of the automatic fit can be poor with many peaks present:



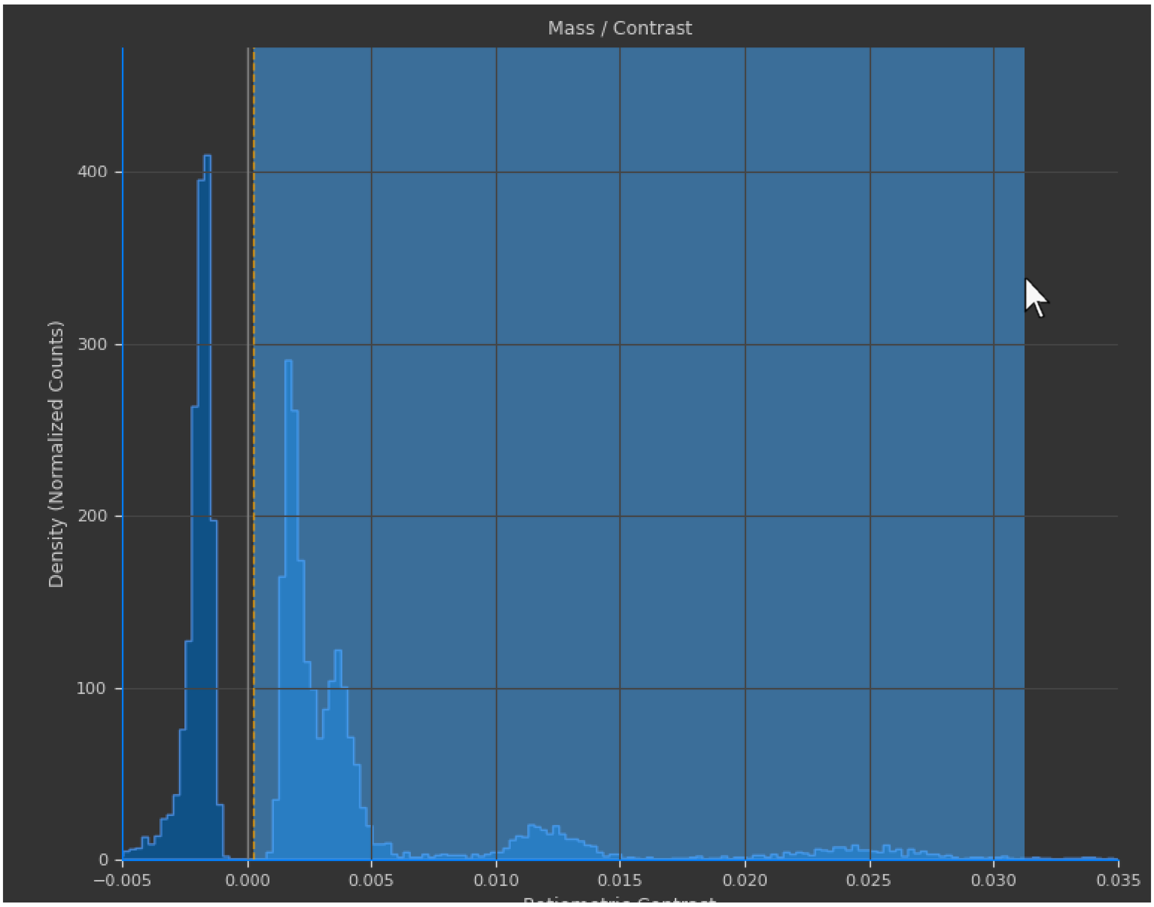
It is therefore recommended to use the advanced preview window to fit peaks in a user specified window. First delete the bad fits using the button in the histogram ribbon above:



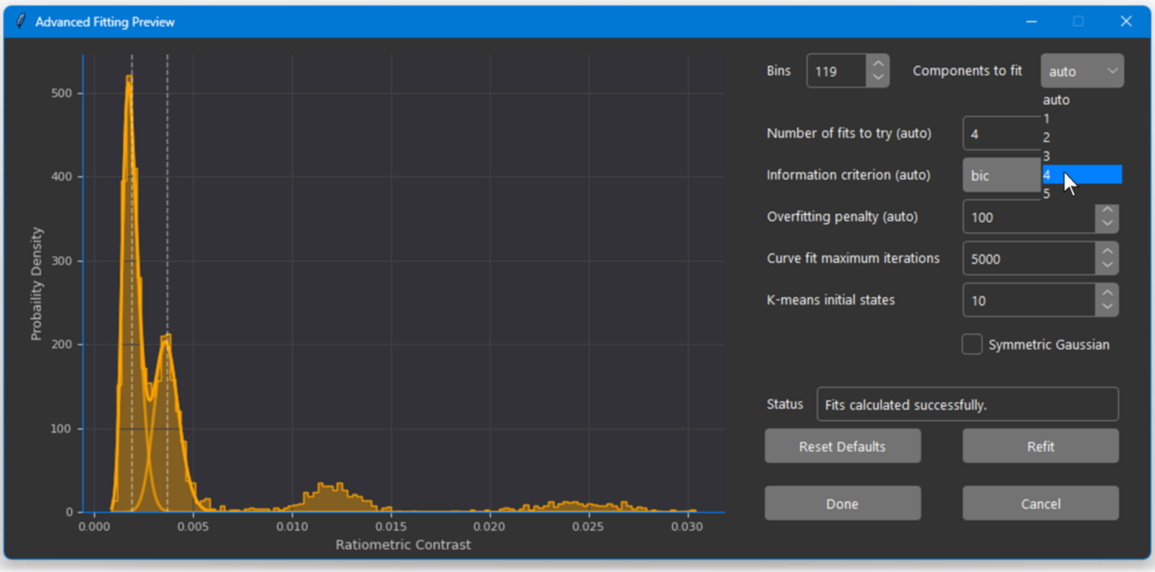
Then select the 'Advanced preview' radio button also located in the histogram ribbon:



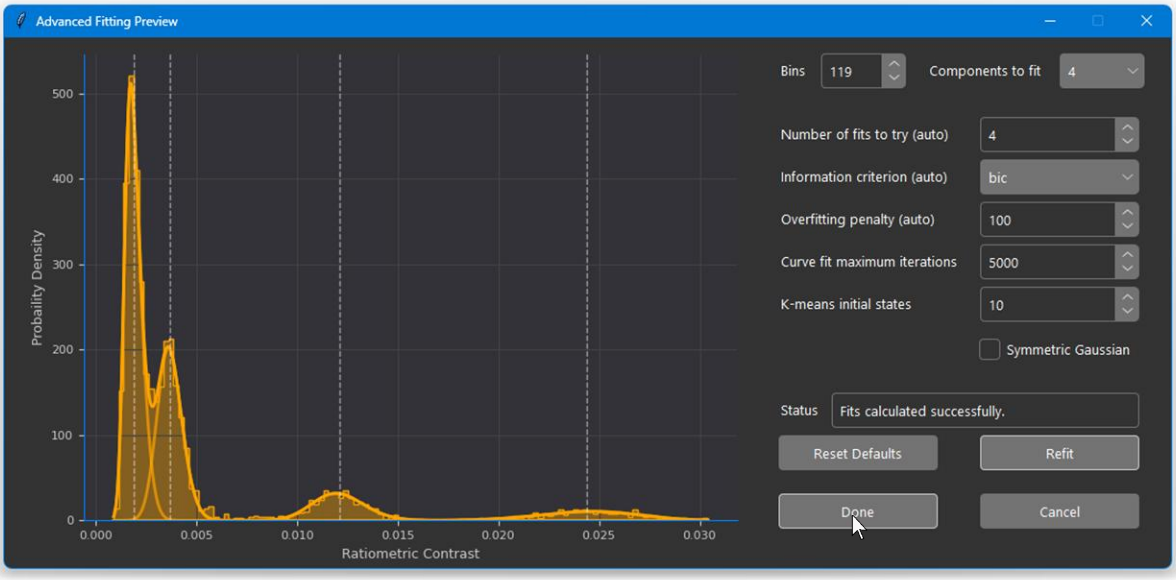
Next, use the mouse to drag over the region containing the peaks you want to fit. It's best to fit binding and unbinding peaks separately:



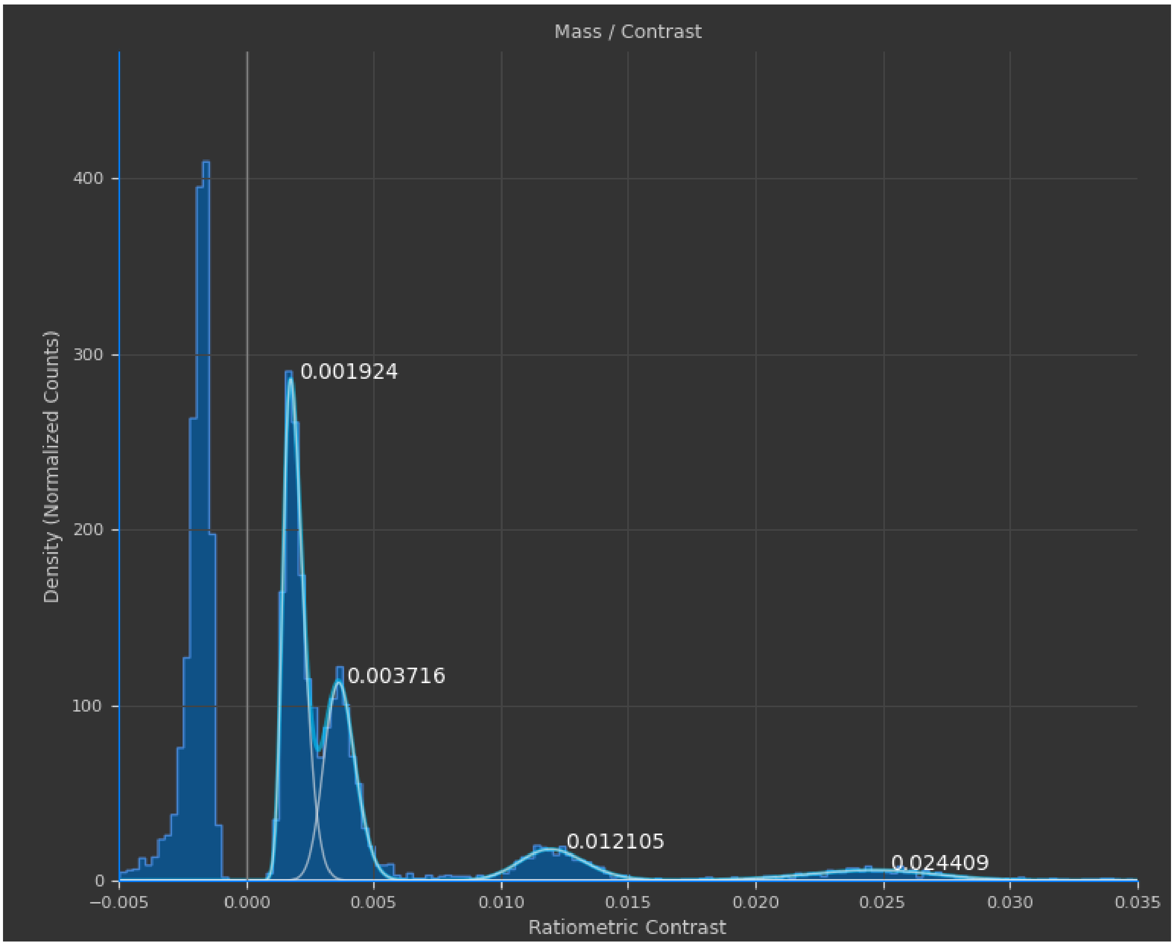
The advanced fitting window will open and again try to fit the region automatically. If the automatic fit fails to fit all peaks correctly, you can manually select the number of peaks to fit:



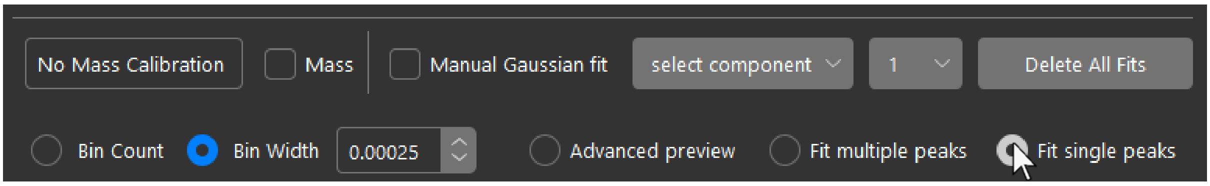
Once the correct number of components to fit has been manually selected, click 'Refit'. Then once the peaks are fitted correctly click 'Done':



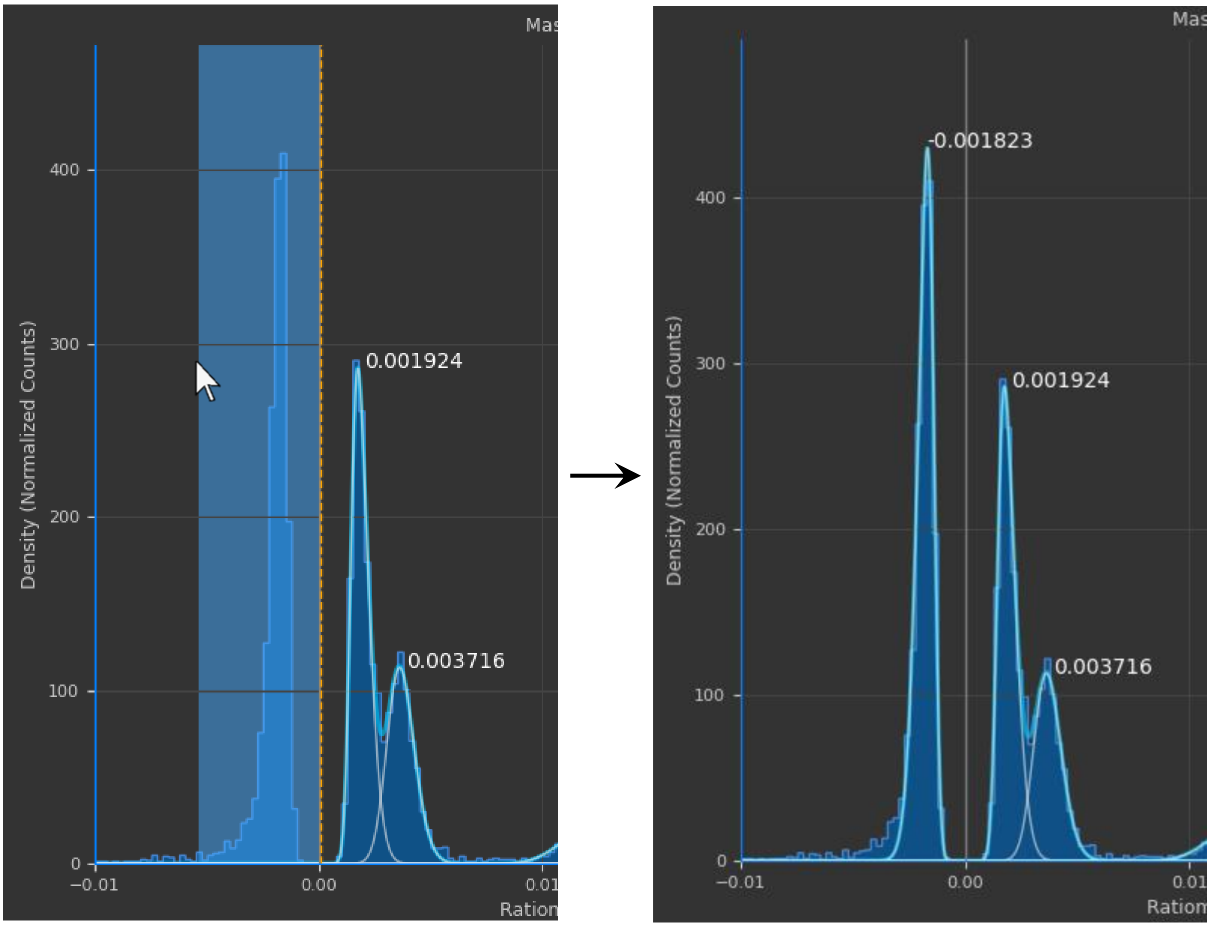
The advanced fitting window will then close and the fits in the histogram in the main UI will be updated:



To fit peaks individually, select the ‘Fit single peaks’ radio button in the histogram ribbon:

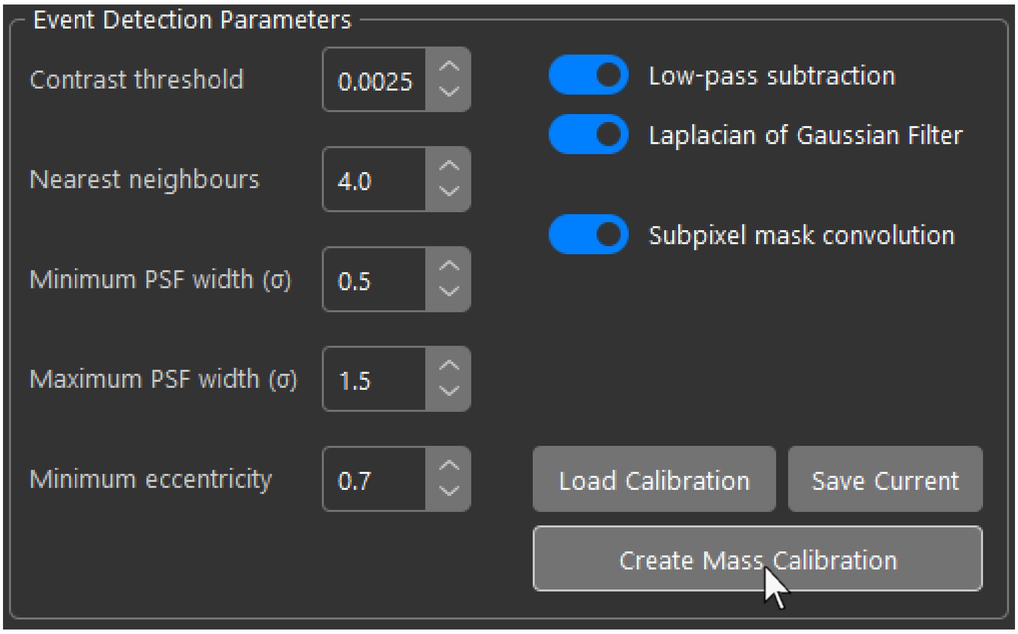


Then, once again, drag the mouse over the peak you want to fit:

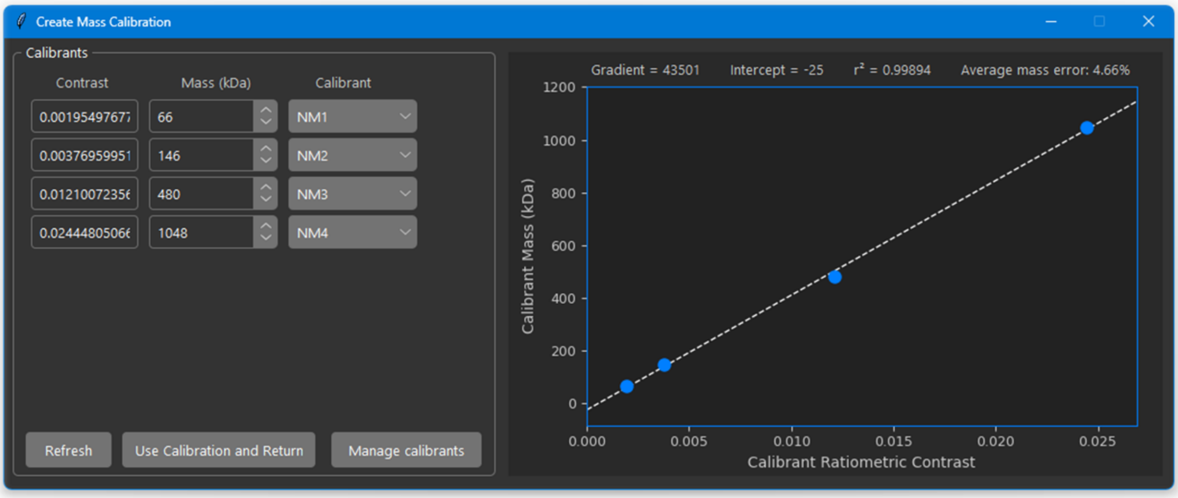


As opposed to DiscoverMP, OpenMASS uses a convention of positive contrasts for landing (adsorption) events and negative contrasts for unbinding (desorption) events for mass calibration, time series analysis, and histogram plotting. Events are on the other hand used in their raw state (negative landing contrast) when viewing ratiometric traces and exporting events or contrasts to an Excel spreadsheet. Mass calibrations use only binding peaks (positive contrast convention).

To create a mass calibration using the fitted peaks, click the 'Create Mass Calibration' button in the centre of the UI under the event detection parameters:

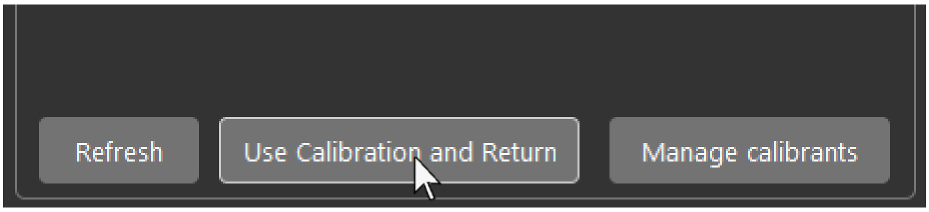


OpenMASS will open the mass calibration interface and automatically assign the expected 4 peaks of the NativeMark:

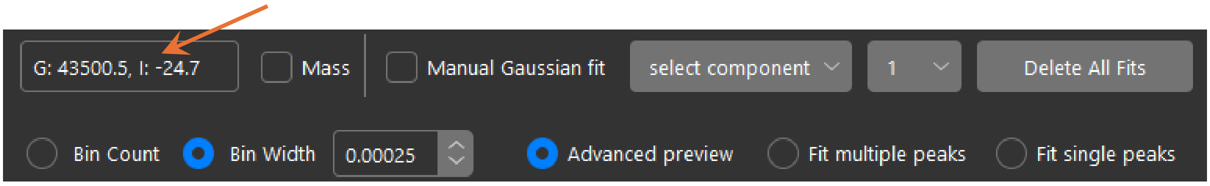


OpenMASS uses the following mass calibration convention. Expected calibrant mass (y) is plotted against measured contrast (x) using the positive landing contrast convention. This gives an intuitive positive gradient relationship where mass is obtained using simple linear regression, y = mx + c, where y is the mass and x is the plugged in contrast value of each event using the positive landing contrast convention. A subtle but important detail is that binding and unbinding events are handled differently. Landing events use y = mx + c and unbinding events use y = mx – c, where m is the fitted gradient of the mass calibration and c is the intercept. This keeps binding and unbinding events symmetrical about the line x=0 such that binding and unbinding events of the same contrast are equal in magnitude when converted to mass.

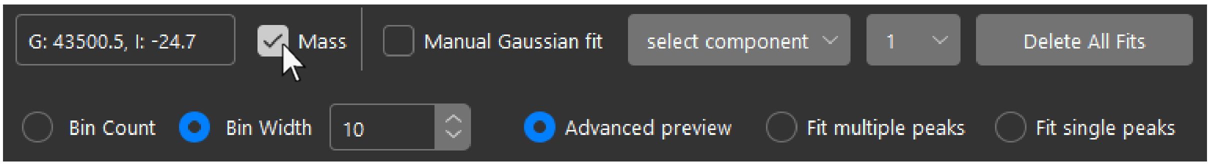
Once happy with the mass calibration, click 'Use Calibration and Return' in the mass calibration window:



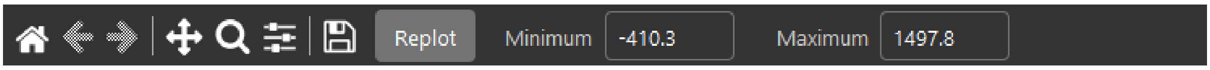
The window will close automatically and apply the mass calibration. The gradient and intercept are displayed in the top left corner of the histogram ribbon:



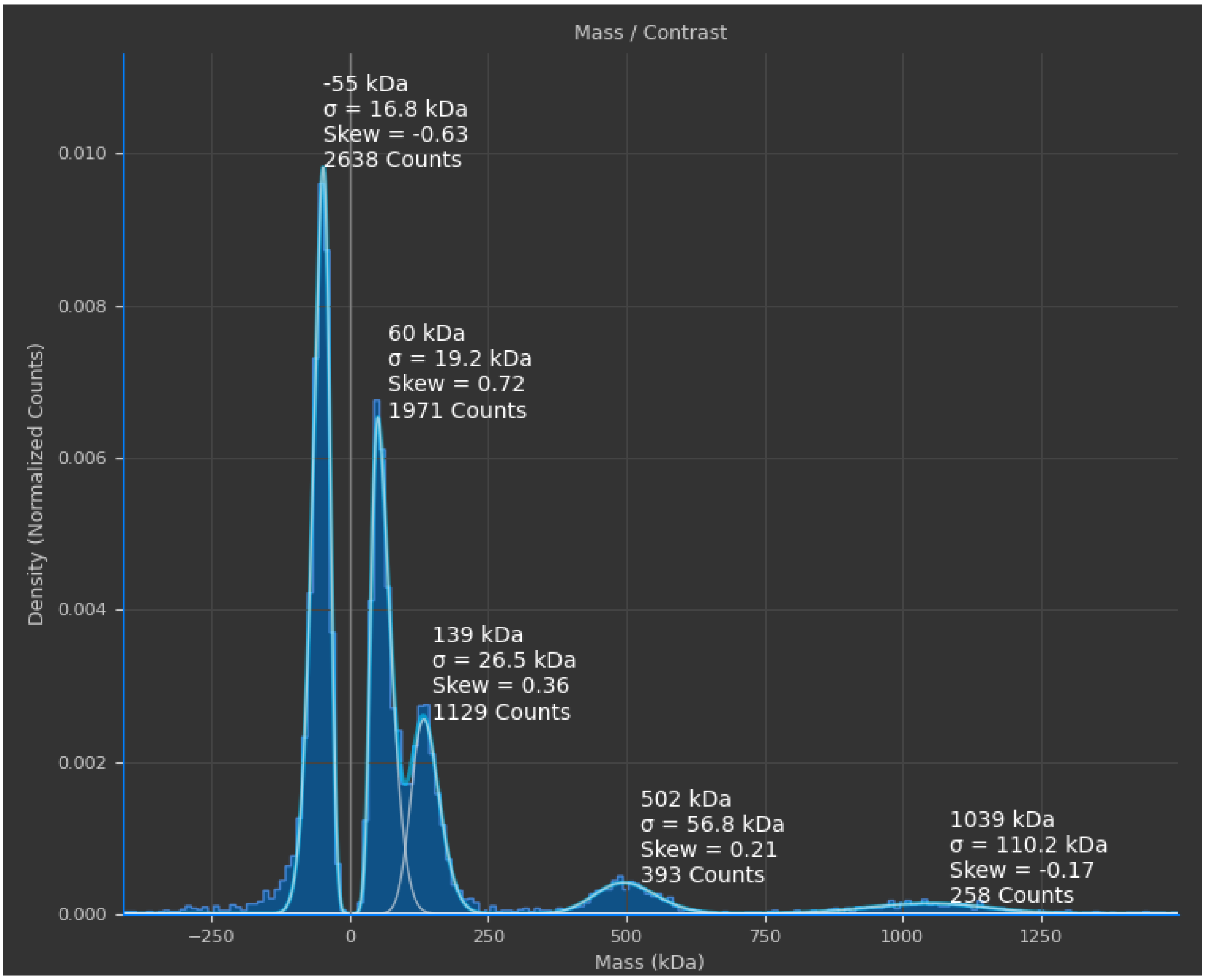
Check the 'Mass' checkbox in the top of the histogram ribbon to switch the histogram from contrast to mass mode:



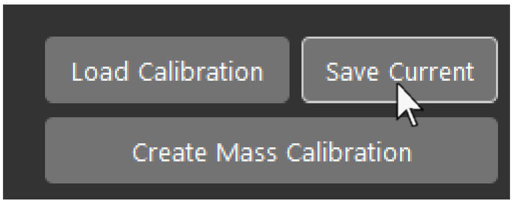
Bin width and the minimum and maximum axis boundaries, located in the histogram toolbar below the plot, will also be automatically converted to masses (kDa):



In mass mode, the fitted peaks have additional annotations including peak modal mass, standard deviation, skew, and integration (counts):



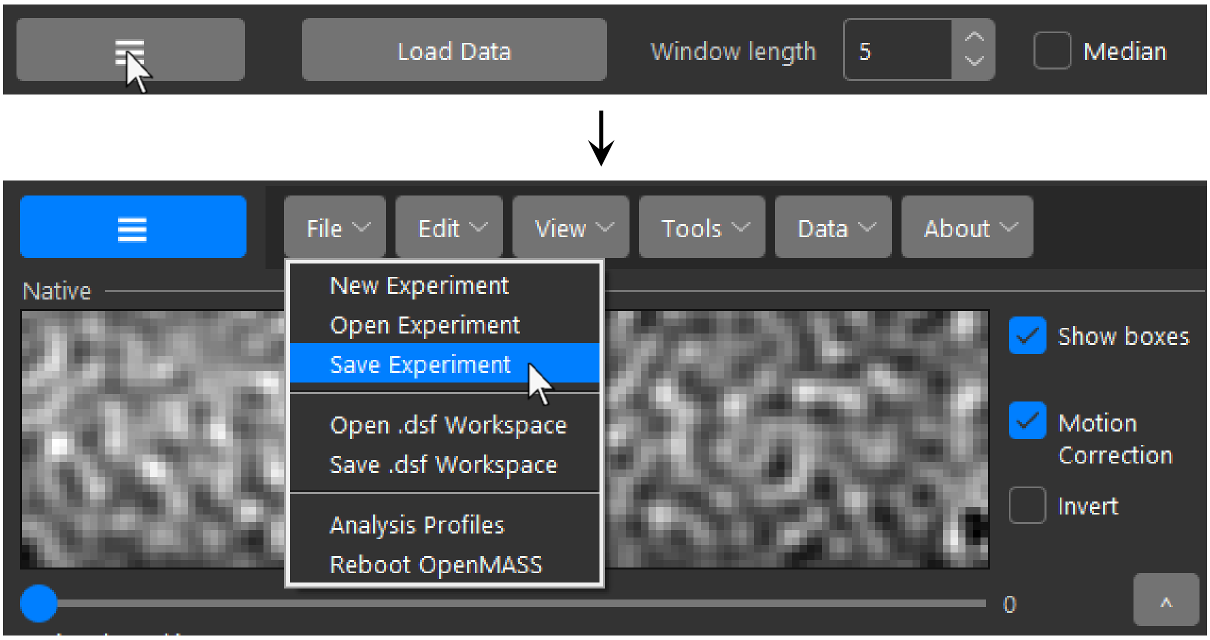
Save the mass calibration by clicking ‘Save Current’ under the ‘Create Mass Calibration’ button:



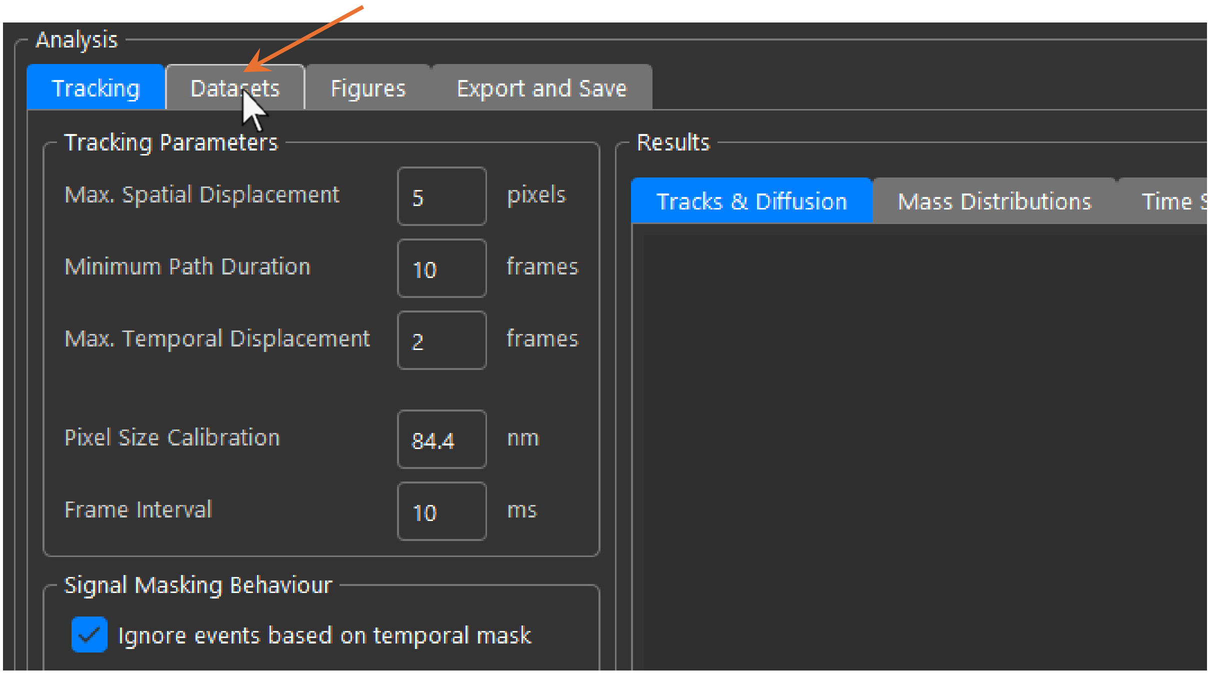
Ensure the name of the mass calibration ends with the ‘.mc’ extension:

It is of importance to note that '.mc' mass calibrations made in OpenMASS are not compatible with Refeyn DiscoverMP and vice versa.

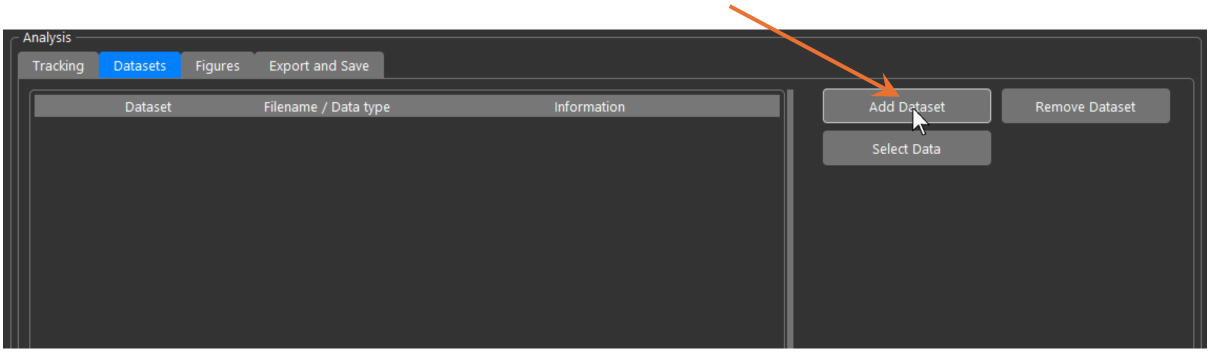
To save the workspace, settings, and mass calibration into a file that can be reopened in the exact state it was saved as, open the Menu Toolbar with the button in the top left of the UI, then 'File' → 'Save Experiment':



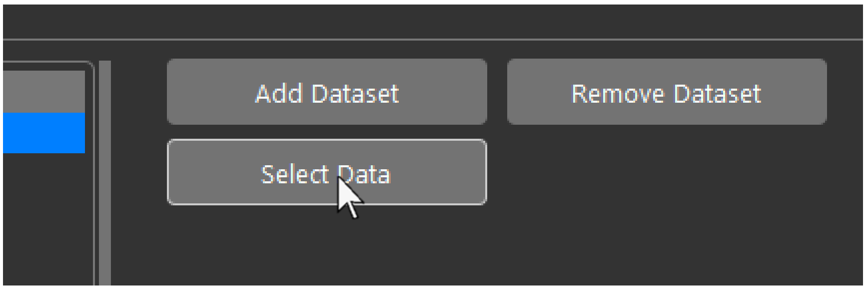
To extract datasets for making figures of the distribution and calibration plot, navigate to the dataset manager by switching to the 'Datasets' tab in the analysis panel of the UI:

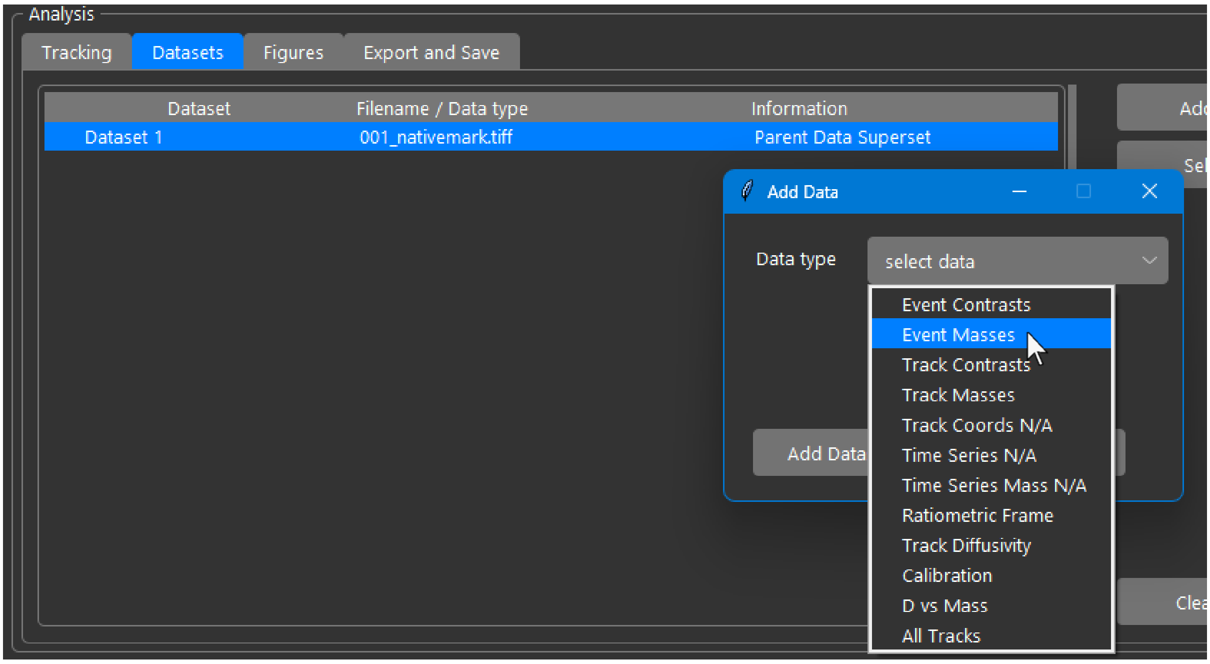


First, create a parent dataset to hold the different data types of the current file being analysed (The NativeMark calibration movie in this case) by clicking 'Add Dataset' in the Datasets tab:

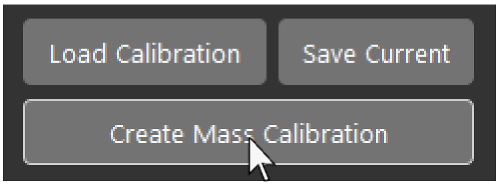


Next, click 'Select Data'. The selection window will appear. Select 'Event Masses' in the drop-down menu and click 'Add Dataset. The event masses data will be added to the currently selected parent Dataset:

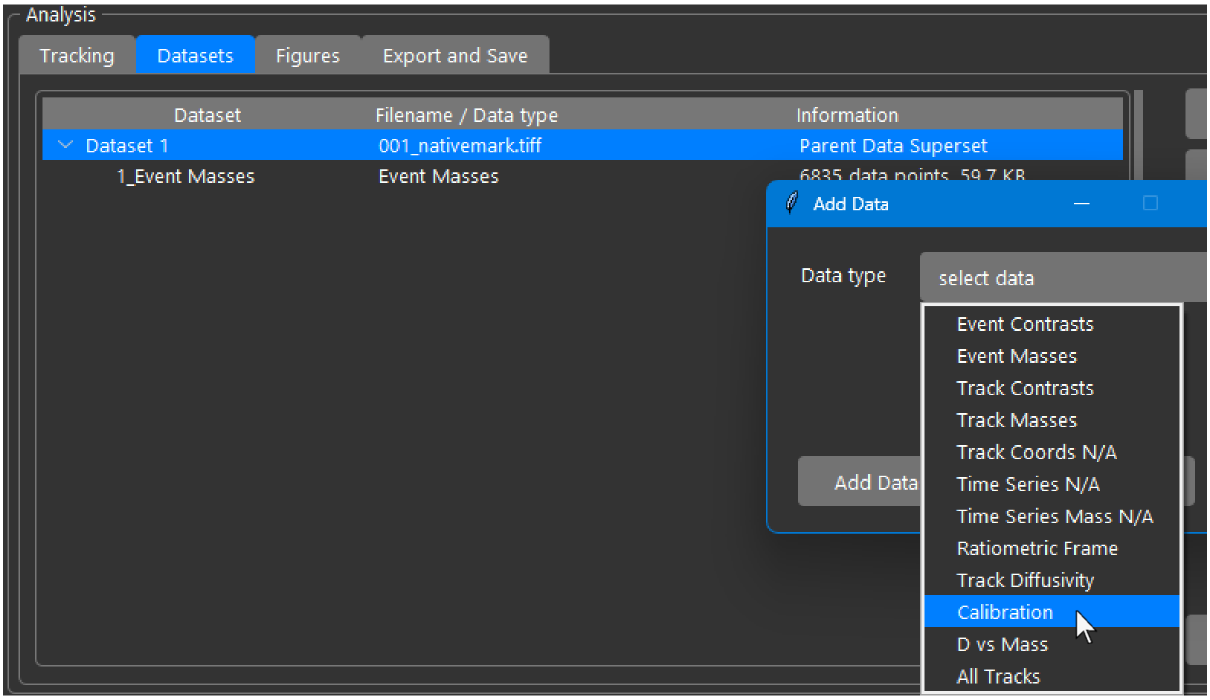




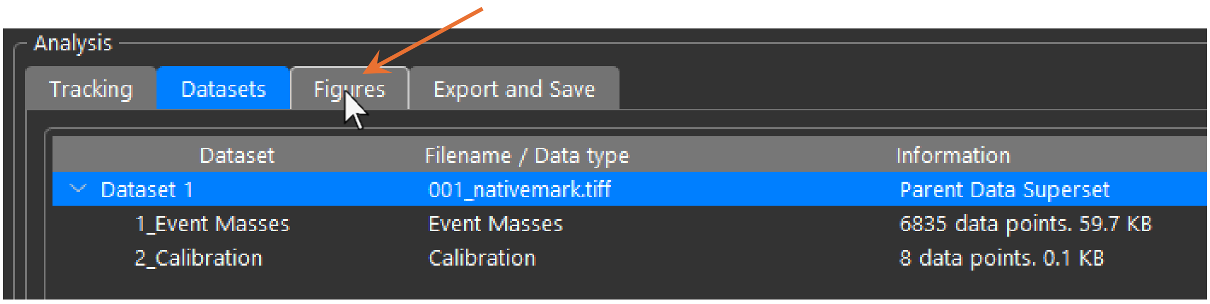
Next, open the mass calibration window again by clicking 'Create Mass Calibration' like before:



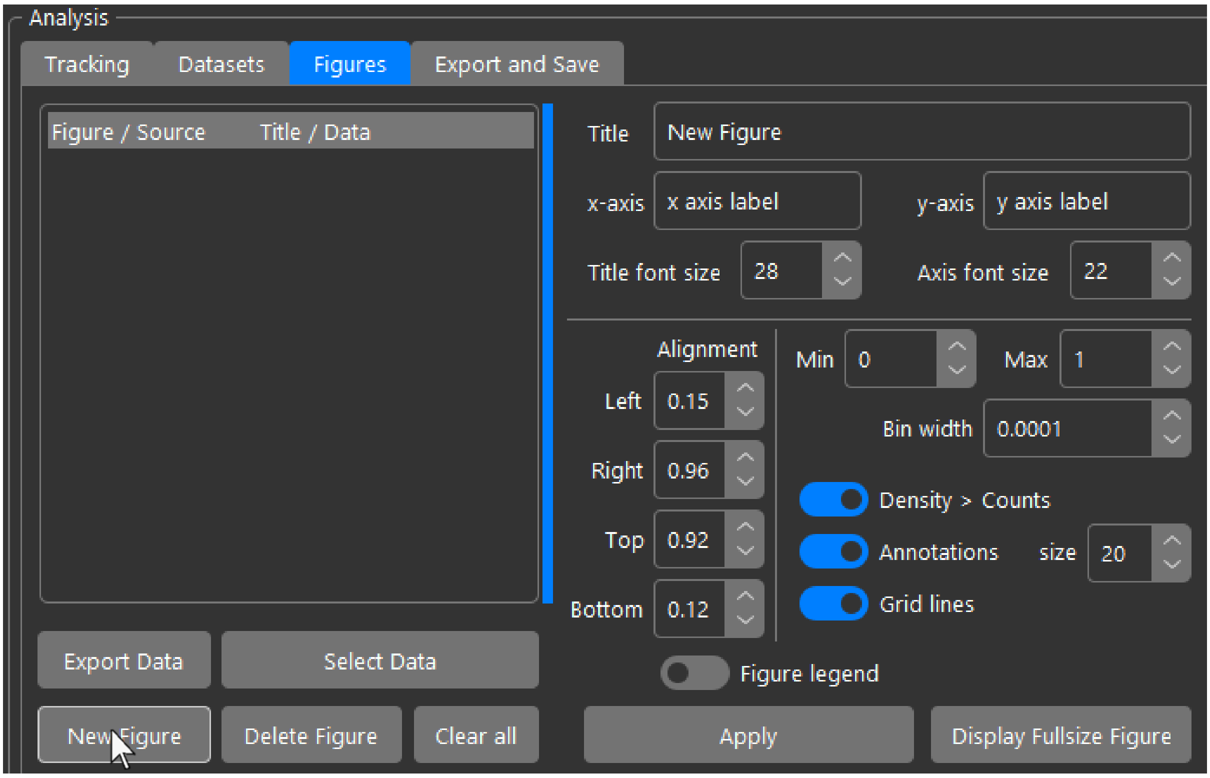
Again, like before, click 'Select Data' in the 'Datasets' tab and select 'Calibration' from the drop-down menu and click 'Add Dataset' to add calibration data to the parent dataset. Then close the mass calibration window:



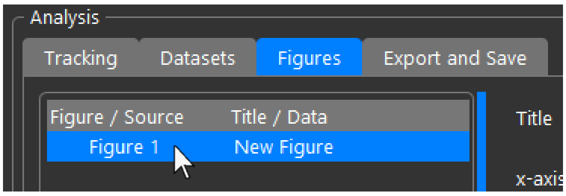
To start creating figures of the datasets, navigate to the 'Figures' tab in the analysis panel of the UI:

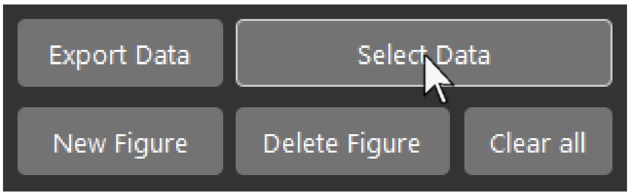


In the ‘Figures’ tab, click ‘New Figure’ in the bottom left corner:

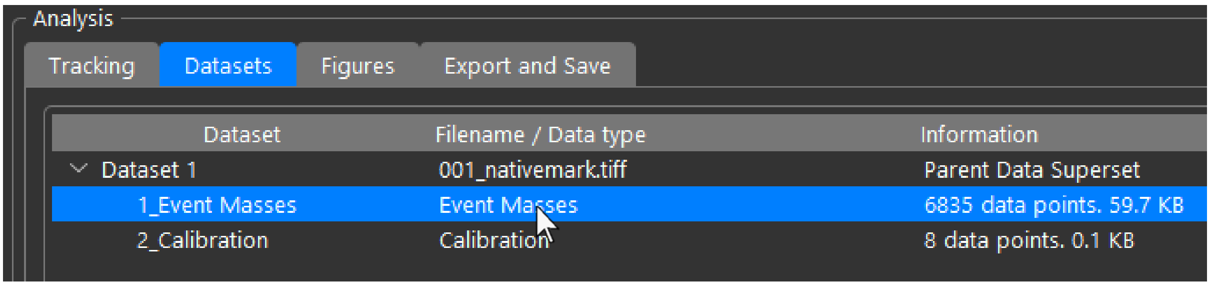


Then, either right click on the newly created figure in the treeview, or click ‘Select Data’ with the new figure selected:

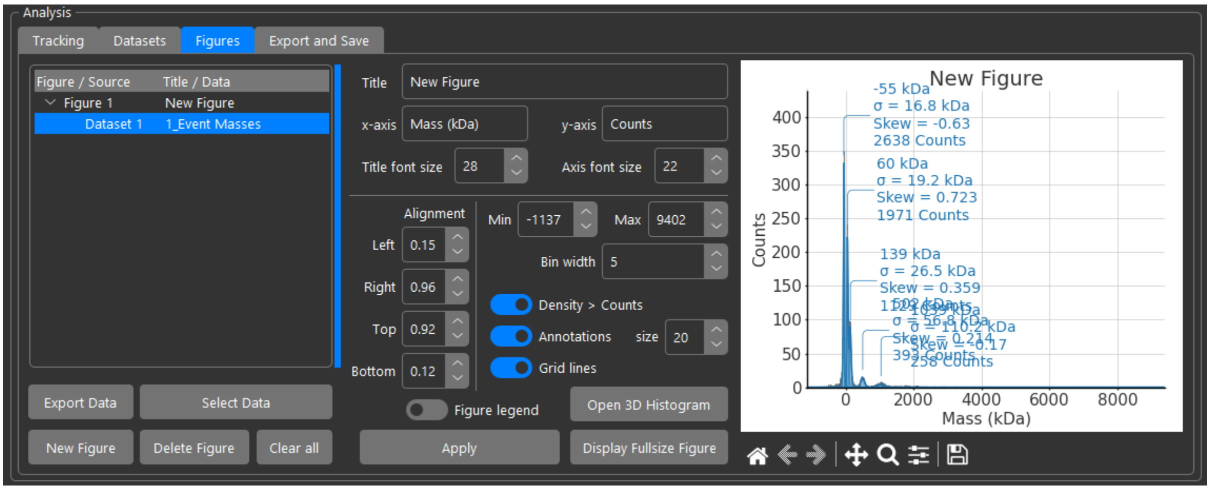




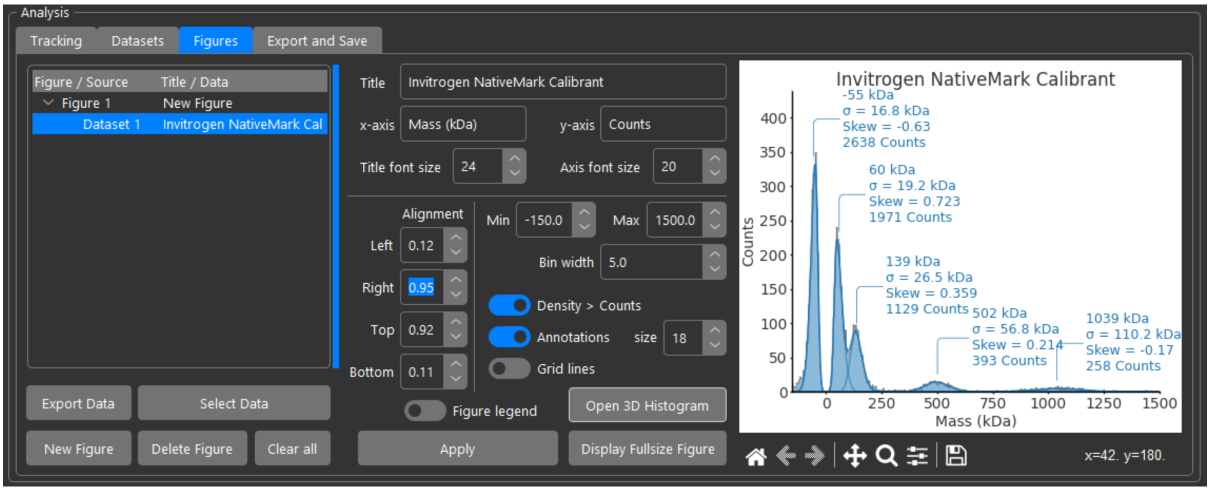
OpenMASS will then automatically bring up the ‘Dataset’ tab. Double click the dataset you wish to add to the figure. Let’s start with the event masses:



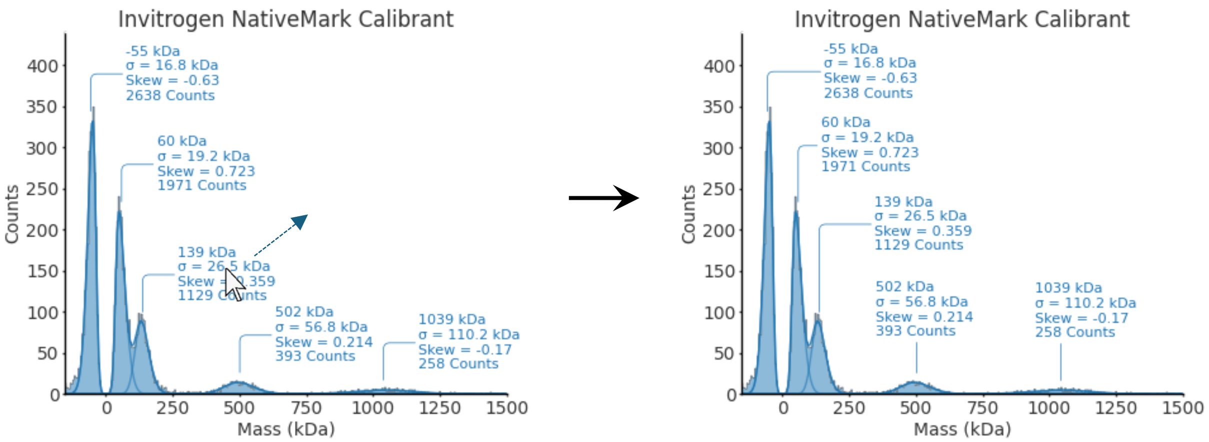
OpenMASS will then automatically return to the ‘Figures’ tab and add the data to the figure in the treeview as well as plot it in the inline canvas:



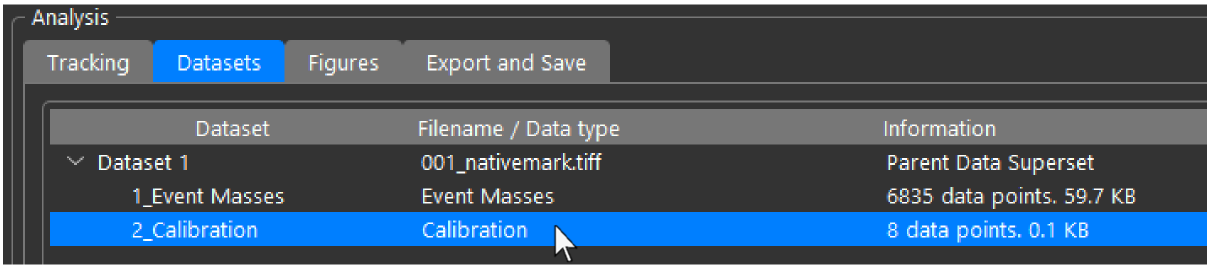
Next, you can customize the font sizes, alignment, axis text, and title etc. as well as setting the axis limits and configuring visual elements such as grid lines:

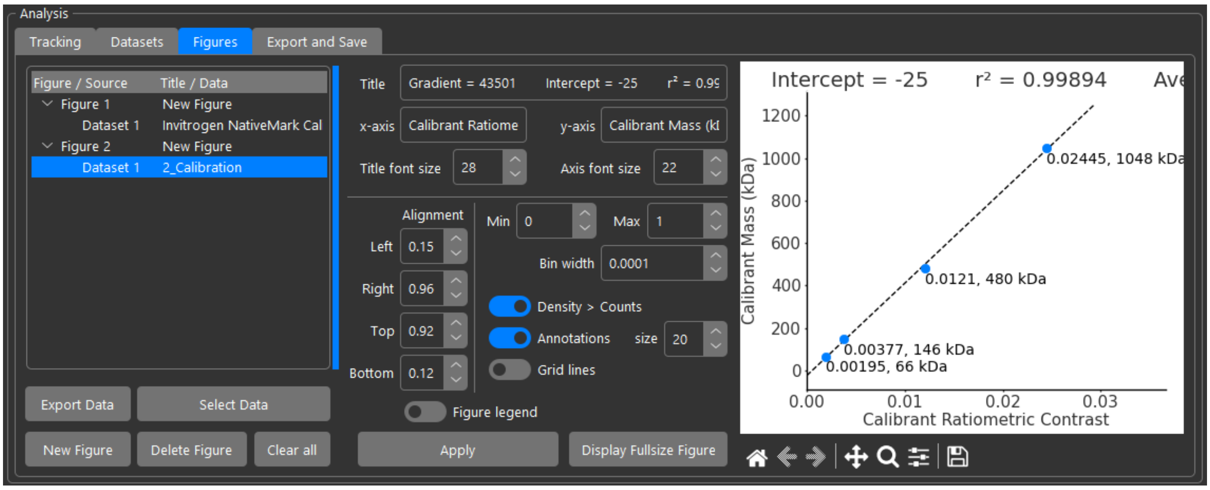


Peak annotations can be resized and dragged to favourable locations:

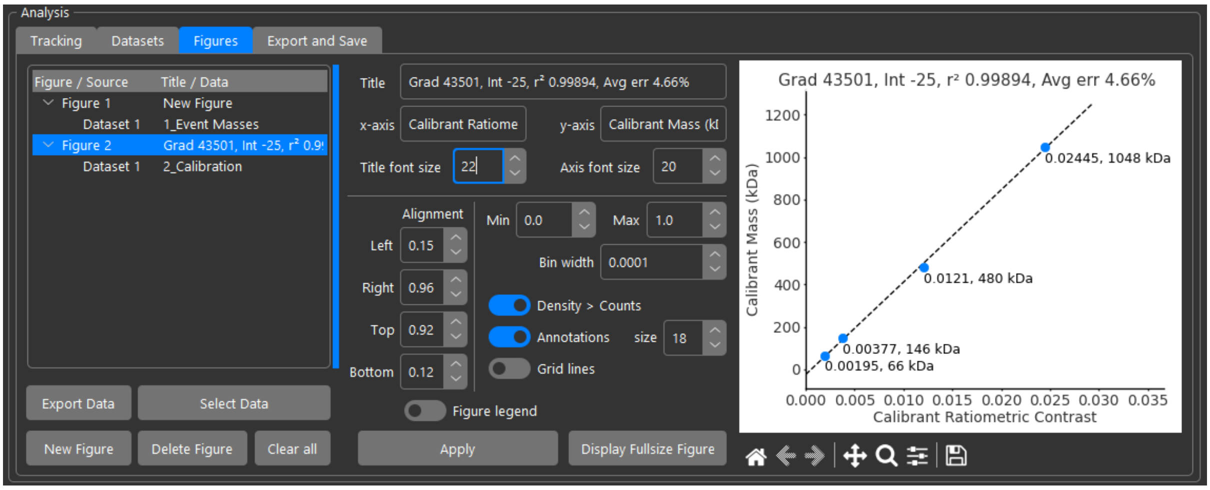


Click ‘New Figure’ again like before. Then click ‘Select Data’ and double click on the Calibration dataset:



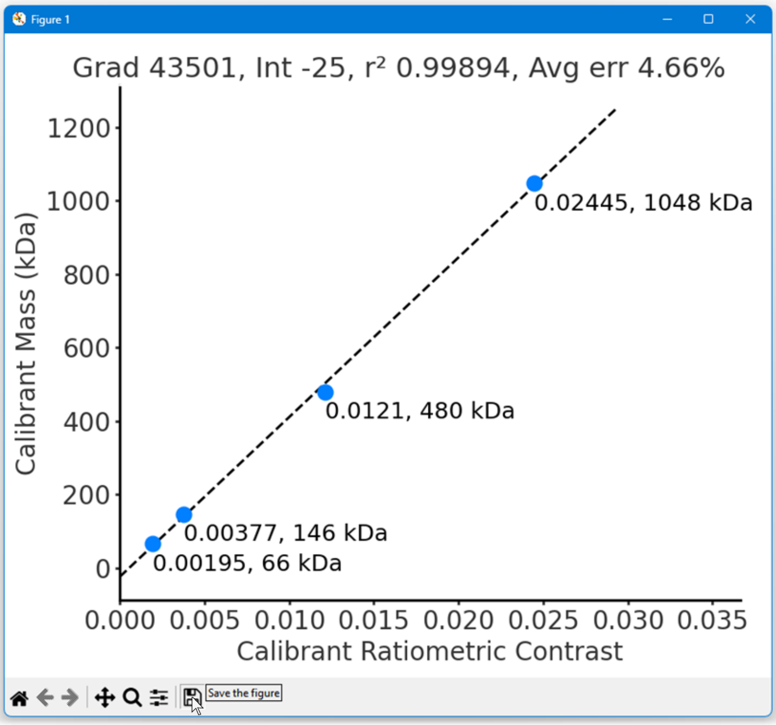


Adjust text, font sizes, and alignment until you’re happy with the result:



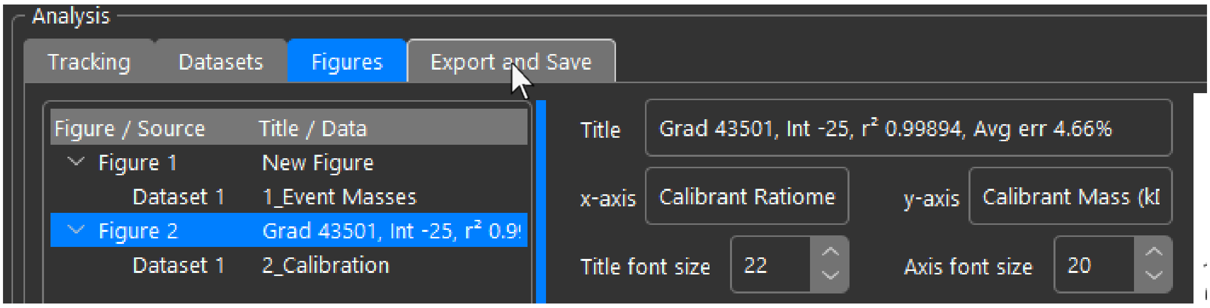
To display the figure in a larger floating window, click ‘Display Fullsize Figure’:

You can then save the figure from the toolbar:

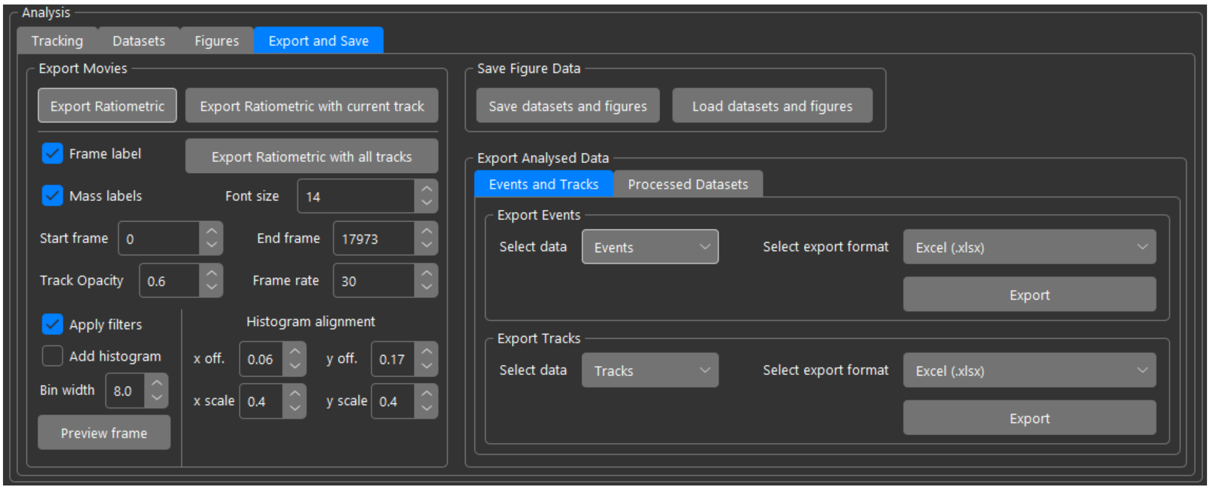


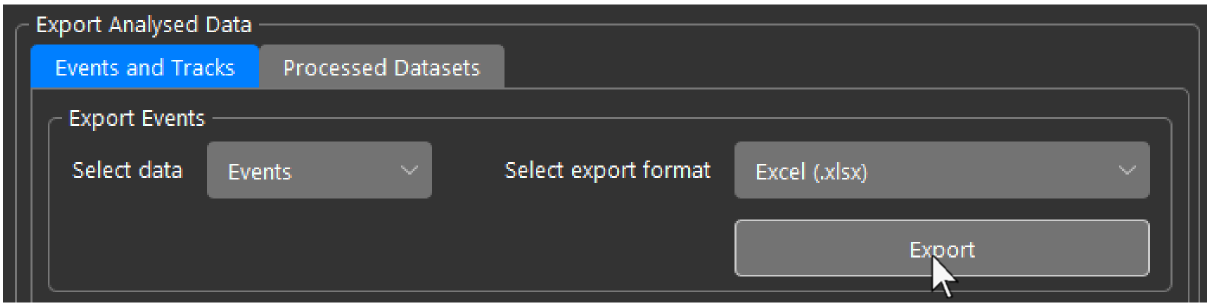
Figures can also be saved directly from the inline preview. This is fine if saving to vector formats such as '.svg'. Saving as '.png' is recommended only from the full-sized figure to maximise resolution. The full-sized figure can also be resized by dragging the corner of the window, although this will break consistency in sizing of the saved images between figures which may be important to preserve if using for publications.

To export data, navigate to the 'Export and Save' tab in the analysis panel of the UI:

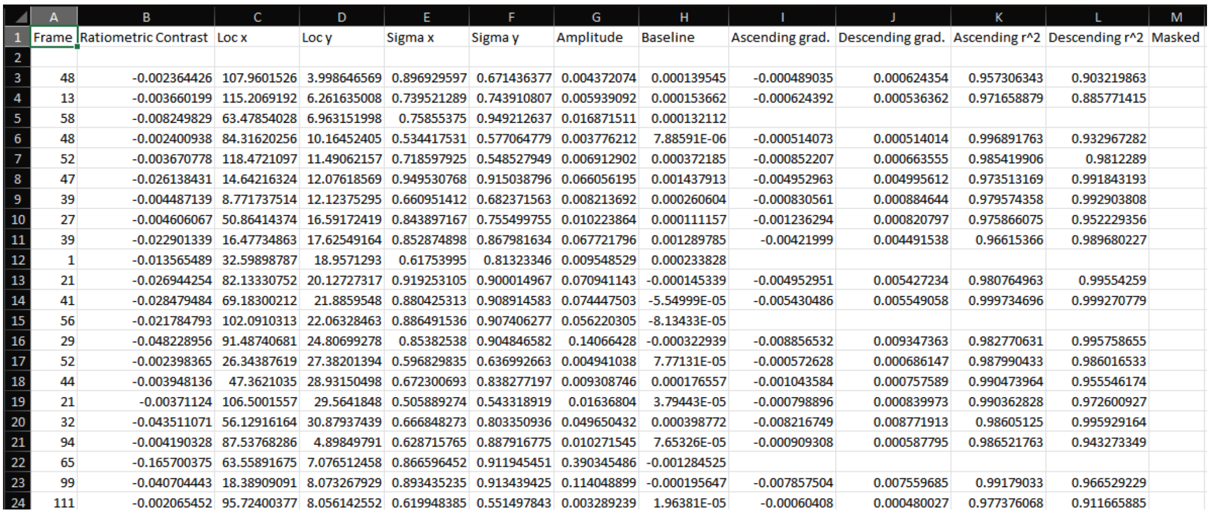


For example, to export all events detected in the movie to an Excel spreadsheet, ensure that the data type selected is ‘Events’ and the export format is set to ‘Excel (.xlsx)’ and click ‘Export’ and save the file using file explorer:





The output looks as follows:



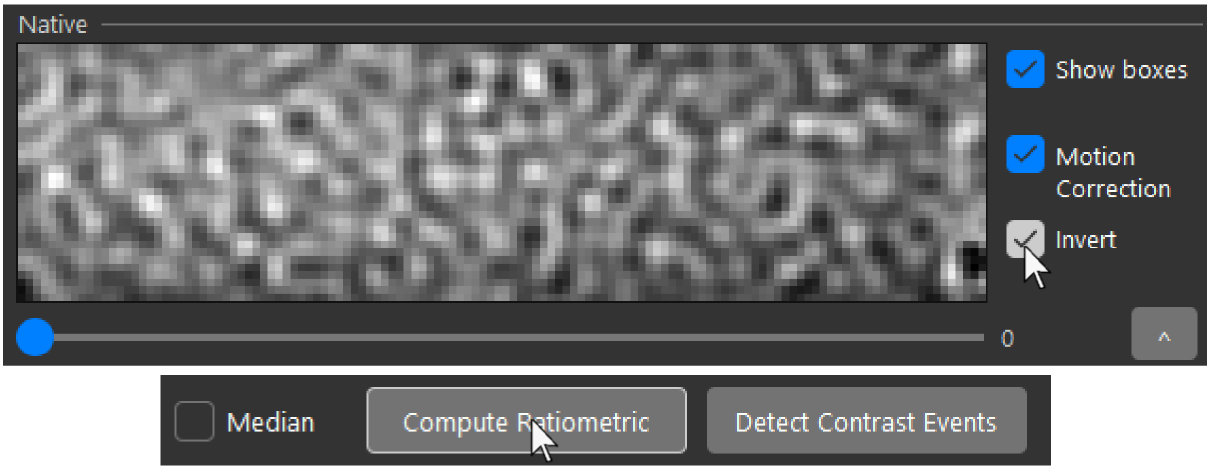
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Frame	Ratiometric Contrast	Loc x	Loc y	Sigma x	Sigma y	Amplitude	Baseline	Ascending grad.	Descending grad.	Ascending r^2	Descending r^2	Masked
2													
3	48	-0.002364426	107.9601526	3.998646569	0.896929597	0.671436377	0.004372074	0.000139545	-0.000489035	0.000624354	0.957306343	0.903219863	
4	13	-0.003660199	115.2069192	6.261635008	0.739521289	0.743910807	0.005939092	0.000153662	-0.000624392	0.000536362	0.971658879	0.885771415	
5	58	-0.008249829	63.47854028	6.963151998	0.75855375	0.949212637	0.016871511	0.000132112					
6	48	-0.002400938	84.31620256	10.16452405	0.534417531	0.577064779	0.003776212	7.88591E-06	-0.000514073	0.000514014	0.996891763	0.932967282	
7	52	-0.003670778	118.4721097	11.49062157	0.718597925	0.548527949	0.006912902	0.000372185	-0.000852207	0.000663555	0.985419906	0.9812289	
8	47	-0.026138431	14.64216324	12.07618569	0.949530768	0.915038796	0.066056195	0.001437913	-0.004952963	0.004995612	0.973513169	0.991843193	
9	39	-0.004487139	8.771737514	12.12375295	0.660951412	0.682371563	0.008213692	0.000260604	-0.000830561	0.000884644	0.979574358	0.992903808	
10	27	-0.004606067	50.86414374	16.59172419	0.843897167	0.755499755	0.010223864	0.000111157	-0.001236294	0.000820797	0.975866075	0.952229356	
11	39	-0.022901339	16.47734863	17.62549164	0.852874898	0.867981634	0.067721796	0.001289785	-0.00421999	0.004491538	0.96615366	0.989680227	
12	1	-0.013565489	32.59898787	18.9571293	0.61753995	0.81323346	0.009548529	0.000233828					
13	21	-0.026944254	82.13330752	20.12727317	0.919253105	0.900014967	0.070941143	-0.000145339	-0.004952951	0.005427234	0.980764963	0.99554259	
14	41	-0.028479484	69.18300212	21.8859548	0.880425313	0.908914583	0.074447503	-5.54999E-05	-0.005430486	0.005549058	0.999734696	0.999270779	
15	56	-0.021784793	102.0910313	22.06328463	0.886491536	0.907406277	0.056220305	-8.13433E-05					
16	29	-0.048228956	91.48740681	24.80699278	0.85382538	0.904846582	0.14066428	-0.000322939	-0.008856532	0.009347363	0.982770631	0.995758655	
17	52	-0.002398365	26.34387619	27.38201394	0.596829835	0.636992663	0.004941038	7.77131E-05	-0.000572628	0.000686147	0.987990433	0.986016533	
18	44	-0.003948136	47.3621035	28.93150498	0.672300693	0.838277197	0.009308746	0.000176557	-0.001043584	0.000757589	0.990473964	0.955546174	
19	21	-0.00371124	106.5001557	29.5641848	0.505889274	0.543318919	0.01636804	3.79443E-05	-0.000798896	0.000839973	0.990362828	0.972600927	
20	32	-0.043511071	56.12916164	30.87937439	0.666848273	0.803350936	0.049650432	0.000398772	-0.008216749	0.008771913	0.98605125	0.995929164	
21	94	-0.004190328	87.53768286	4.89849791	0.628715765	0.887916775	0.010271545	7.65326E-05	-0.000909308	0.000587795	0.986521763	0.943273349	
22	65	-0.165700375	63.55891675	7.076512458	0.866596452	0.911945451	0.390345486	-0.001284525					
23	99	-0.040704443	18.38909091	8.073267929	0.893435235	0.913439425	0.114048899	-0.000195647	-0.007857504	0.007559685	0.99179033	0.966529229	
24	111	-0.002065452	95.72400377	8.056142552	0.619948385	0.551497843	0.003289239	1.96381E-05	-0.00060408	0.000480027	0.977376068	0.911665885	

Chapter 2 – Ratiometric Mode

2.1 Landing Assays

For landing assays, standard ratiometric image background removal should be used. The window length should be small (3-10 frames) and the ‘Median’ checkbox should be unchecked. The default window length in the landing assays preset is 5 frames:

If the movie was acquired with the focus at a nearby local maximum of sharpness and landing events have inverted contrast (light centre, dark outline PSFs) instead of the normal (dark centre, light outline PSFs), check the ‘Invert’ checkbox to the right of the native stack, then recompute the ratiometric movie:



2.2 Dynamic Tracking

For tracking experiments, median background subtraction with a long window length (100-400 frames) should be used. The ‘Median’ checkbox should be checked. The default window length of the ‘Dynamic Tracking’ preset is 100 frames:

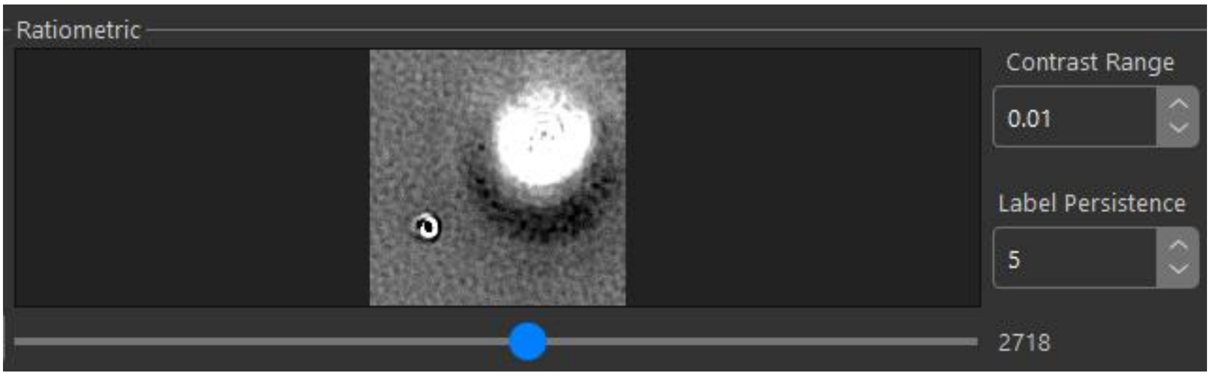
If contrast was acquired inverted, the ratiometric movie can be inverted back in the same way as shown above for landing assays.

Chapter 3 – Signal Masking

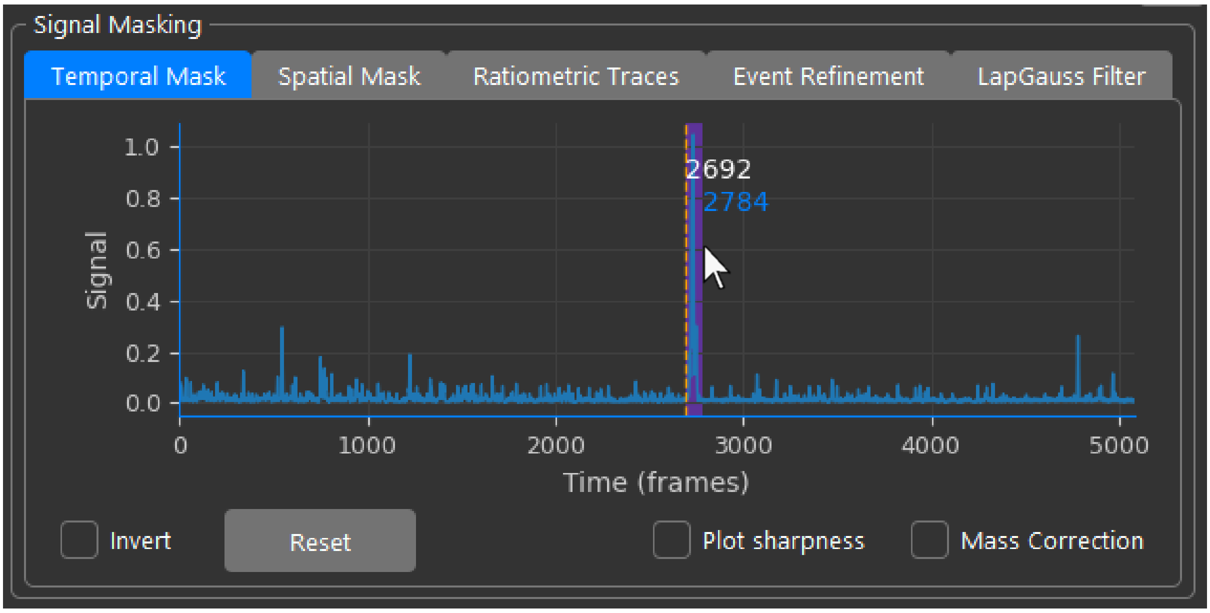
Signal masking is applied in a similar way to Refeyn DiscoverMP, allowing spatiotemporal selectivity of region in the recording as well as nearest neighbour exclusion (covered in event detection), and inclusion / exclusion of individual events. The logic for combining masks is 'OR' unless masks are inverted in which case they must be included in both.

3.1 Temporal Mask

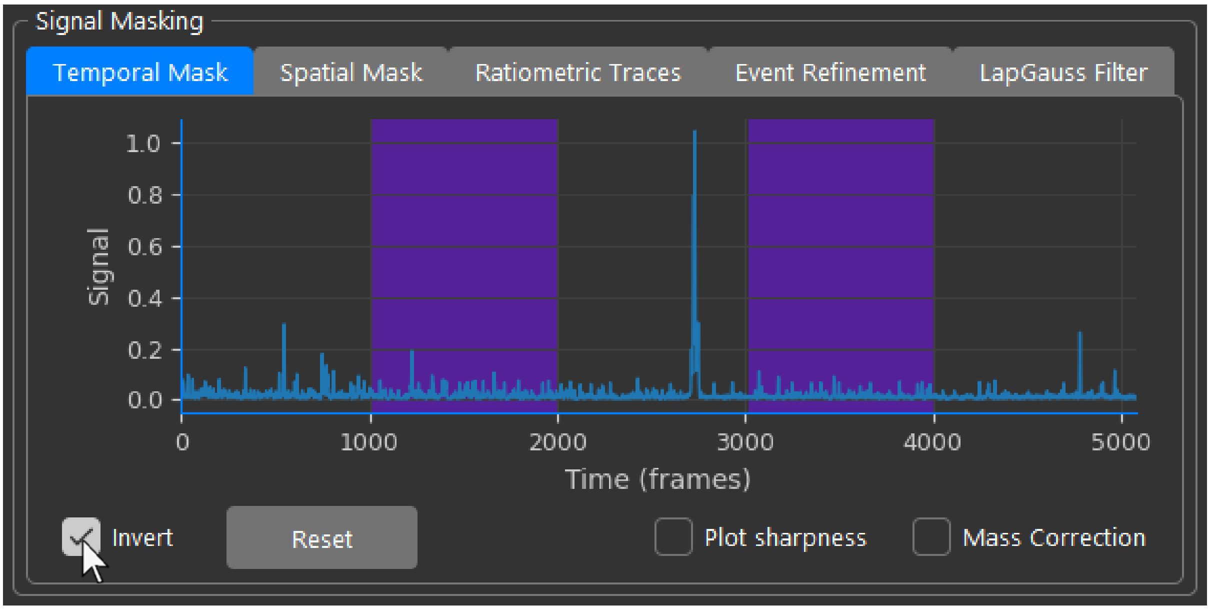
Temporal masking can be used to selectively exclude regions of time in the recording, for example the presence of large aggregates that pollute sections of the recording with distortions and artifacts:



Regions can often be identified by aberrantly strong signal strength. To apply a temporal mask, click and drag the mouse over the region of the signal vs time graph that you want to mask out:

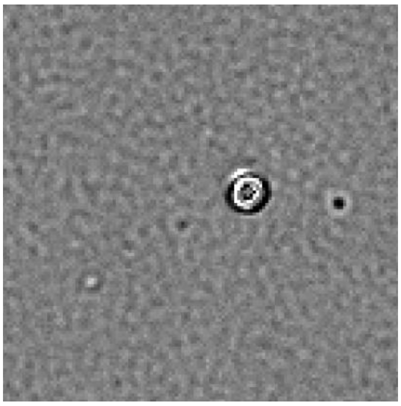


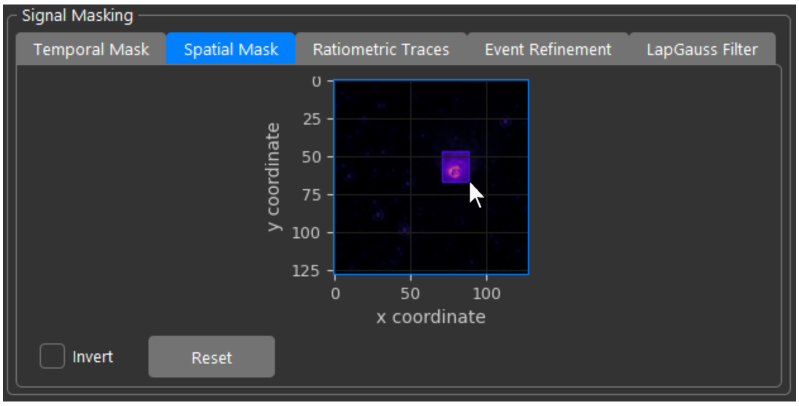
You can select as many regions as you wish, as the software creates a binary mask from the selections. Alternatively, if you want to keep only a specific region in time, you can invert the behaviour of the mask, this instead excludes the non-highlighted region:



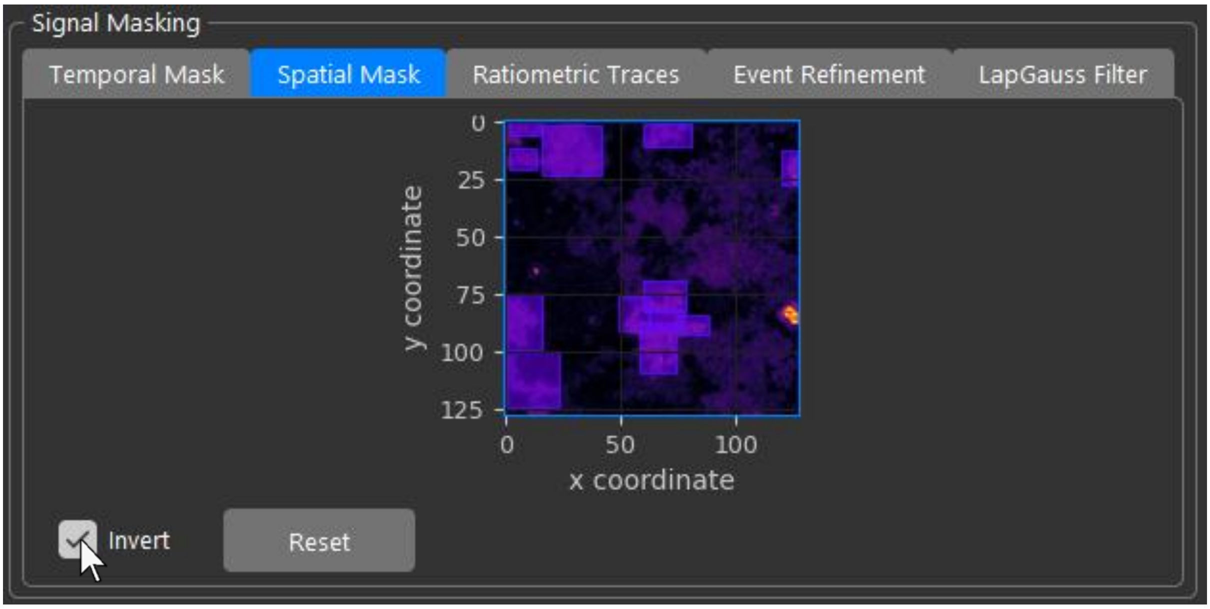
3.2 Spatial Mask

Sometimes, artifacts aren't just limited to a small region in time but may be present for the entire duration of a movie but localised in a small number of pixels in each frame. This can occur when exceptionally heavy particles such as aggregates land on the surface especially if their size is on par or larger then the PSF or they wobble. In this case, simply draw a rectangle over the affected are in the 'Spatial Mask' tab:



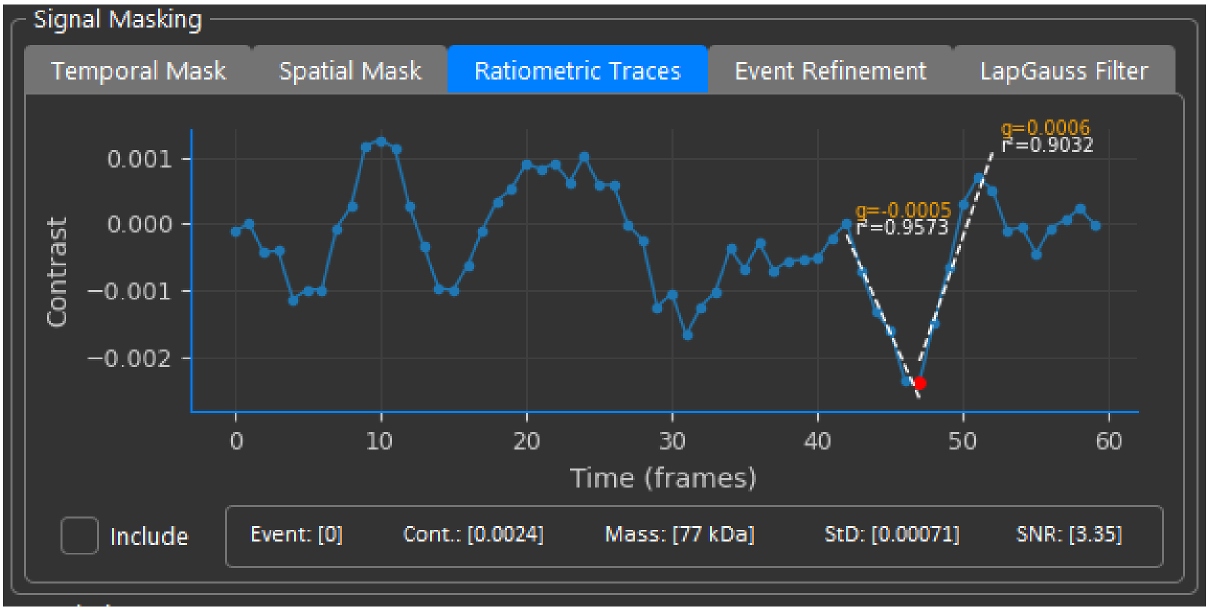


Just like the temporal mask, multiple selections can be made and the selection inverted to include only selected regions:

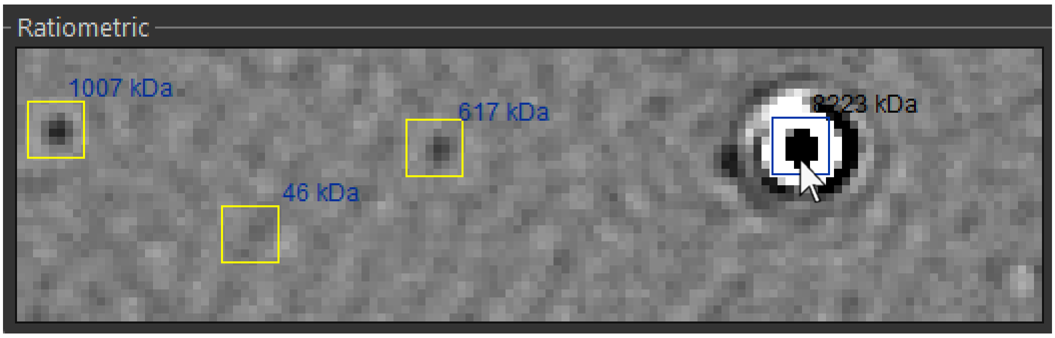


3.3 Event Mask

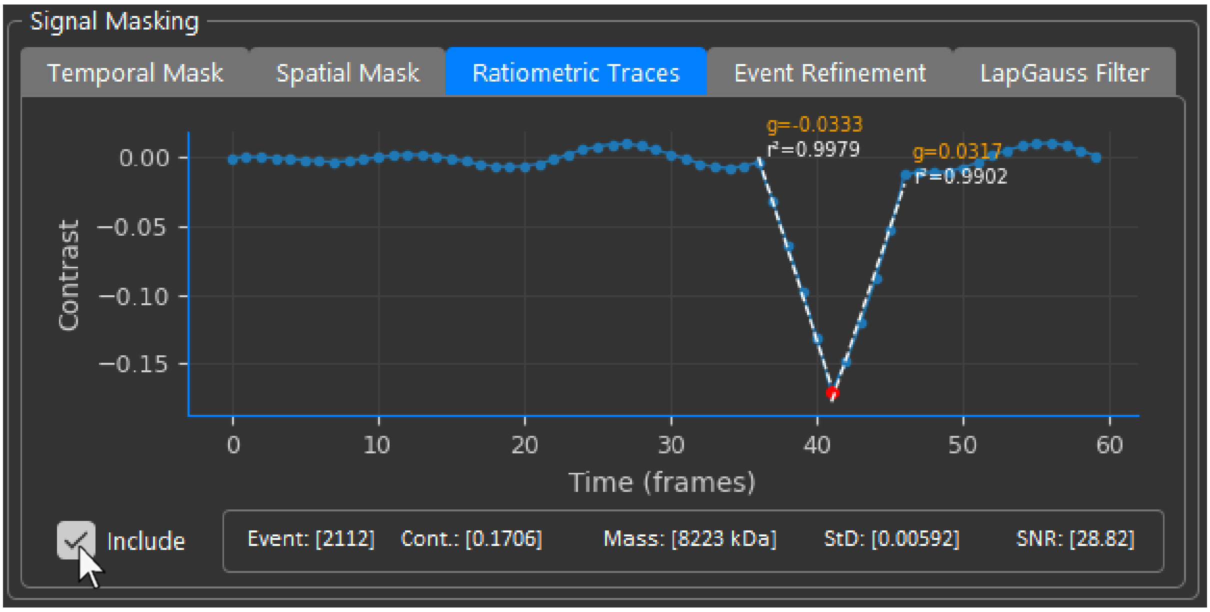
The event mask can only be used once events have been detected. Navigate to the 'Ratiometric Traces' tab:



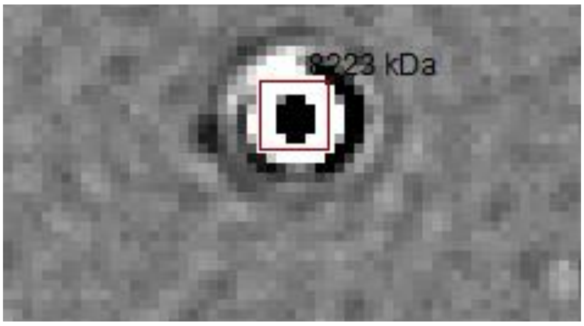
Next, click on the event of interest in the ratiometric view; the box colour will change from yellow to blue when selected:

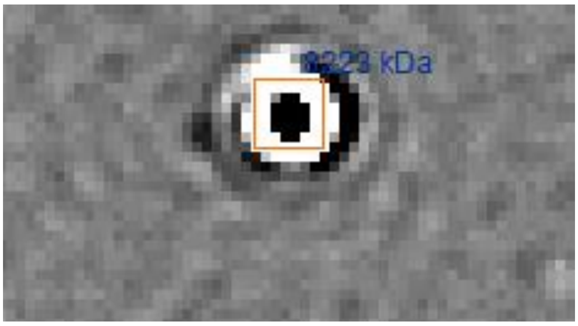


The ratiometric trace will update with the selected event. You can include or exclude the event by checking or unchecking the 'Include' checkbox:



Excluded events will show up in the ratiometric view in dark red when selected and orange when not selected:



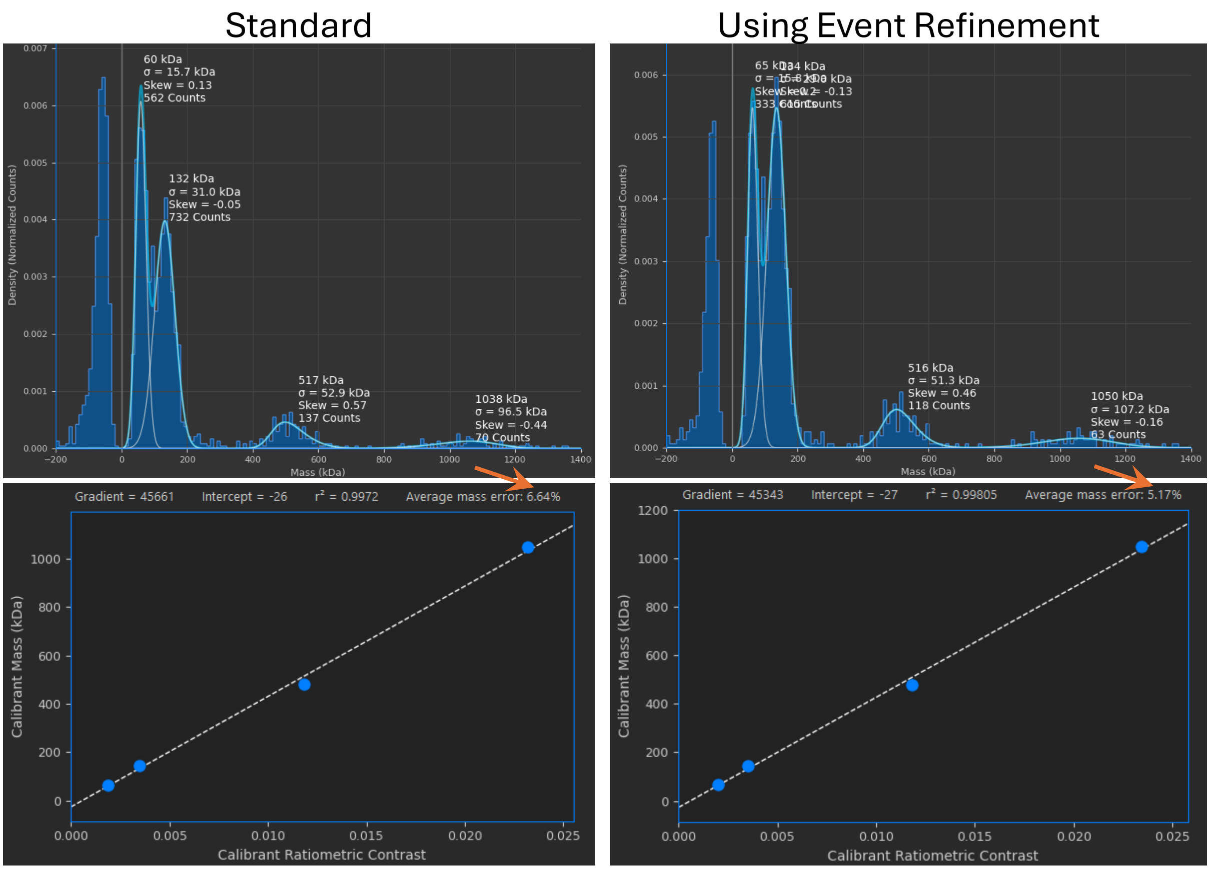


Care should be taken not to abuse this tool. It was specifically designed for removing individual outlier events where using the temporal and spatial masking systems over general areas was deemed too aggressive. It should not be used to cherry-pick data.

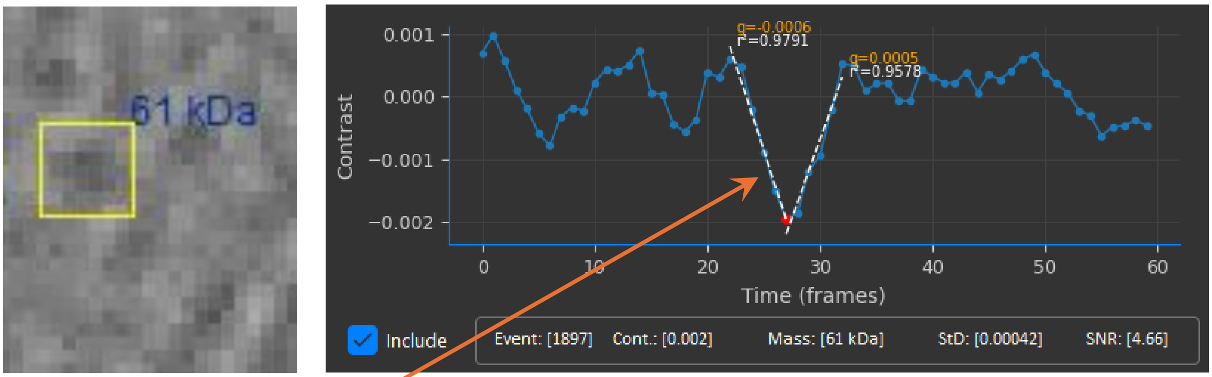
3.4 Event Refinement

While motion correction aims to remove oscillating background patterns, sometimes they are unavoidable, for example from low concentrations of detergents required for membrane proteins. This means the detection limit of the instrument is increased as the noise floor is higher, resulting in background fluctuations sometimes being mistaken for real events. This can result in an additional peak appearing between 40-60 kDa or peak broadening of low mass peaks in the 50-80 kDa range which can affect mass calibrations.

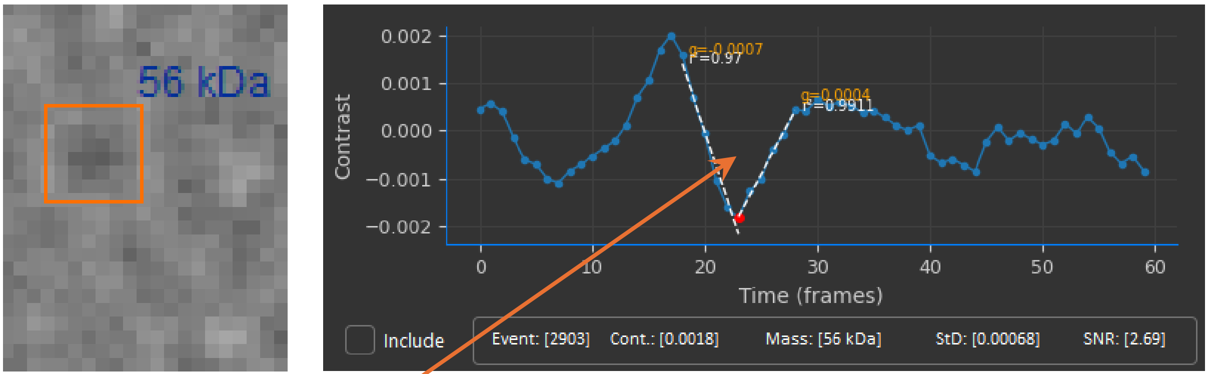
Event refinement uses thresholds based on ratiometric trace fitting and signal-to-noise ratio (SNR) to try and clean up events with poor fits and improve histogram resolution in the low mass regime:



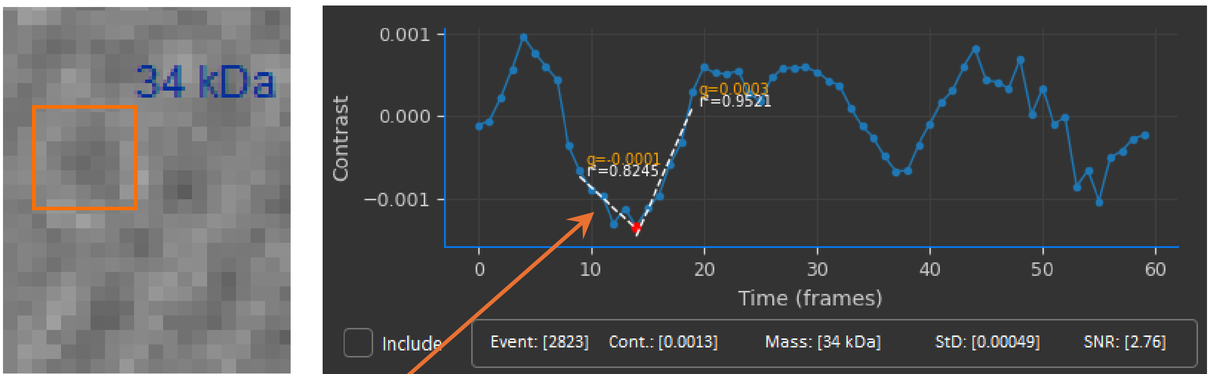
Events excluded by event refinement are coloured orange in the ratiometric view just likes manually excluded events. By clicking on included (yellow) or excluded (orange) events we can inspect their ratiometric traces and understand why an event was rejected:



Good quality of fit. Descending and ascending gradients closely match. Good SNR.



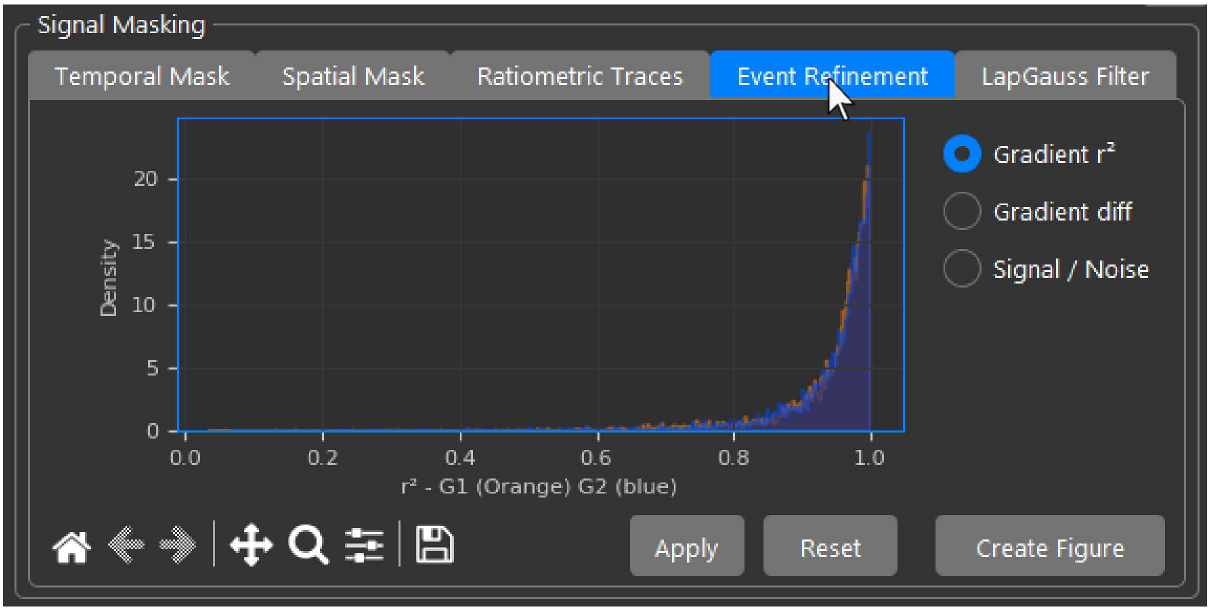
Good quality of fit. Descending and ascending gradients poorly match. Poor SNR.



Poor quality of fit. Descending and ascending gradients poorly match. Poor SNR.

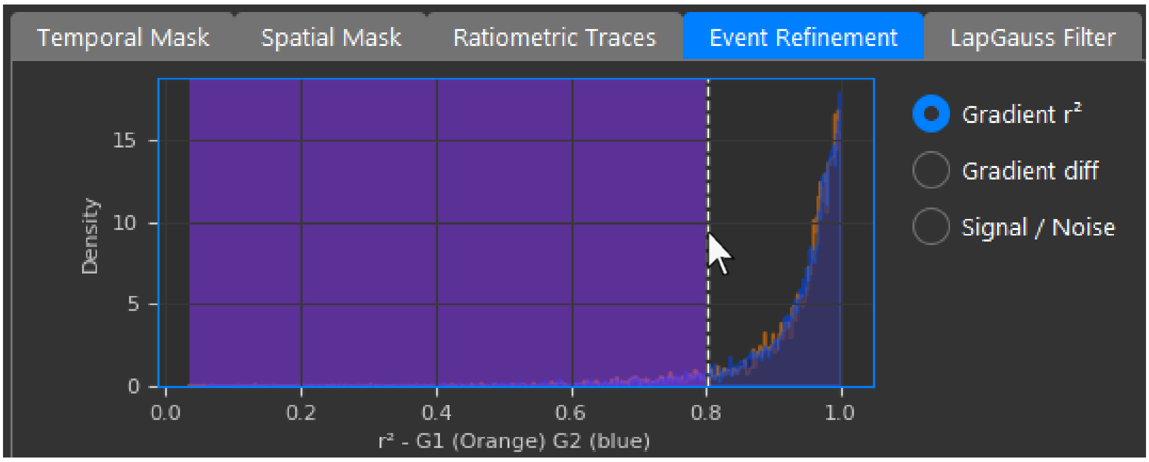
Event refinement is designed for landing assays and is not available for tracking as it requires fitting of time traces, not instantaneous contrast of moving PSFs.

To enable event refinement, navigate to the ‘Event Refinement’ tab:

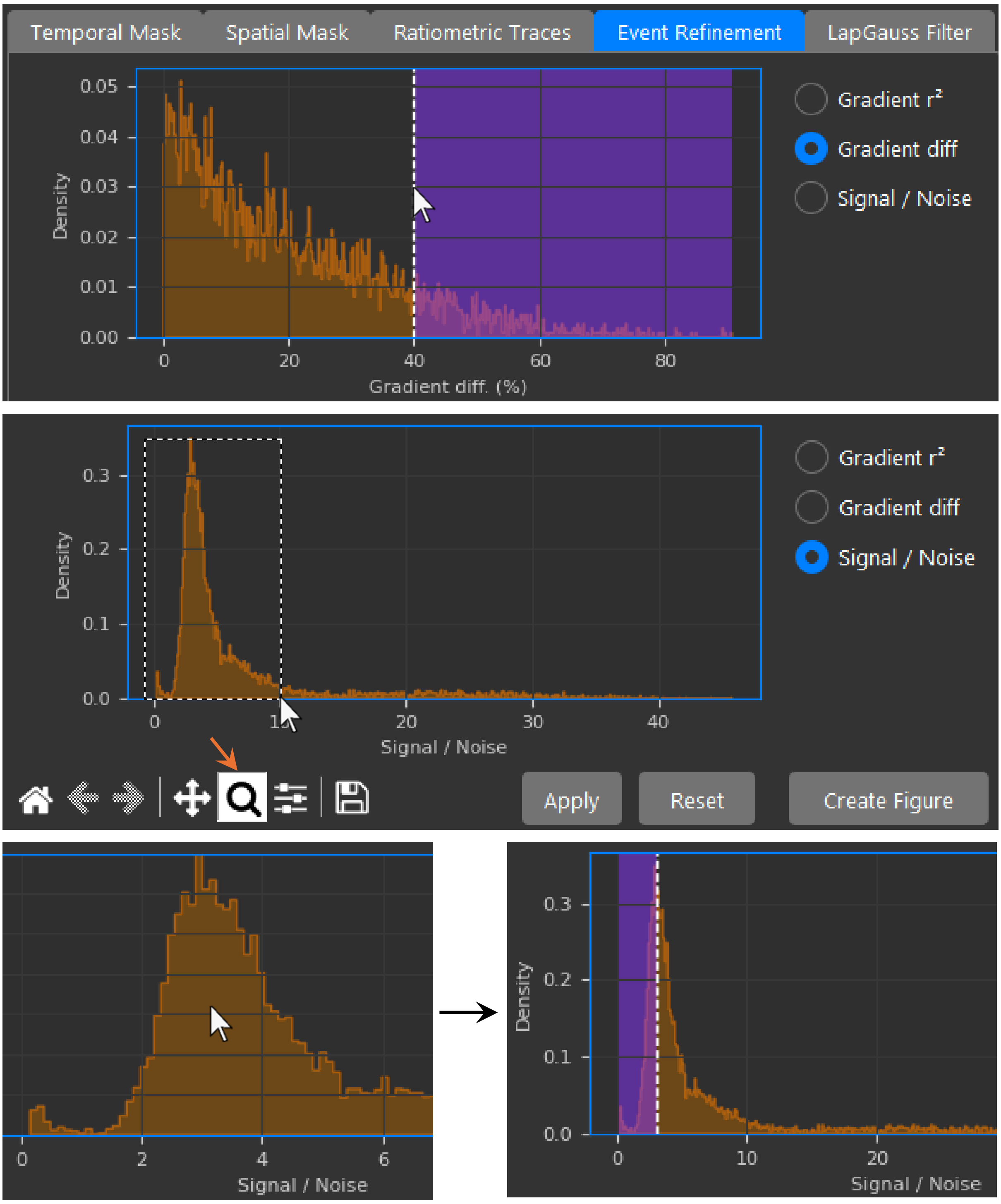


Any of the thresholds can be used independently from each other in any combination and combine using ‘OR’ logic. For example, if only an r^2 threshold is used then events will only be refined using this fitting metric. If ‘Gradient diff’ and ‘Signal / Noise’ are used, events will be rejected if the descending and ascending gradients poorly match or if the SNR is too low. When a threshold is set, events will be refined automatically. When a mask, such as the spatial or temporal mask, is adjusted, event refinement will not be applied immediately but the thresholds will be remembered. Event refinement will be applied again after the ‘Apply’ button is clicked.

To set the threshold for ‘Gradient r^2’ click anywhere on the histogram and events where either the descending or ascending gradient is lower than the selected threshold will be rejected. The recommended threshold is at least 0.8:



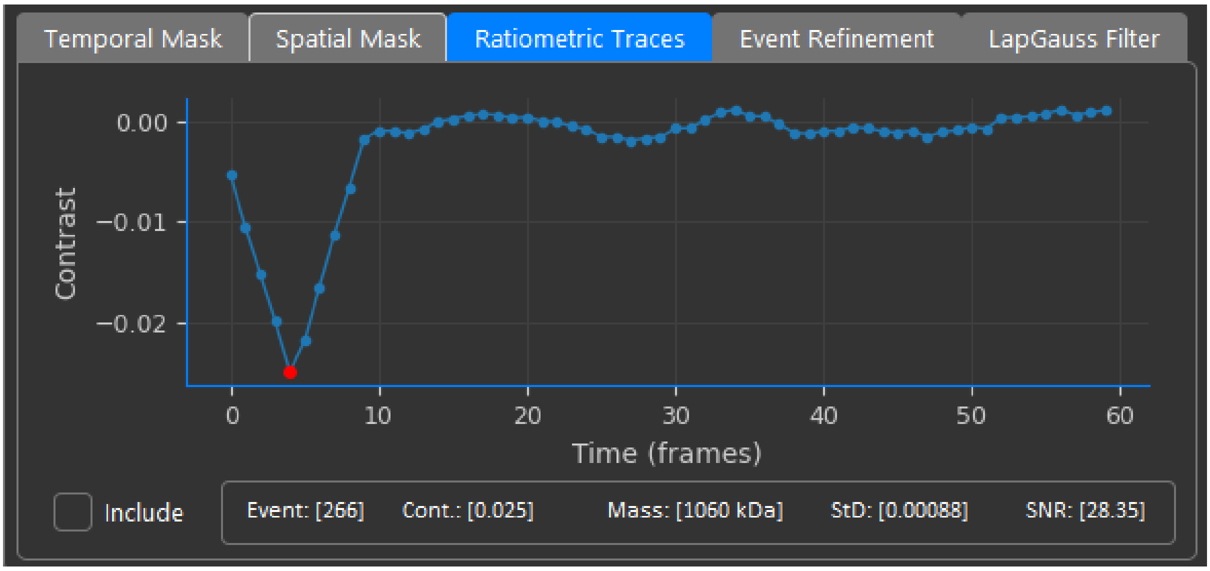
‘Gradient diff’ and ‘Signal / Noise’ thresholds can be set in much the same way. Navigate to the respective graph with the radio buttons on the right and click on the graph to set the threshold. For SNR, you may first need to use rectangle zoom in the toolbar first before clicking to improve the precision of the selection. The recommended ‘Gradient diff’ threshold is 40% which will reject events where the difference between ascending and descending gradients is greater than 40% (min grad / max grad). The recommended SNR threshold is 3. Thus, events with an SNR of less than 3 will be rejected:



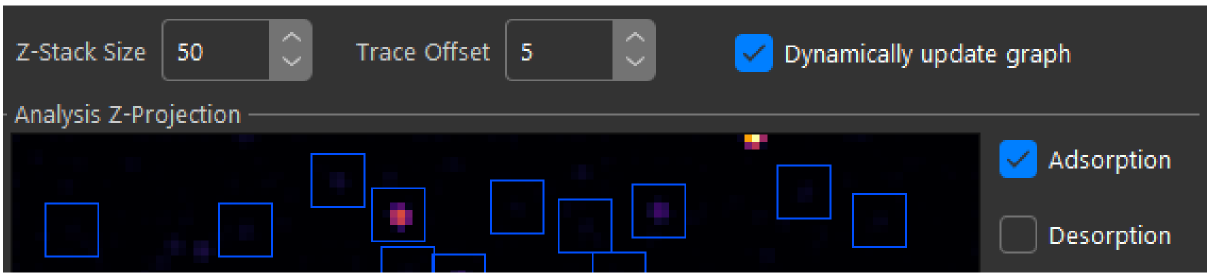
Chapter 4 – Event Detection Settings

4.1 Z-Stack

The Z-stack is the pre-binning of frames for faster detection of events. For landing assays, instead of detecting events in each frame, a stack of frames is averaged according to the Z-stack size. Events are then detected from the Z-max projection of these binned frames. This works because the sparse distribution of PSFs in single frames. To prevent duplication, each set of binned frames is padded at the start and end by a number of frames where no events are detected. This parameter 'Trace Offset' should be set the same as the half window length used for computing the ratiometric stack. I.e. if 'Window length' is 5, then 'Trace Offset' should also be 5. Whenever event refinement is used, in addition to thresholding events based on gradient fit thresholding, even when no thresholds are set, leading and trailing events that overlap between the Z-stack and the padding will be removed since a fit could not be calculated:

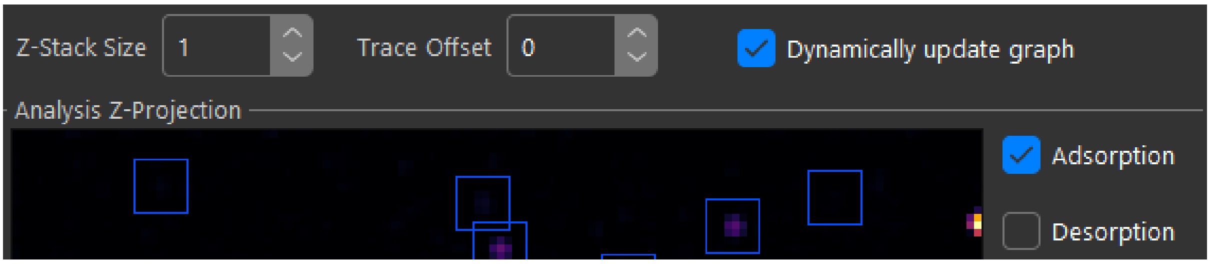


The default 'Z-stack size' and 'Trace Offset' are 50 and 5 frames respectively for landing assays:



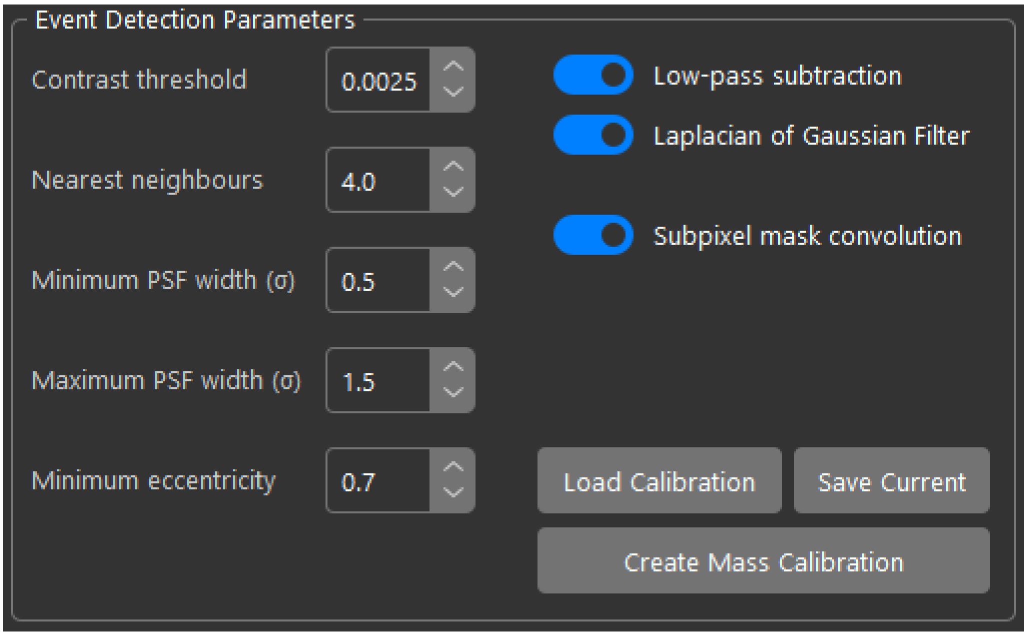
If the density of events is high (high analyte concentration), the Z-stack size should be reduced (to a minimum of 20) and vice versa, if the concentration is very low and events are sparse the Z-stack can be increased to speed up detection.

For dynamic tracking experiments, PSFs are detected in each frame individually, and the Z-stack size should be set to 1 frame, and the padding set to 0 frames:



4.2 Event Detection Parameters

Event detection parameters are found in the centre of the UI under the ratiometric view:

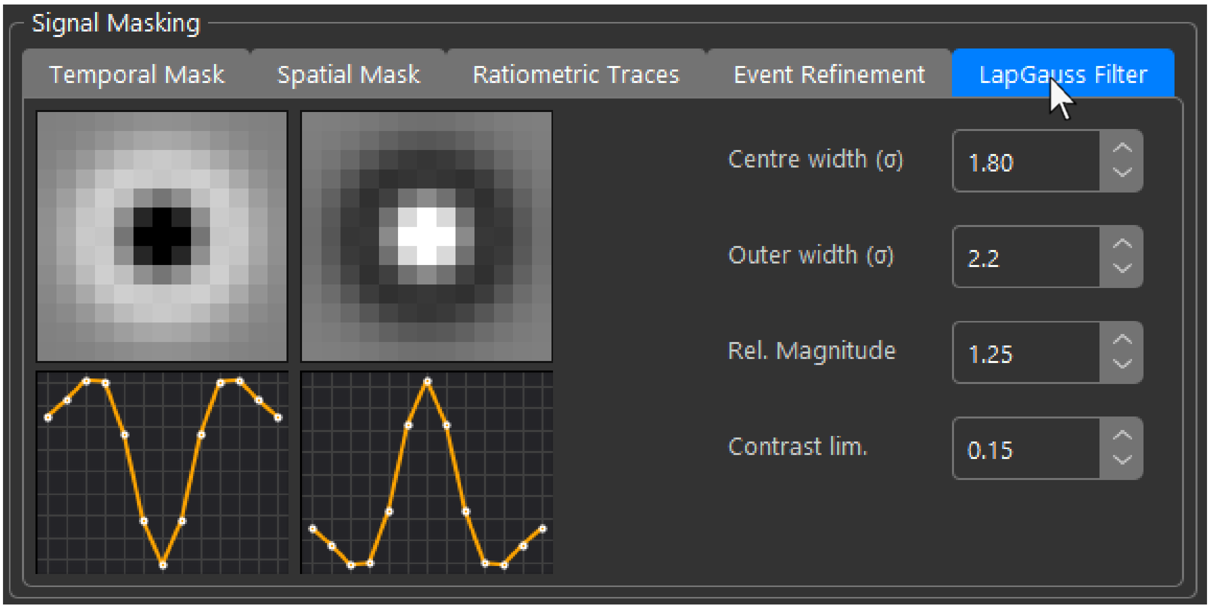


On the right, processing steps can be toggled on or off. 'Low-pass subtraction' when enabled, subtracts low frequency background from the z-stack. 'Laplacian of Gaussian Filter' when enabled, applies an approximated (Using difference of Gaussians) LoG filter to enhance PSFs. 'Subpixel mask convolution' resamples the 3x3 pixel binary mask used to extract contrast information from the raw ratiometric movie using the fitted subpixel coordinates of the PSF fit.

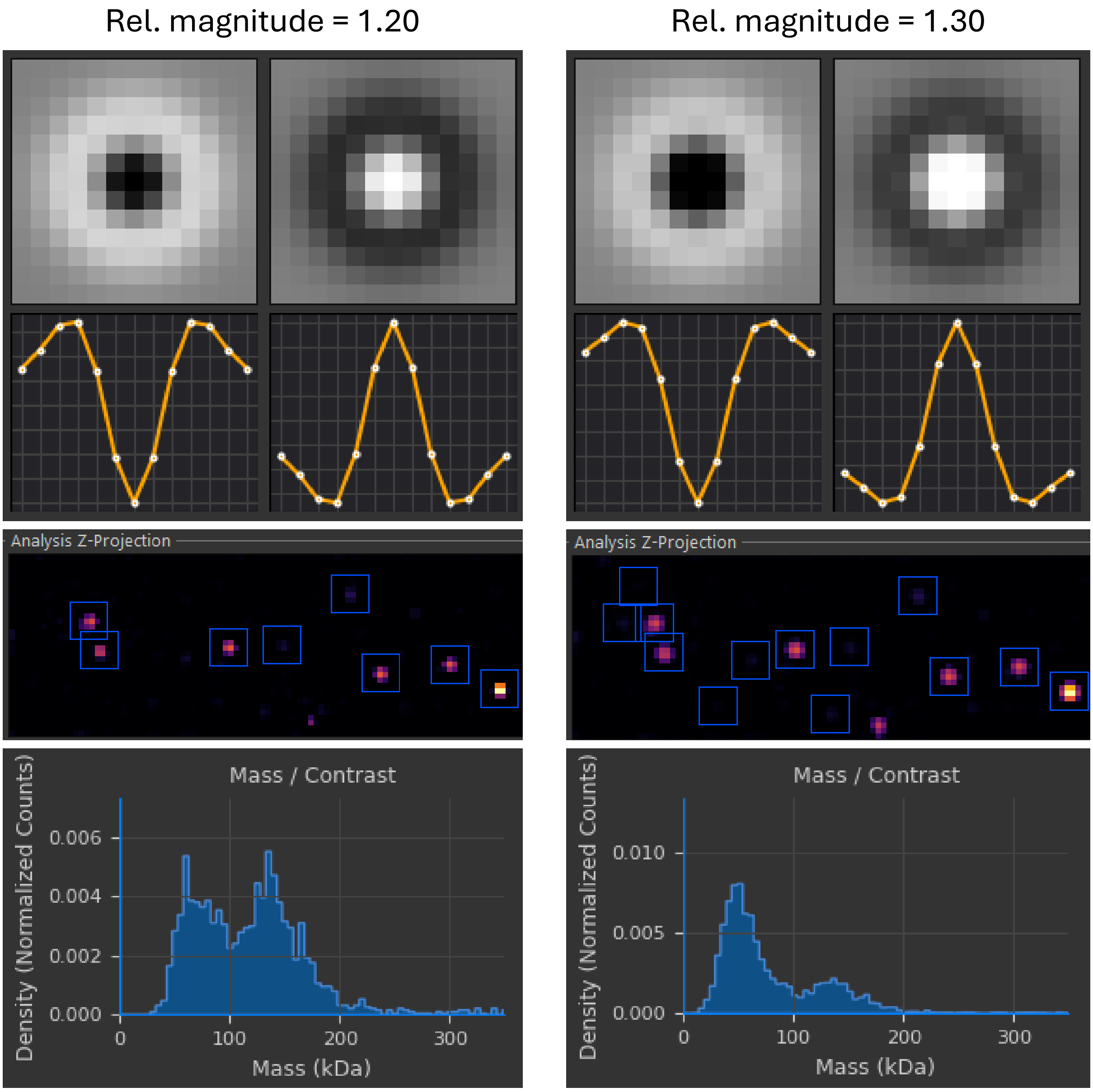
The parameters on the left control thresholds for Gaussian fitting of the enhanced PSFs. ‘Contrast threshold’ is the minimum allowed contrast of the most intense pixel of a candidate PSF. ‘Nearest neighbours’ will reject events with PSFs that have centroids with less than or equal to this distance in pixels between them within the same Z-stack. ‘Minimum PSF width (σ)’ and ‘Maximum PSF width (σ)’ control the range of allowed PSF widths as a function of the standard deviation of the Gaussian fit in pixels. ‘Minimum eccentricity’ rejects non-round PSFs with a min. / maj. eccentricity of less than the threshold.

4.3 LapGauss Filter

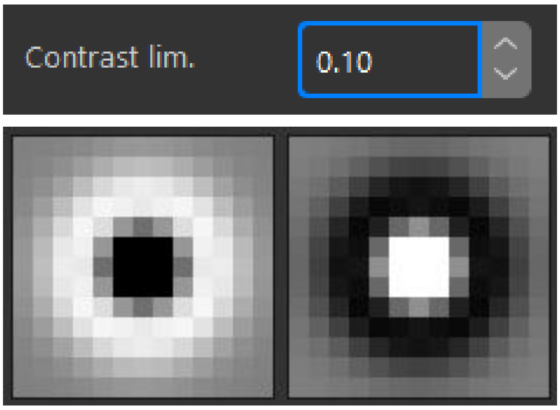
Navigate to the ‘LapGauss’ tab in the signal masking panel of the UI to access the controls for adjusting the PSF model:

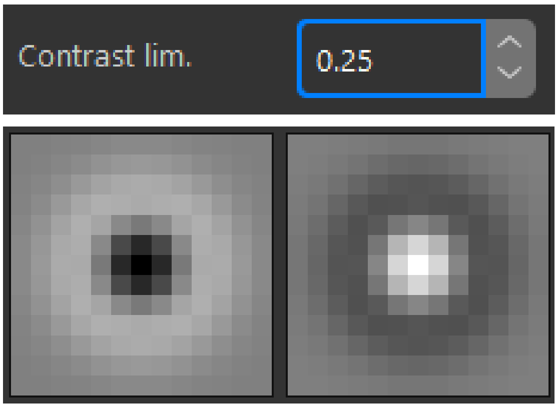


‘Centre width (σ)’ and ‘Outer width (σ)’ control the standard deviation of the inner and outer Gaussians in pixels respectively and should match PSFs. ‘Rel. magnitude’ is the ratio of intensities between the inner and outer Gaussian. Changing this value modulates the frequency response of the filter, with lower values producing sharper features and vice versa. When the Rel. magnitude is lower, frequency response is higher and low contrast events which don't exactly match the shape of the filter due to background are more likely to be rejected. When the Rel. magnitude is higher, frequency response is lower and low contrast events are more likely to make it through. This system is an additional way to filter out false positives due to background fluctuations, the other being event refinement.



‘Contrast lim.’ is just for visualization and sets the normalization / clipping of the plot:



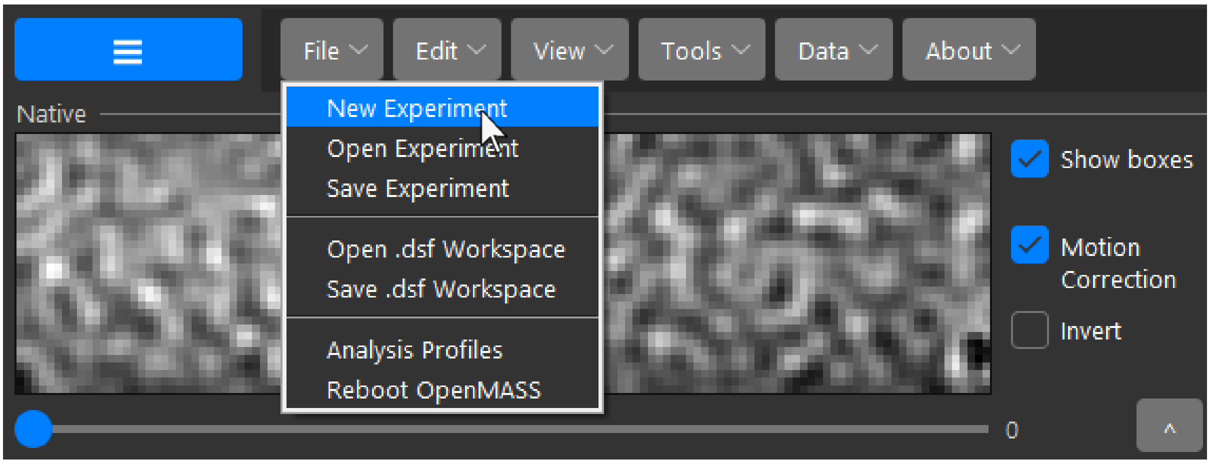


Note that while event refinement is unavailable for tracking due to the requirement for time trace gradient fitting, adjustments to the PSF model using ‘Rel. Contrast’ to tune the frequency response of the filter is a valid approach for refining PSF picking in tracking experiments.

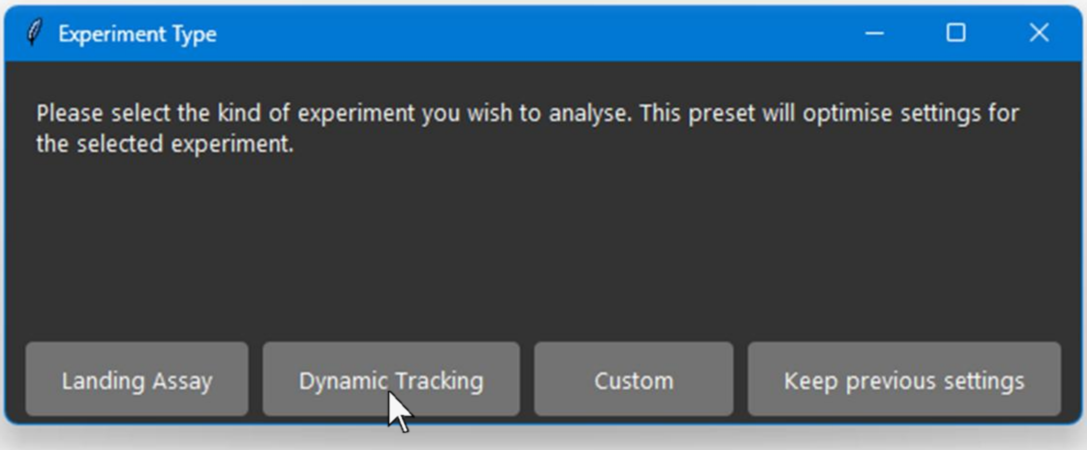
Chapter 5 – Dynamic Tracking

5.1 Getting Started with Tracking

Open the Menu Toolbar and Click 'File' → 'New Experiment':



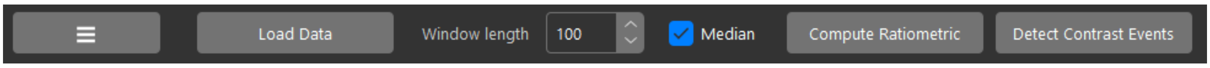
Then select 'Dynamic Tracking' when prompted to choose an experiment type:



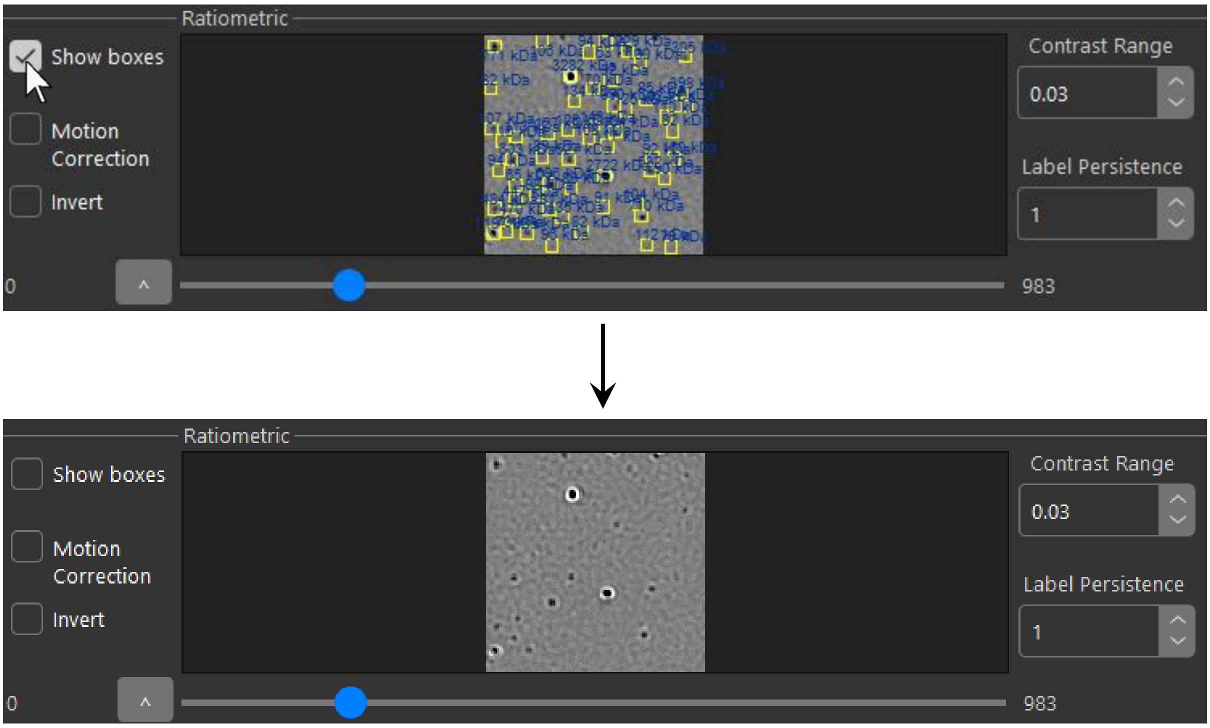
The preset will be applied, changing all settings to those optimised for tracking experiments. Ratiometric background subtraction mode will be set to median with a default (half) window length of 100 frames. Z-stack size and Trace Offset will be set to 1 and 0 respectively. Eccentricity will be relaxed to 0.65 to allow small amounts of PSF deformation due to motion blur.

Next, load and process data as you normally would for a landing assay:

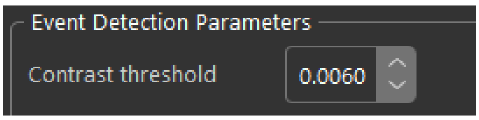
Load Data → Compute Ratiometric → Detect Contrast Events



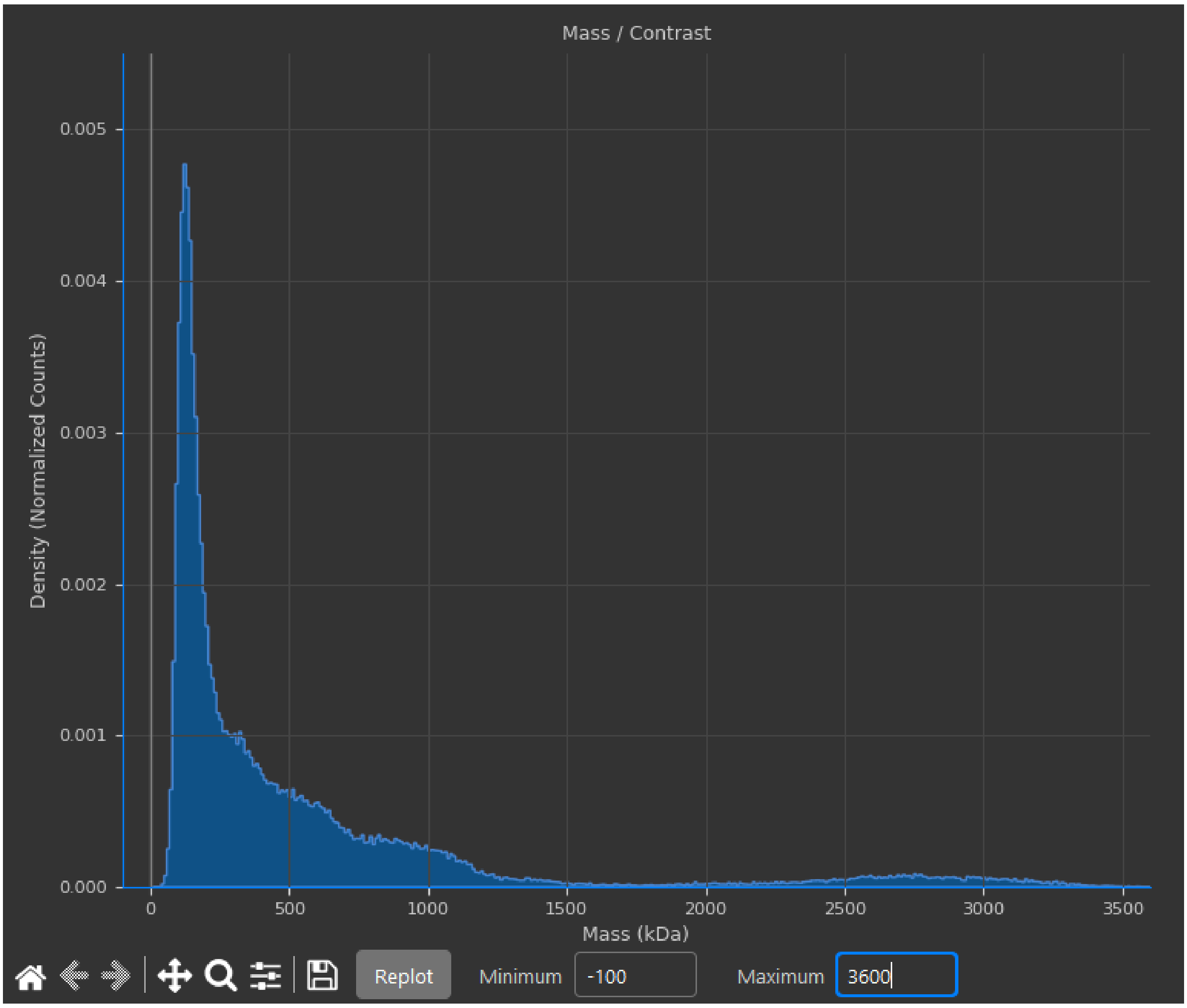
It is recommended to turn off ‘Show boxes’ in the UI when using scrolling through the ratiometric stack as the number of events detected in tracking experiments can exceed 100,000 and slow down the UI when rendering boxes:



During tracking, each frame is enhanced, filtered, and events fitted with Gaussians and contrasts calculated from sub-pixel localizations. Thus, the process can be quite slow. For example, for a movie of size 128x128x5093 (large image preset on OneMP mass photometer recorded for 1 minute) with over 300,000 events, the event detection process takes around 10-15 minutes. If particles are known to be of high mass, the contrast threshold in the event detection parameters can be increased, for example, if the minimum mass of particles is known to be at least 200 kDa, the ‘Contrast Threshold’ can be increased to 0.0060 to speed up the process by ignoring low contrast background:



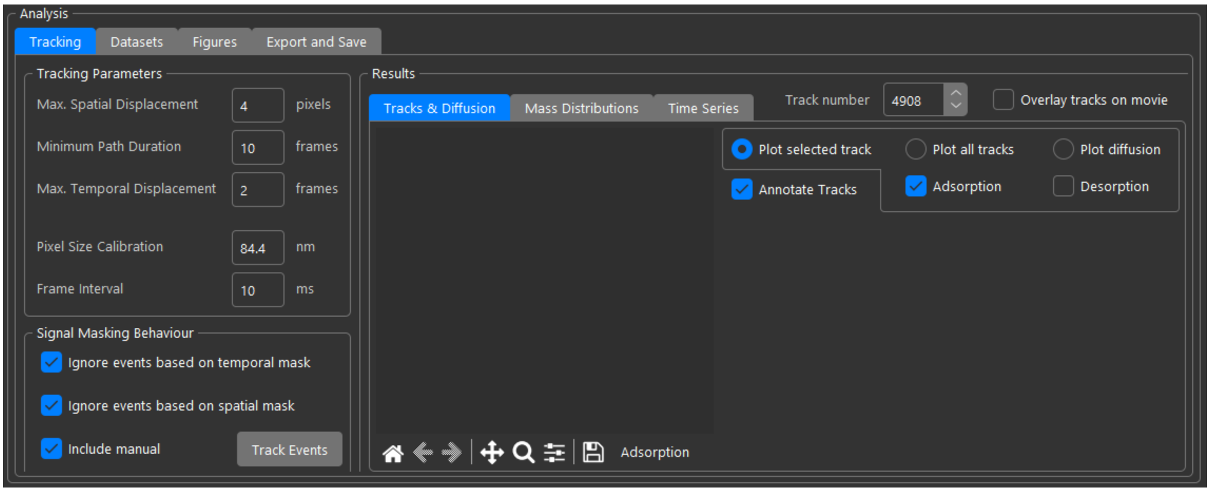
Note that at this stage the histogram will show individual events as they have not been connected into tracks yet. In tracking experiments, event detection will only look for negative contrast PSFs as the presence of a particle causes scattering that manifests as a reduction in signal (thus negative in ratiometric mode). Also, unlike landing assays, in tracking mode, OpenMASS will not attempt to automatically fit the histogram when event detection completes:



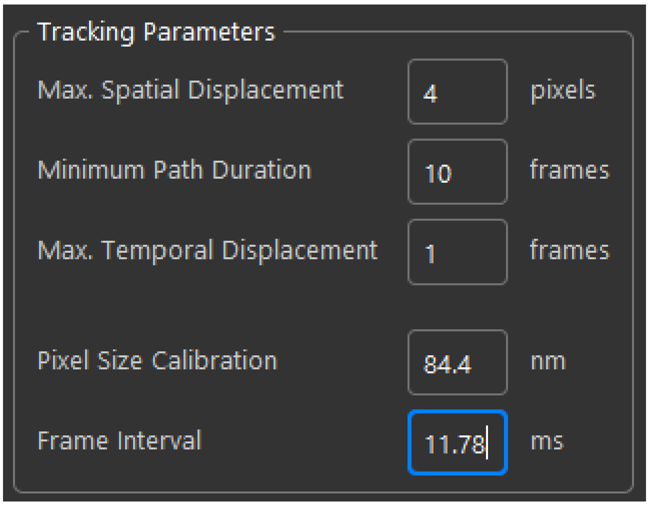
Peaks in the individual event contrast / mass distributions are expected to be wide and overlapping as they represent the instantaneous contrast of the PSF at a single frame with the median background subtracted. On the contrary in landing assays the contrast measured is more precise as it comes from several frames (equal to the half window size) averaged together.

5.2 Tracking Parameters

Tracking analysis is undertaken from the analysis panel in the lower left of the main UI:

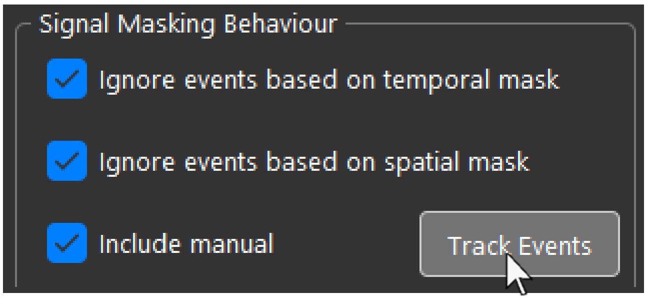


This is where individual events are connected into tracks and downstream data analysis of tracks happens. First, ensure the tracking parameters are set up correctly:

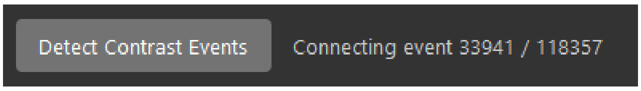


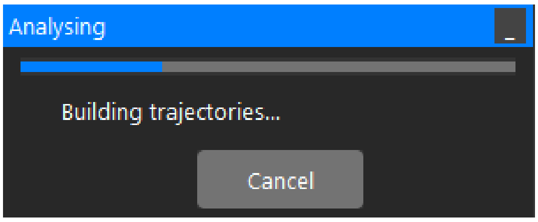
The pixel size is configured to the default pixel size from OneMP mass photometers with pixel binning set to 4. The frame interval can be calculated by dividing the recording duration in milliseconds by the number of frames. In the example, the movie was recorded using the OneMP large preset of 128x128 pixels for 1 minute yielding 5093 frames. Thus, the frame interval = 60,000 / 5093 = 11.78 milliseconds per frame. The 'Max. Spatial Displacement' parameter sets the largest distance a PSF can move from time-point to time-point before being considered as belonging to a different track. This parameter should not be set too high to reduce the chance of track mixing in dense environments. ‘Minimum Path Duration’ is the minimum number of time points that need to be connected for the path to be considered a track and not be rejected. ‘Max Temporal Displacement’ is the maximum difference in frames between events to be connected. When set to 1, the track must be contiguous with every event being connected between adjacent frames. When set to 2, a 1 frame gap is allowed, and tracks can be built from non-contiguous events. This may be useful for noisy data or low contrast events where a particle may be visually present, but some frames may not contain a detected event from the particle due to a bad local PSF fit etc. When set to 3, the particle can be dropped from 2 frames etc. Though it is recommended not to go beyond this point to prevent tracks from linking artifactual events or a nearby PSF if closer than the ‘Max. Spatial Displacement’.

Check on uncheck boxes in the ‘Signal Masking Behaviour’ to control whether events that will be linked into tracks should be selected via temporal or spatial masks applied earlier. These boxes don’t update the events in real time, thus if they are changed, or masks are changed, the changes don’t apply in real time, but instead trajectory linking needs to be repeated with the current state. To start linking events into tracks, click ‘Track Events’:



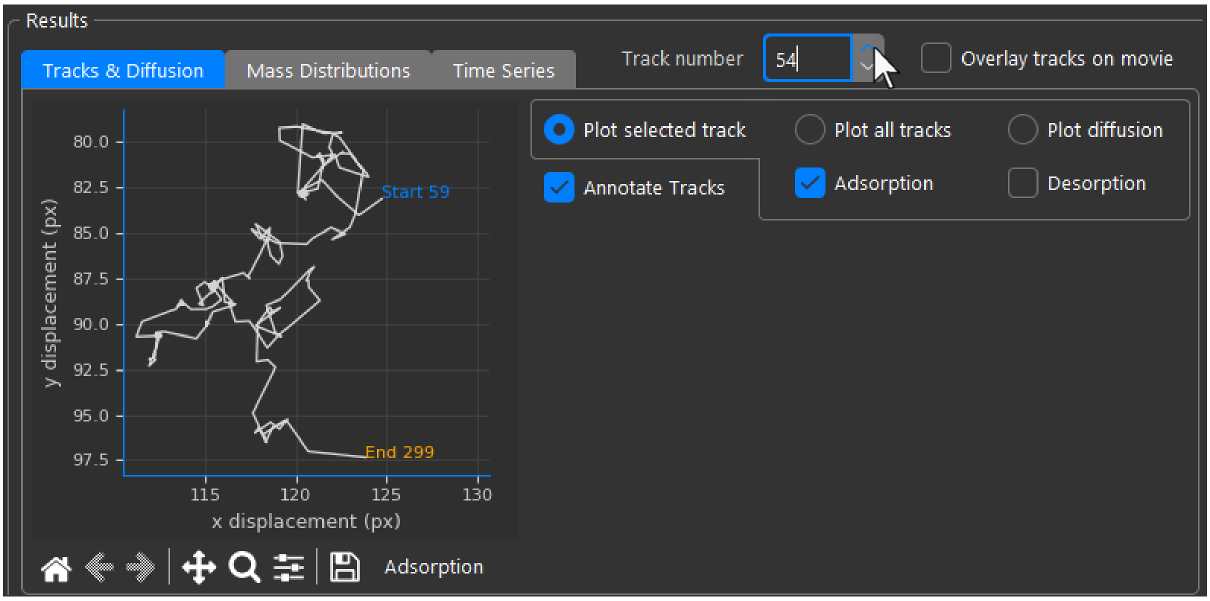
This process may take some time if there are more than 100,000 events. A progress bar window and a label at the top of the UI will provide progress updates:



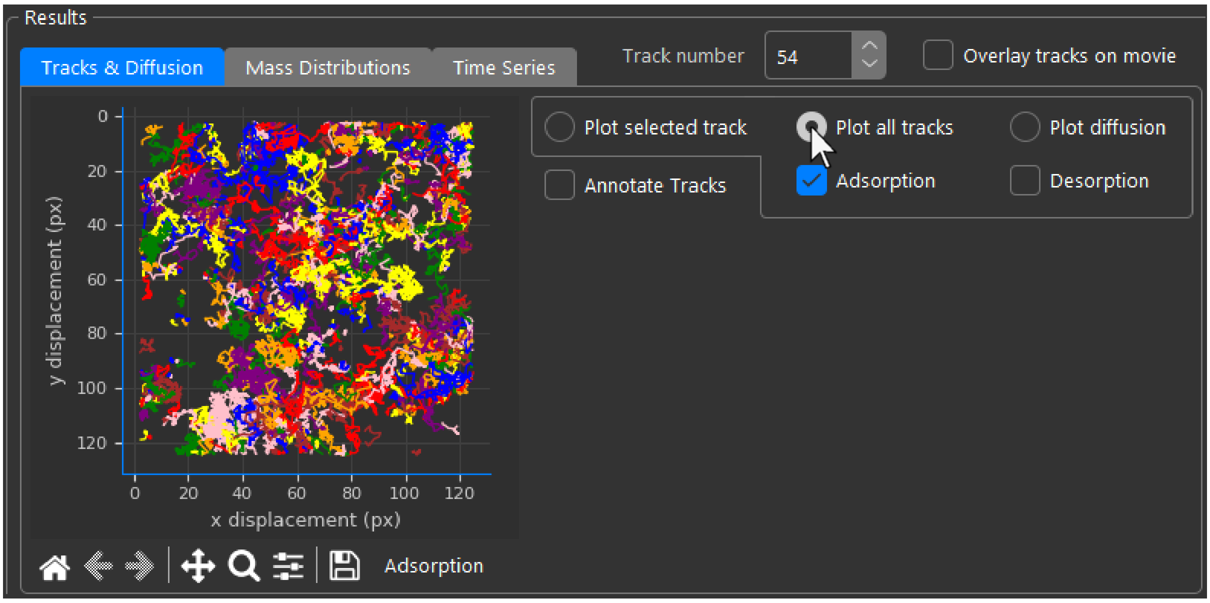


5.3 Tracking Results

When the linking process completes, tracks will be displayed in the results panel:

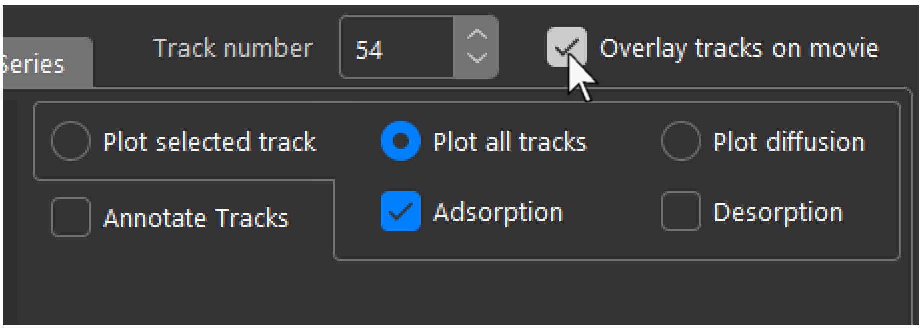


The currently displayed track can be changed using the 'Track number' spinbox. By default, tracks are annotated with their start and end frame number, though this can be disabled with the checkbox. Disable annotations and switch to plotting all tracks using the radio button:

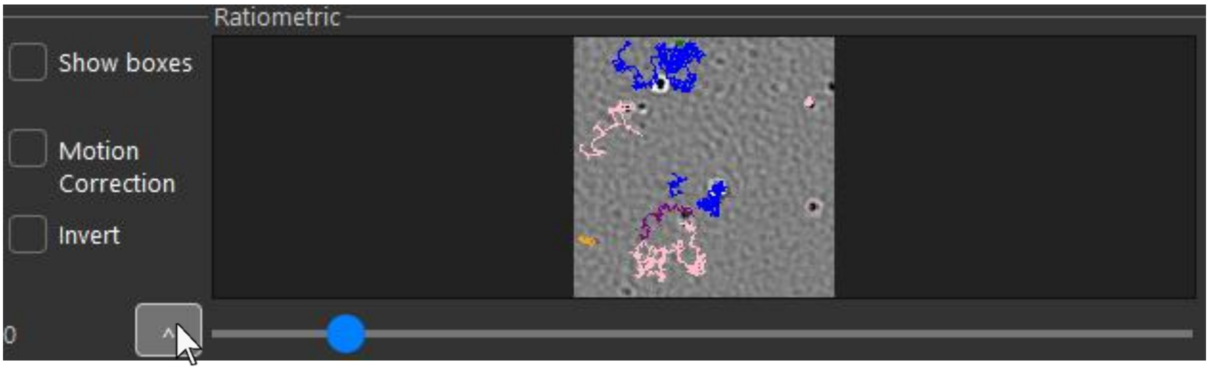


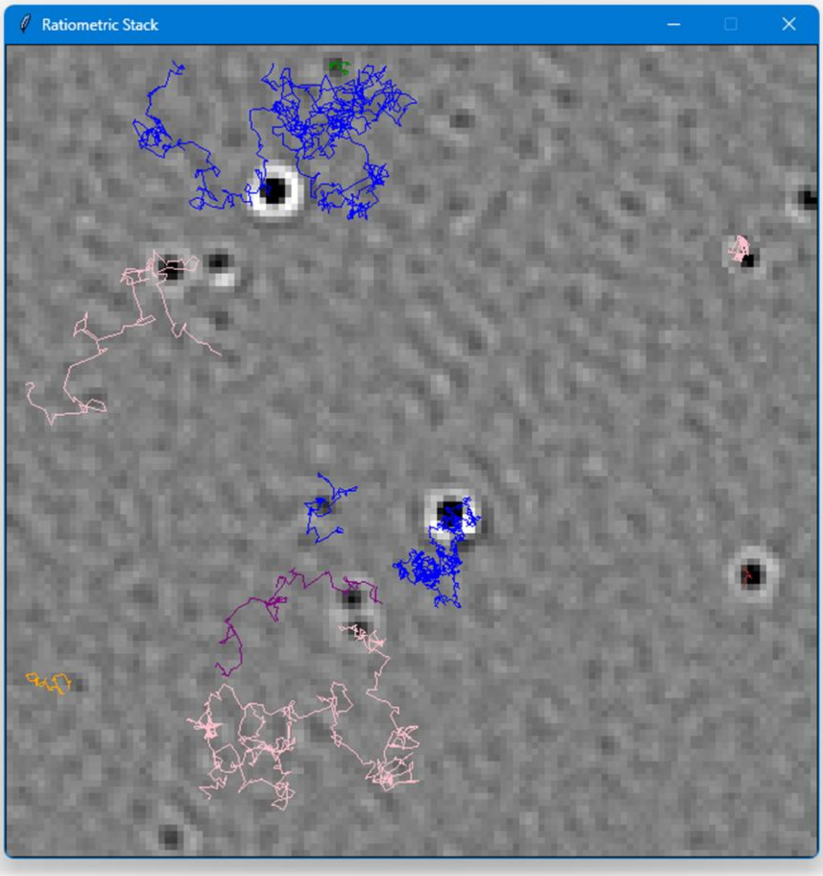
All tracks selected by the current filter will then be displayed here. Ignore the 'Adsorption' and 'Desorption' checkboxes; their functionality is deprecated.

Check the 'Overlay tracks on movie' checkbox to display an overlay of the tracks on top of the ratiometric view:

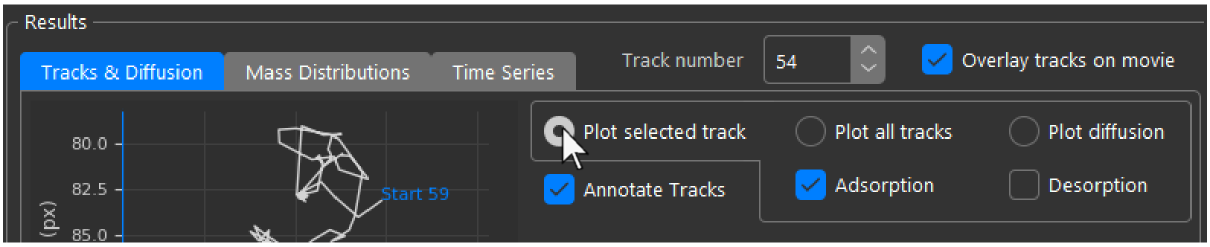


Use the '^' button to float the ratiometric view in a new large window:

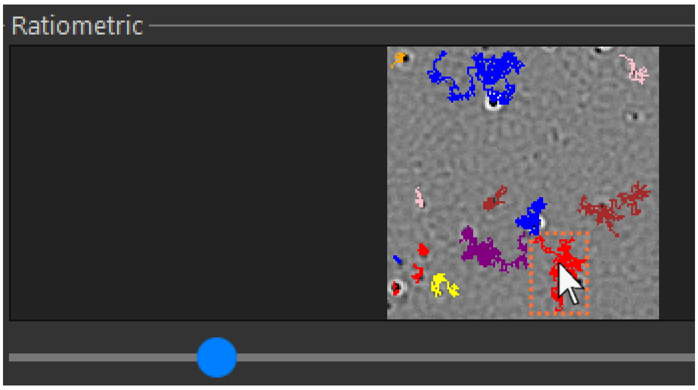


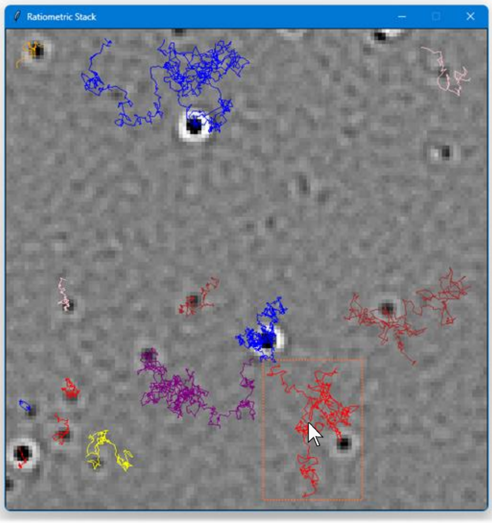


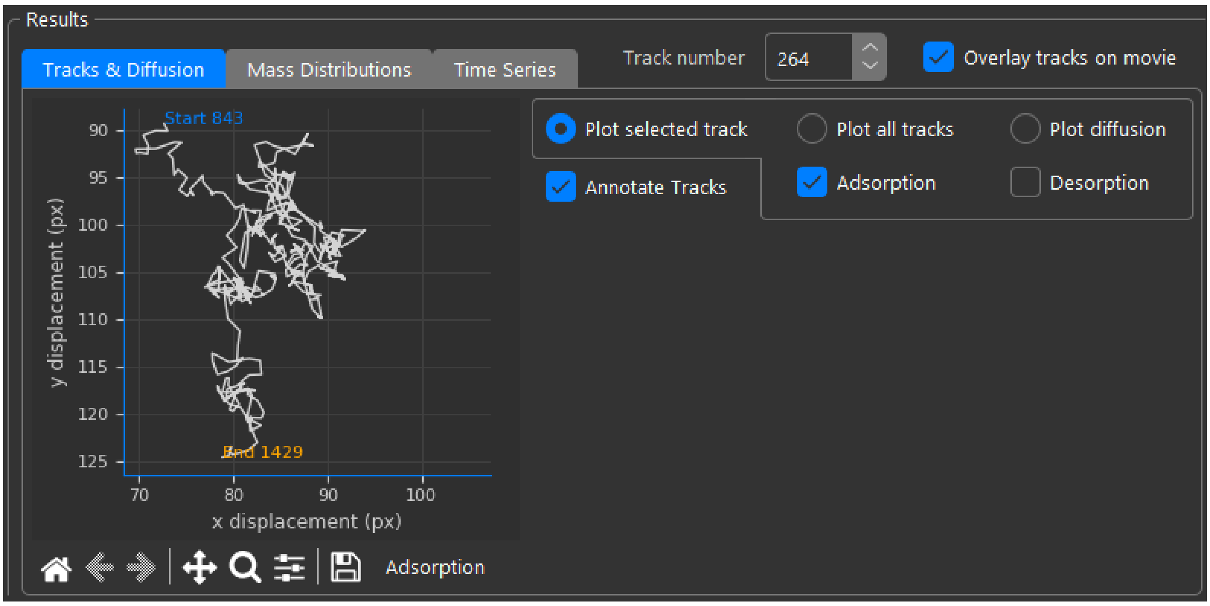
Switch back to 'Plot selected track' and turn track annotation back on:



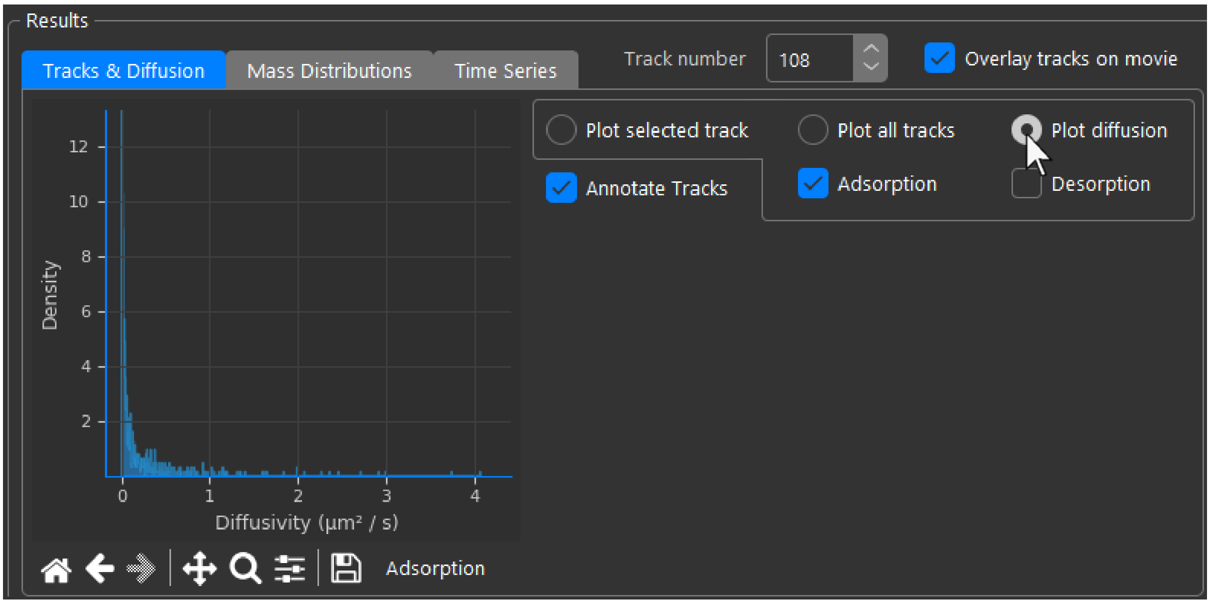
Now, clicking on a track in the ratiometric view will select it, and the selected track will be plotted in the results panel. Tracks can be clicked either in the inline ratiometric view or when floating it in a top-level window:





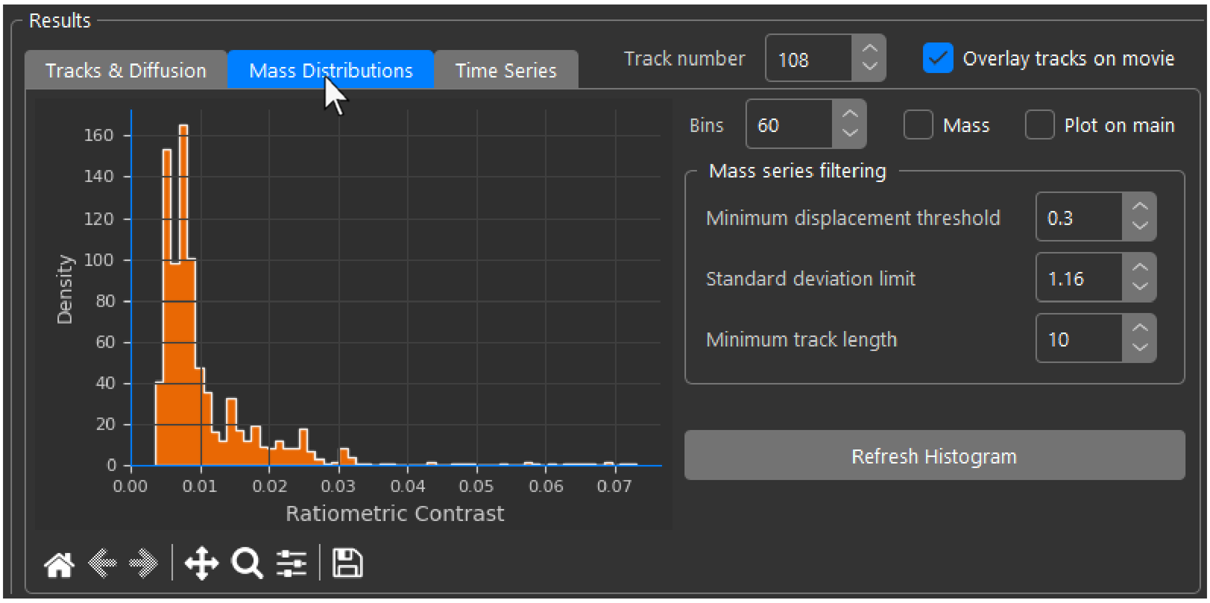


Plot the distribution of diffusion coefficients by selecting the 'Plot diffusion' radio button:



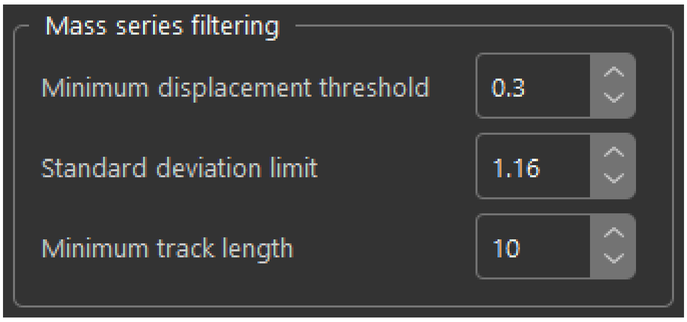
5.4 Track Mass Distributions

Navigate to the 'Mass Distributions' tab in the Results panel. You may wish to increase the number of bins to see the distribution more clearly. The default is 60:



The histogram displayed here is built not from individual events, but from the mean contrast of the time series contrast trace of each track.

Mean track contrasts or masses (depending on mode) displayed in the histogram are filtered using the parameters in the 'Mass series filtering' panel:

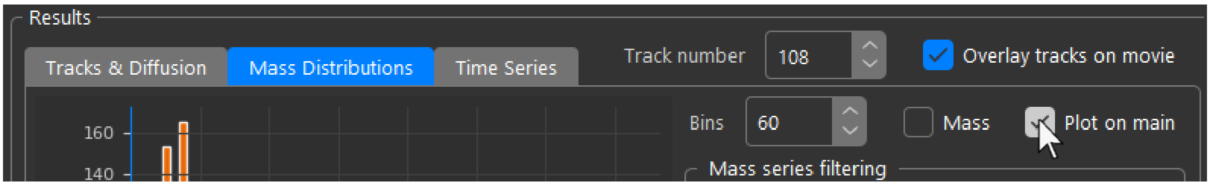


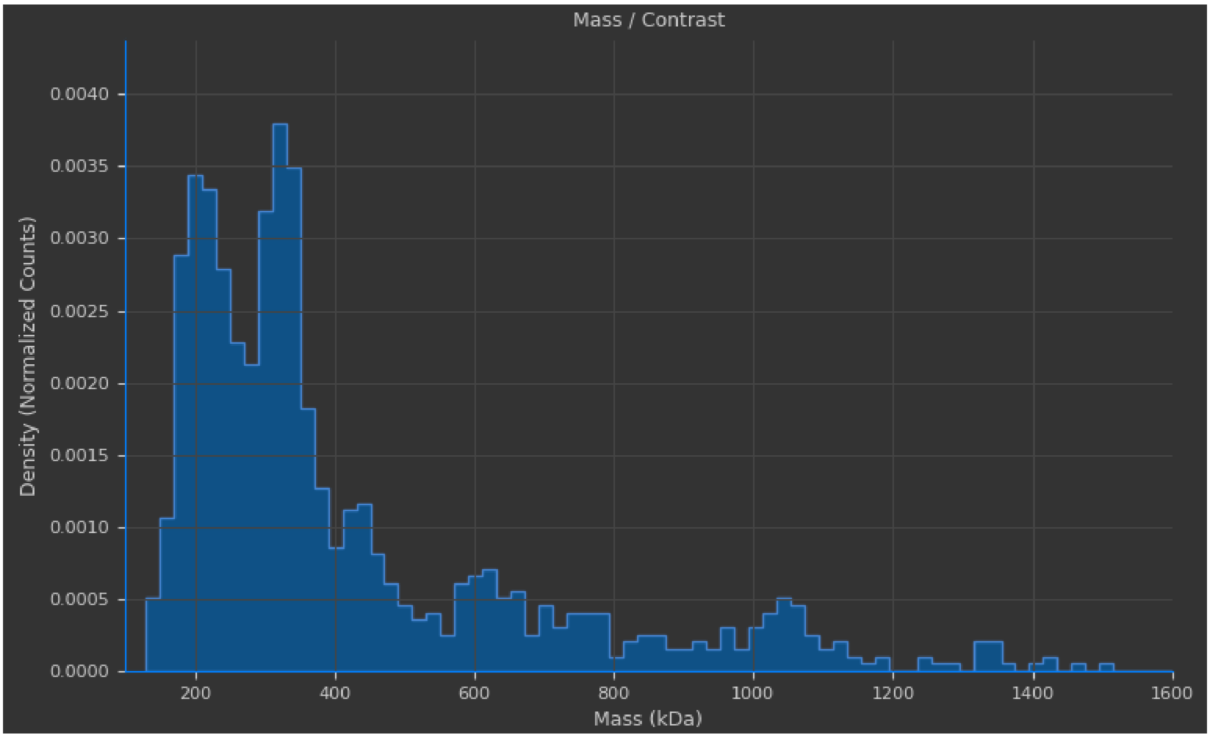
'Minimum displacement threshold' is the minimum allowed mean displacement per frame. Thus, tracks with a lower mean displacement per frame will be excluded from the track mass distribution. This threshold is designed to reject stationary or very slow moving PSFs which would contribute artifactual contrast information.

'Standard deviation limit' is a threshold around the distribution of the time series signal. The mean track contrast is calculated from data points in the time series centred at the original mean of all data points but limited to those within this threshold. The default is ± 1.167 standard deviations (full width at half maximum). This sharpens the peaks in the distribution of mean track masses by rejecting outlier data points in the time series data of the track.

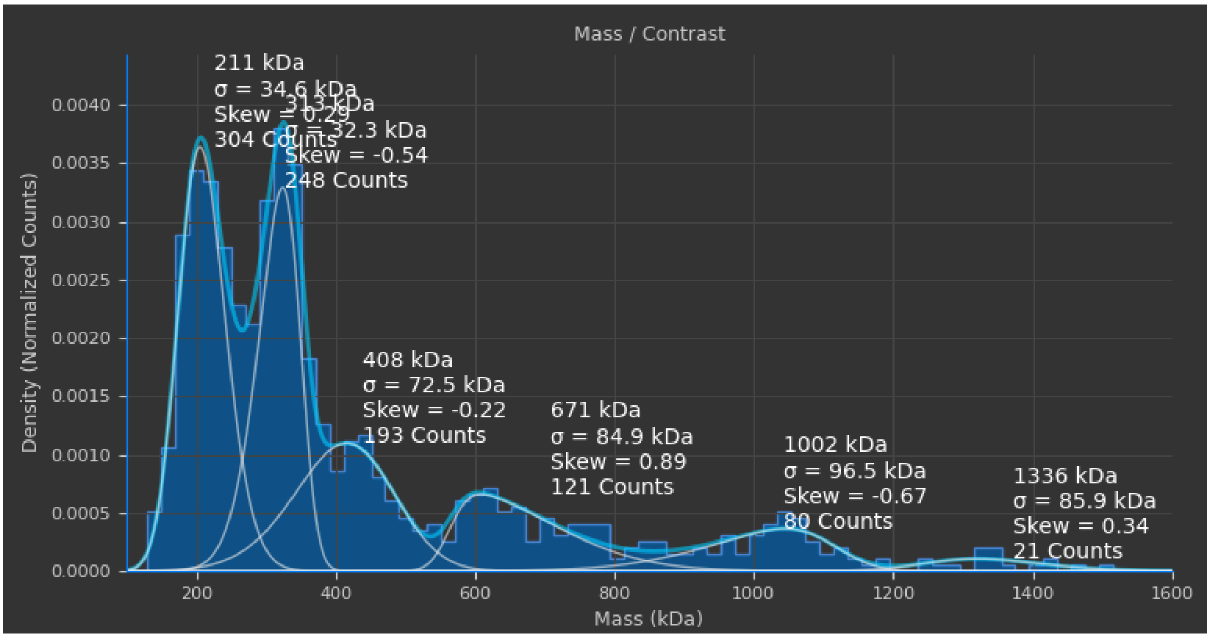
'Minimum track length' is the minimum number of time points a track can have for its mean contrasts or masses to be included in the distribution. Increasing this threshold tends to improve mass precision of peaks in the distribution since only tracks with more data points and thus closer agreement to the true mean are used. The trade-off is that lower contrast peaks will lose counts as track sections from with lower contrast particles are usually shorter. The reason for this is that high contrast particles have very little variability in contrast relative to the background and can be tracked continuously for long durations as long as they remain within the FOV and the PSF doesn't overlap with another particle. Low contrast particles on the hand may vary in contrast from frame to frame enough that every so often in certain frames they are lost in the noise and when they reappear, they will be considered as a new particle. This can be improved by increasing the 'Max. Temporal Displacement' parameter and re-tracking events.

Click ‘Plot on main’ to plot the track mass distribution in the main histogram panel:



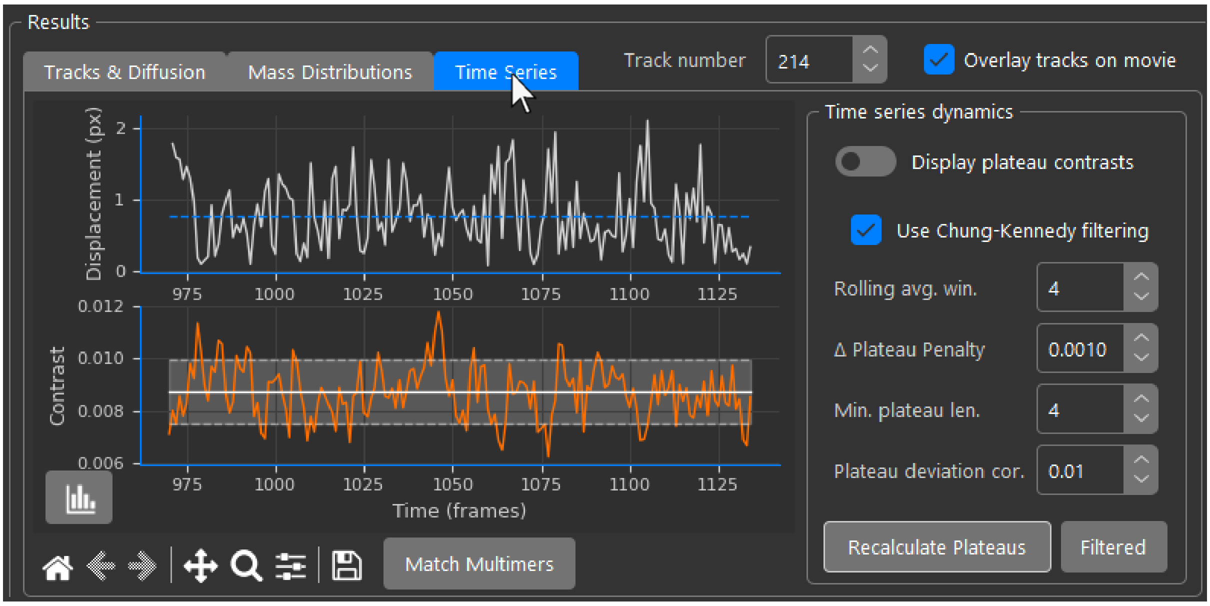


The distribution can be fitted in the same way as detailed in the quick-start guide for landing assays:



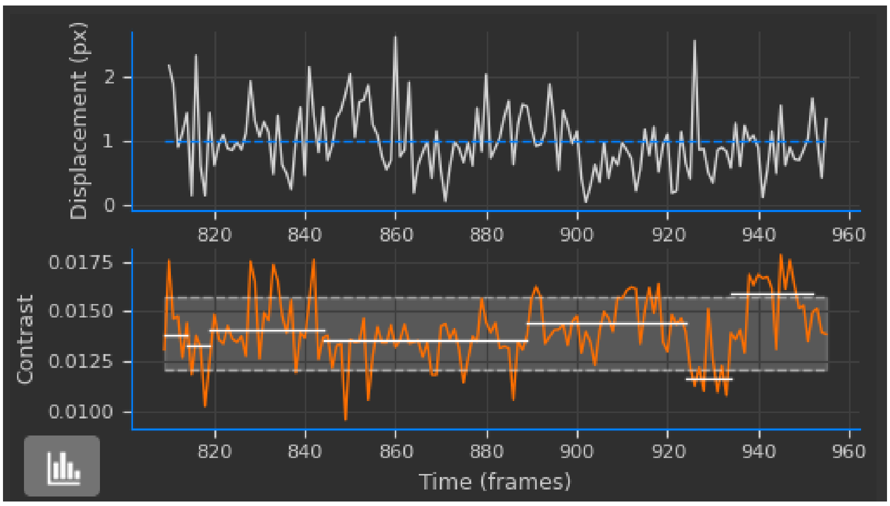
5.5 Time Series Analysis

Navigate to the 'Time Series' tab in the Results panel:

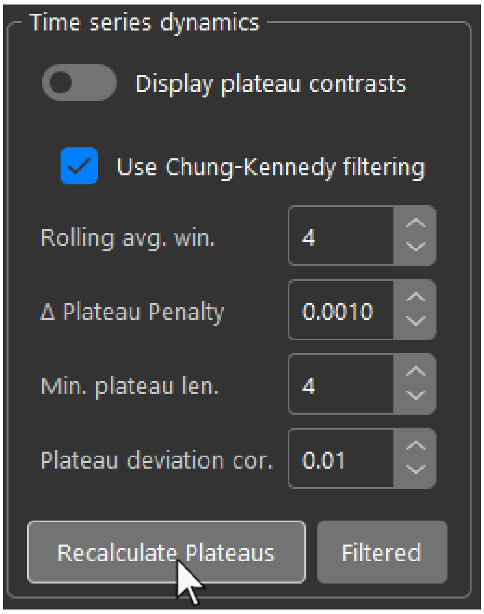


On the top, in white, is the displacement (pixels) vs time (frames) graph. The blue dashed line shows the mean displacement value. Aligned with it below, in orange, is the ratiometric contrast vs time (framed) graph. A convention of positive contrasts (described in section 1.2, pg. 10) is used for the contrast plot so contrast changes if dynamic are present are intuitive (e.g. increase in contrast magnitude looks like a step up). Plateau fits are shown in white. The grey transparent fill region reflects the current standard deviation threshold used to reject outlier points when calculating the mean contrast of the track.

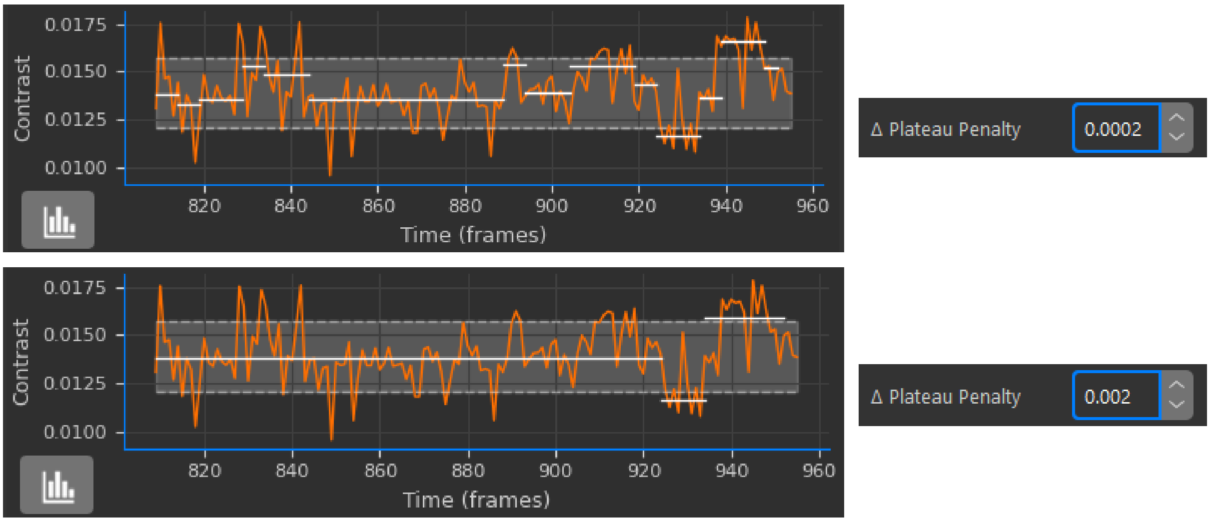
When events are tracked, contrast traces for all tracks are fitted automatically to see if contrast dynamics are present:



Changepoint fitting uses the python ruptures library. Trace fitting parameters can be edited, and traces can be manually refitted at any time in the 'Time series dynamics' panel to the right of the trace plots:



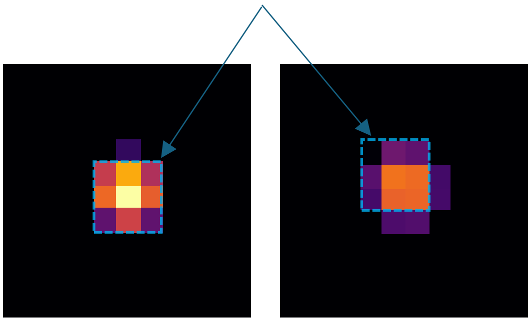
If 'Use Chung-Kennedy filtering' is checked, the time trace will be pre-processed using a Chung-Kennedy filter. Additionally, it is prefiltered with a rolling average window. The window size can also be adjusted here. 'Δ Plateau Penalty' is a regularization parameter for the ruptures changepoint model. The higher it is, the less sensitive it is to noise at the cost of potentially missing changepoints and vice versa:



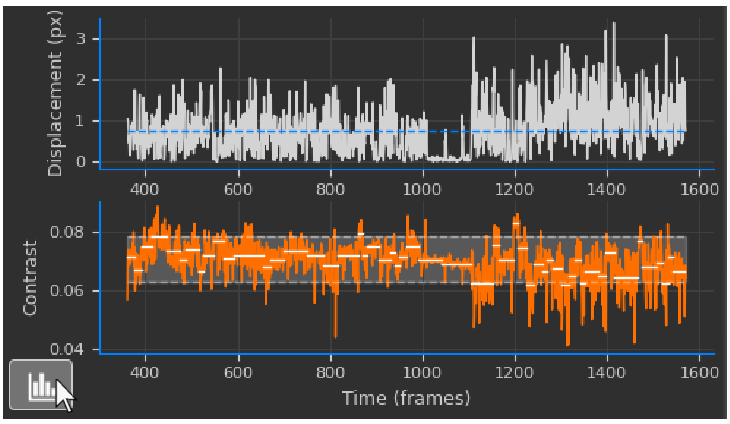
'Min. plateau len.' is the threshold that sets the minimum number of datapoints each fitted plateau can have.

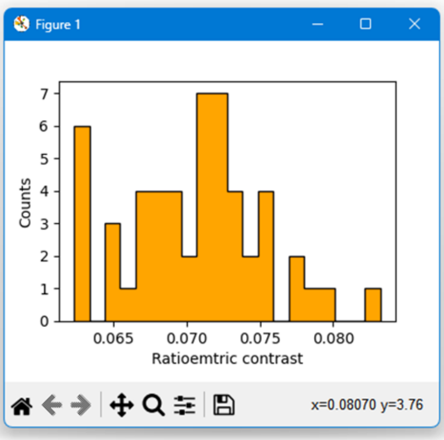
'Plateau deviation cor.' is a parameter that compares the $\frac{\text{Standard Deviation}}{\text{Mean}}$ for each trace. If this metric is less than the threshold the variability in signal is considered too small to be the result of state-switching dynamics and instead a single plateau mean is fitted between the start and end of the trace. The metric is theoretically invariant with scale as the primary source of error in instantaneous contrast measurement (variability in contrast / time signal) is PSF segmentation error not shot noise. PSF segmentation error is directly proportional to signal strength as opposed to shot noise which is proportional to the square root of the signal.

Contrast calculated from central 3x3 pixels of PSF. Sub-pixel mask convolution attempts to resample the mask with subpixel localization, but error remains based on sub-pixel centroid position of the PSF.

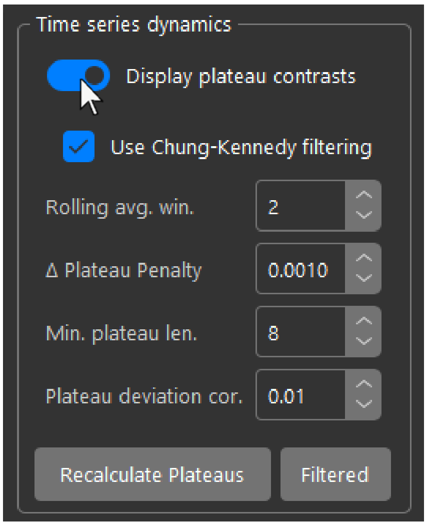


Click the histogram icon in the bottom left corner of the plot to open a histogram of the plateau states:

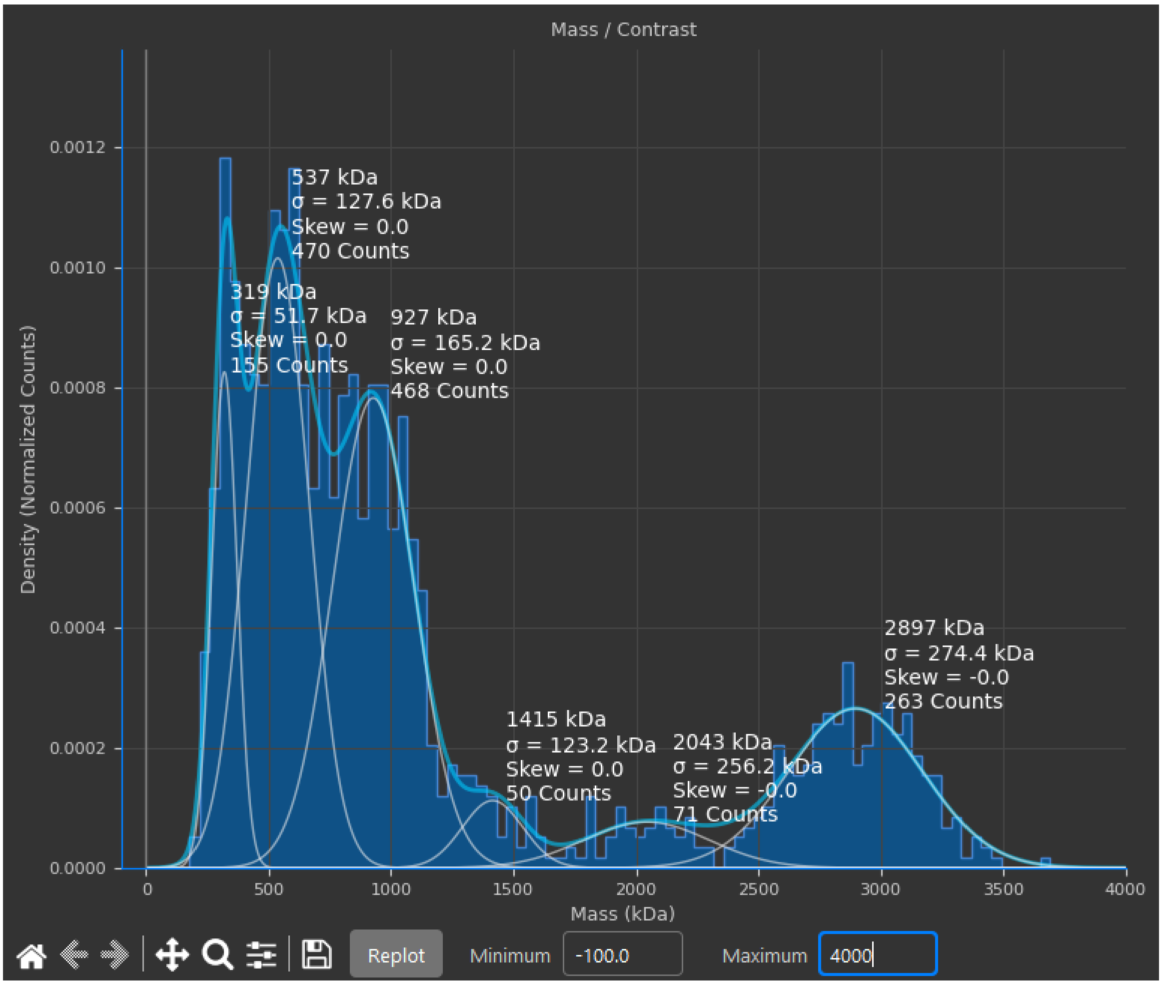




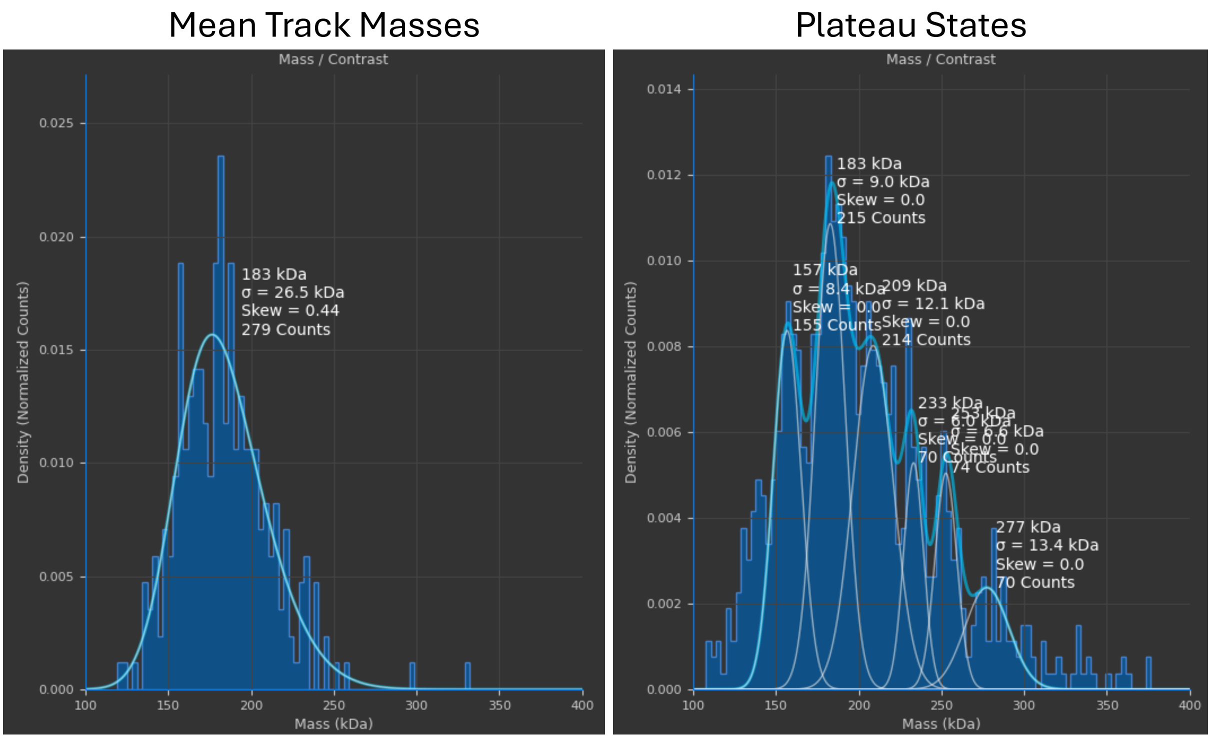
You can plot a distribution of all fitted plateau states in the main histogram panel by clicking 'Display plateau contrasts' in the 'Time series dynamics' panel:



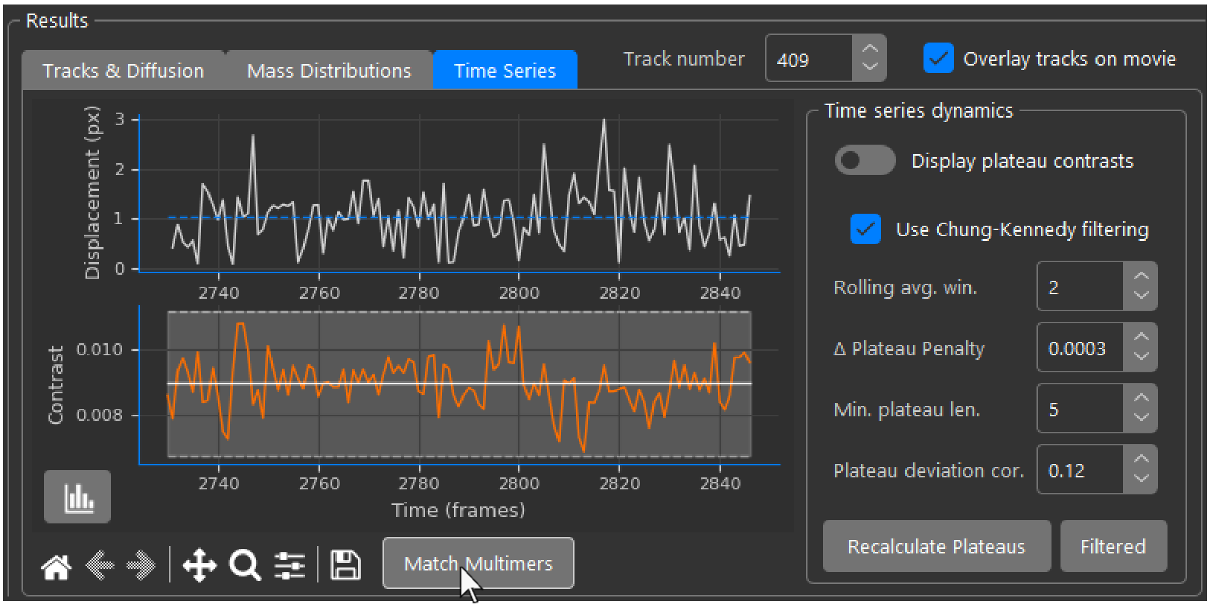
The distribution can be fitted as described in section 1.2 and may reveal otherwise hidden states of the contrast or mass dynamics of your system that aren't present when just plotting the mean track contrasts or masses:



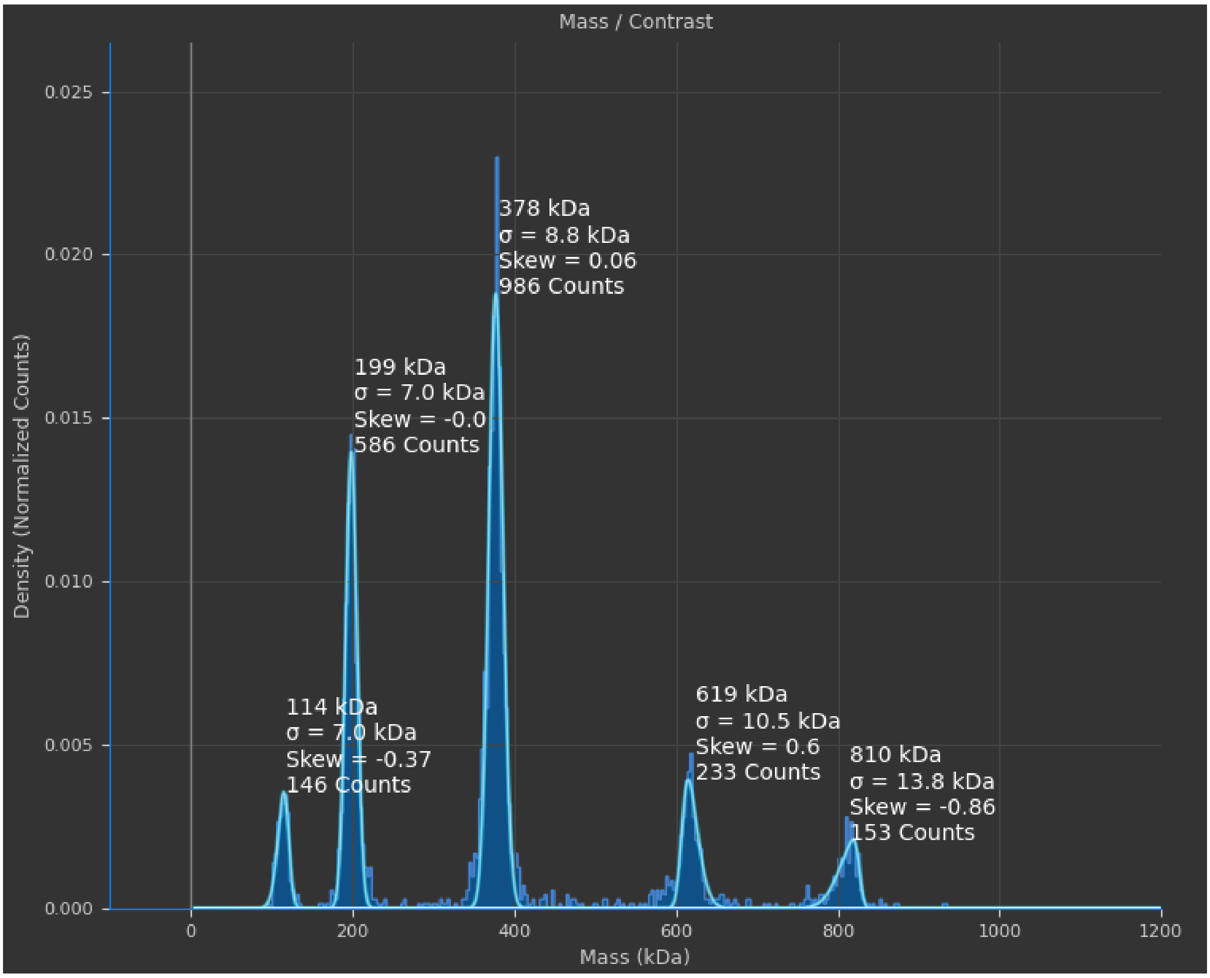
A comparison of track means vs plateau states for tracked RNA molecules is shown below as an example:

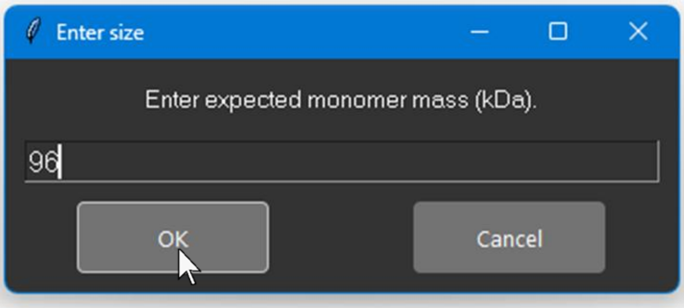


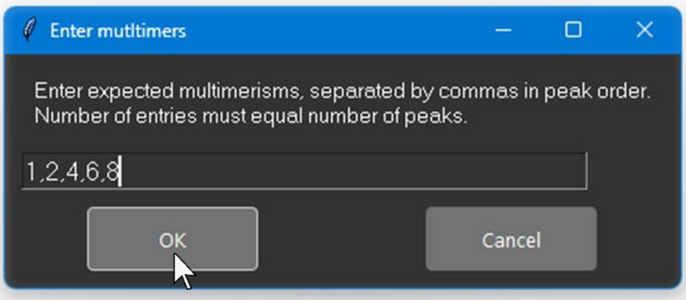
You can check if fitted peaks in the main histogram match your predicted or expected multimerization states and monomer masses for multimeric proteins or complexes. To do so, ensure the peaks you're interested in are fitted, then click 'Match Multimers' in the results panel:



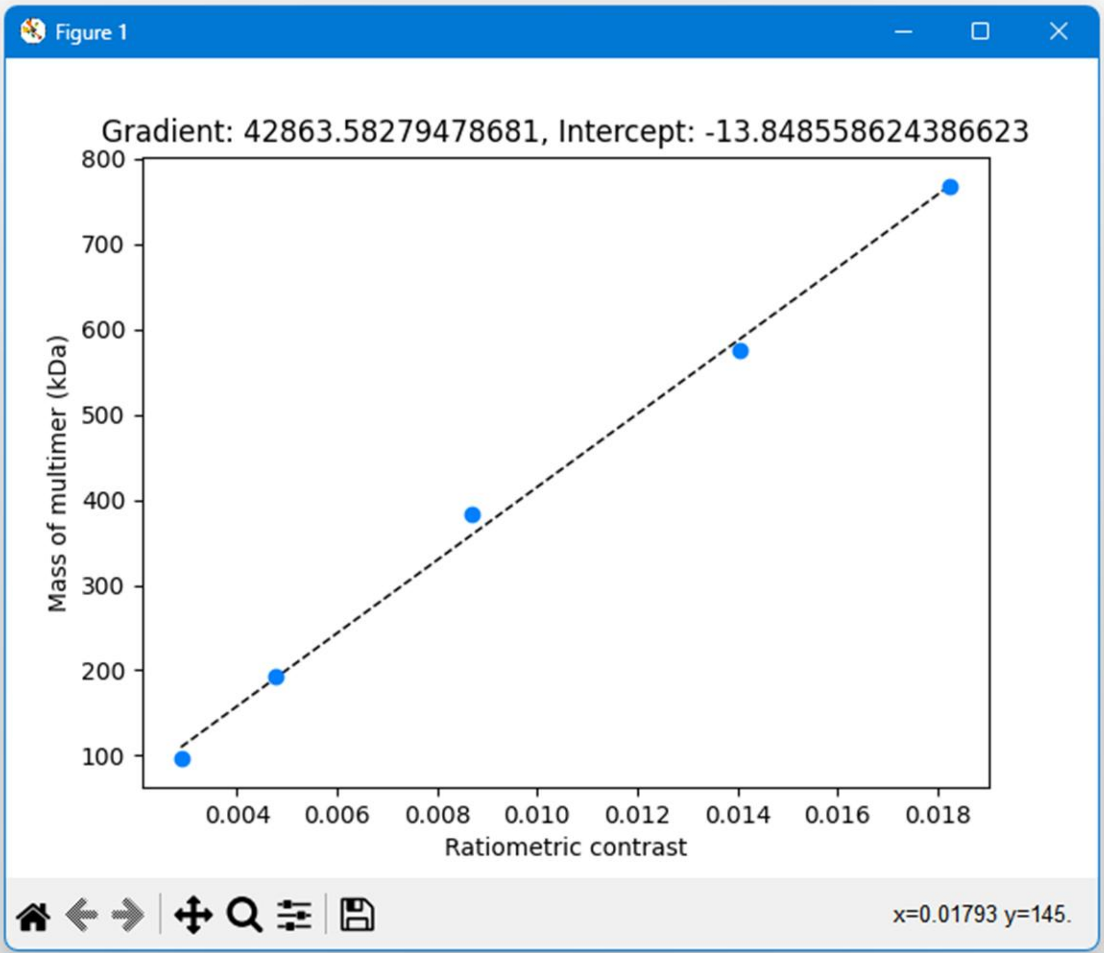
You will then be prompted to enter the monomer mass and expected multimerizations, the example shown here is for simulated dynamin:







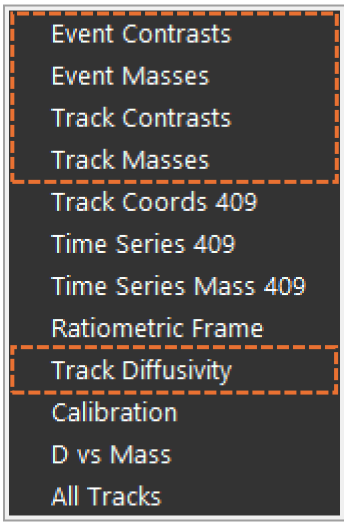
A plot similar to a mass calibration will be generated, correlating ratiometric contrast of the fitted peaks and expected weight of multimers. If the expected masses correlate well (close to diagonal with gradient closely matching the current mass calibration) and the intercept is close to the origin, your model is likely correct:

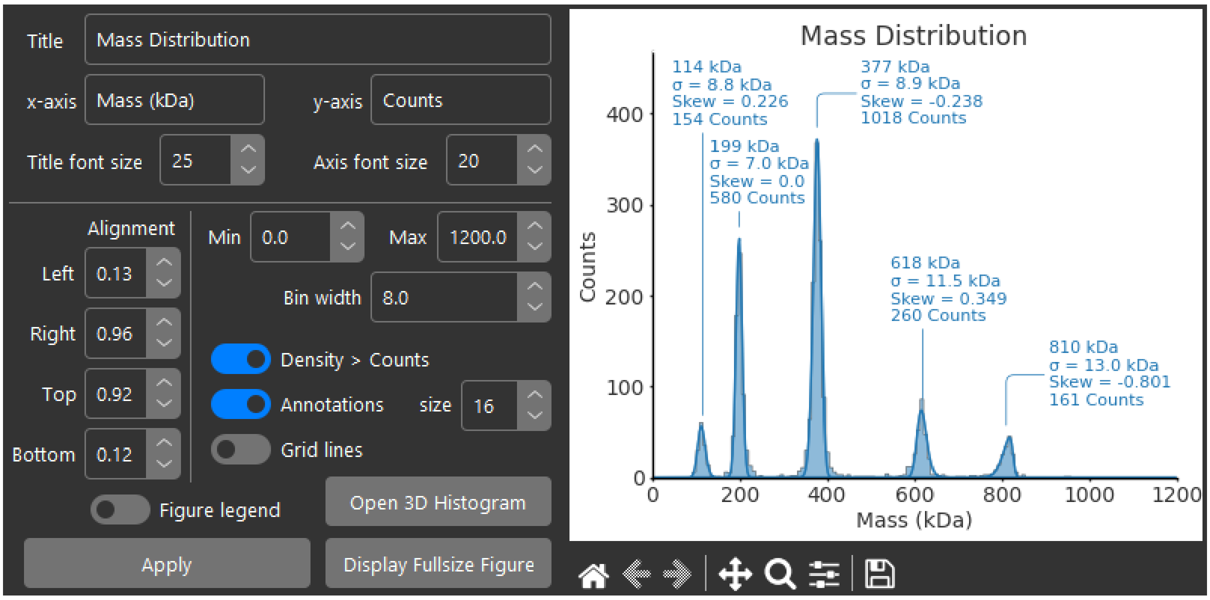


Chapter 6 – Data and Figures

6.1 Histograms

The outlined data types are plotted as histograms by the figure manager:

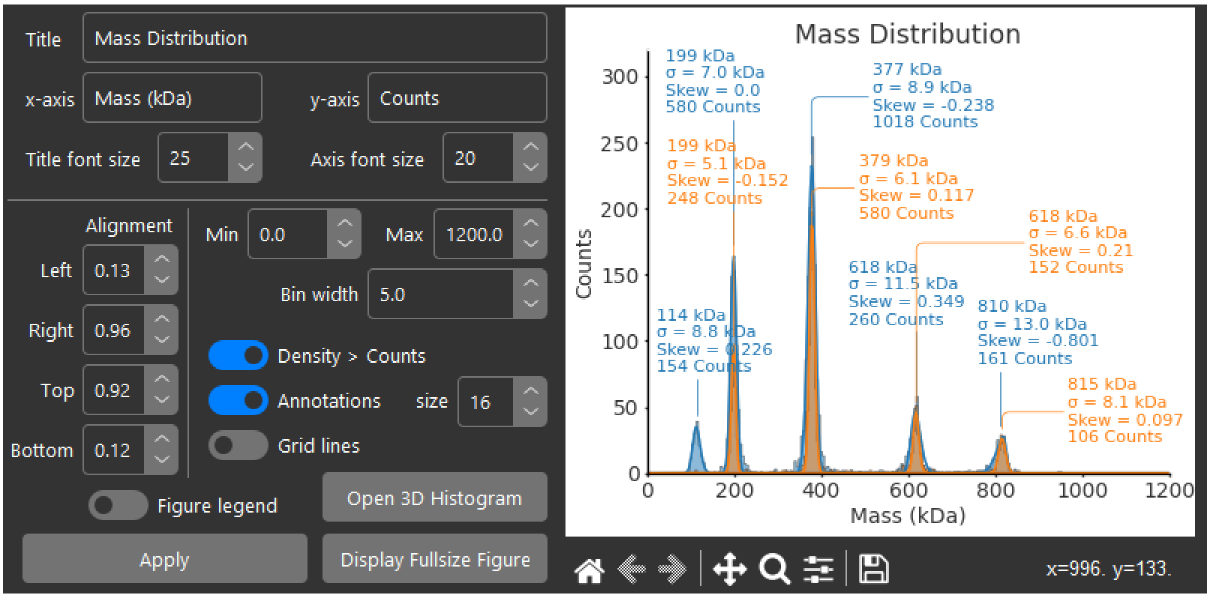


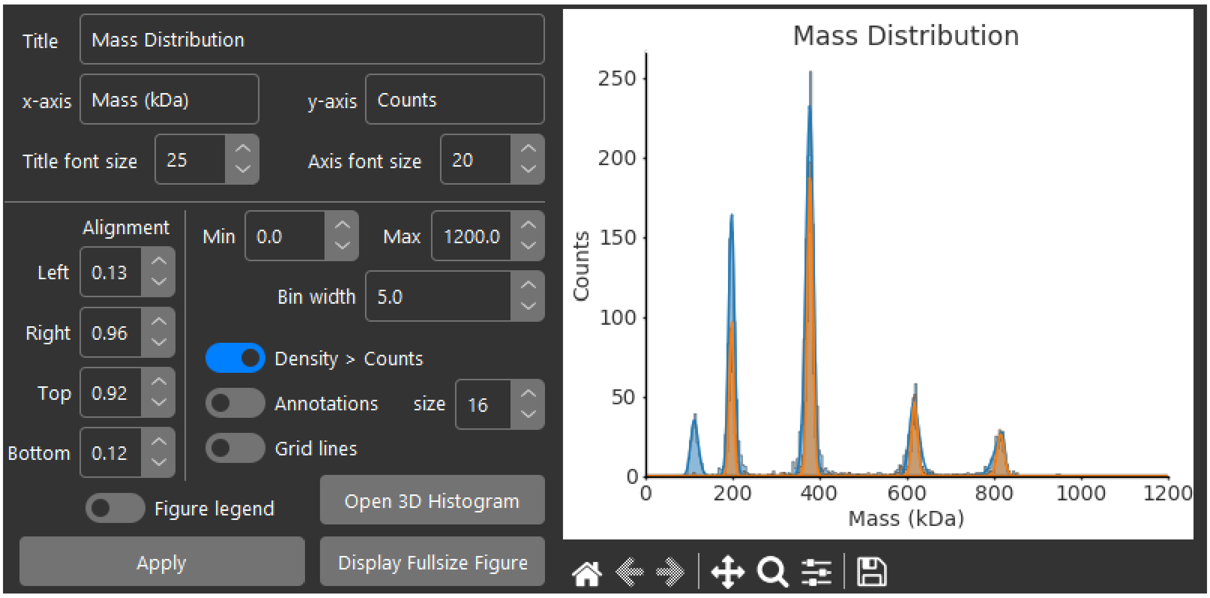


By default, the 'Density > Counts' switch is enabled. This means the distribution will plot counts on the y axis and the histogram is not normalised. Take care that if this switch is disabled and the histogram is normalised to a probability density, that you change the y-axis label to reflect that since it is not updated automatically.

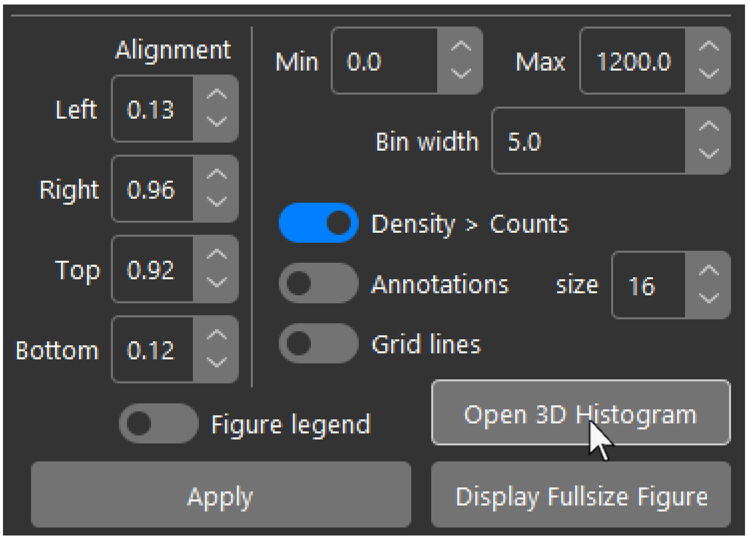
When plotting histograms with any of the available data types, it is possible to add any number of additional datasets to the same figure. You must ensure that the data types are compatible though. Don't try and plot mass, contrast, or diffusivity in the same figure as the x-scales are incompatible and different by orders of magnitude.

When plotting multiple datasets in the same figure, unless the total number of peaks is low, it is recommended to disable annotations as they may clutter the figure space:

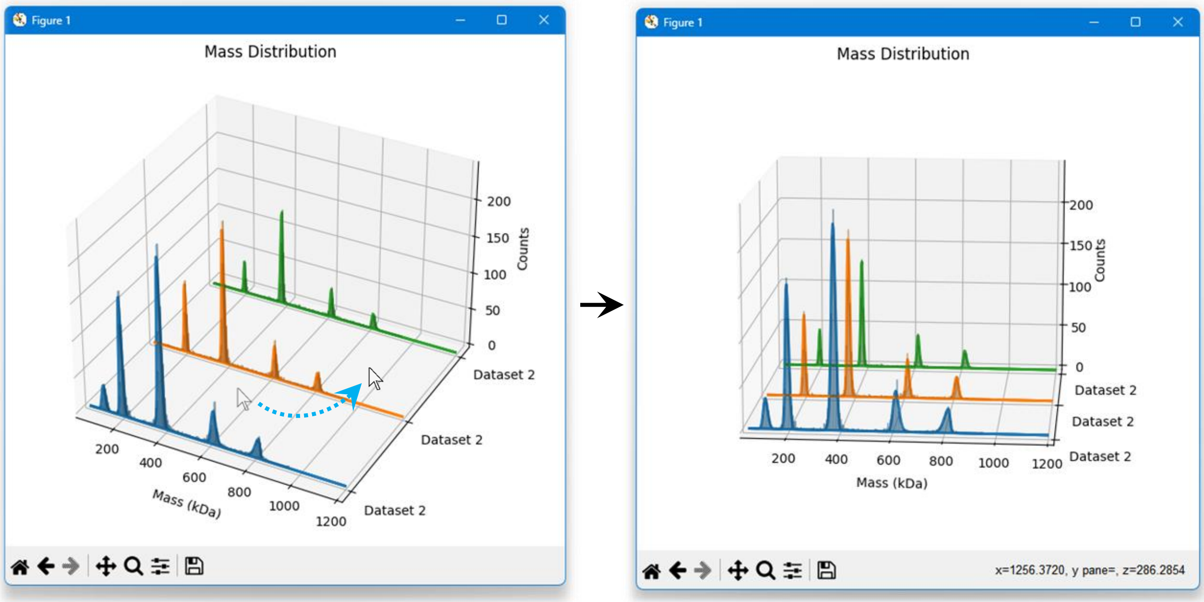




Multiple histograms can also be plotted in a 3D axis. To do so, click 'Open 3D histogram':

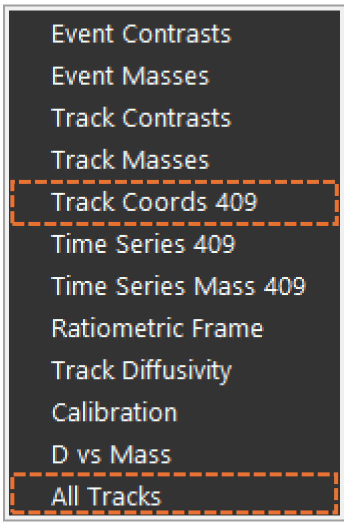


The 3D histogram will then be plotted in a new window. You may need to manually edit the axis labels after exporting the figure since the dataset names only support using the parent dataset name from the dataset manager at this time. The figure can be rotated by dragging over the plot with the mouse:

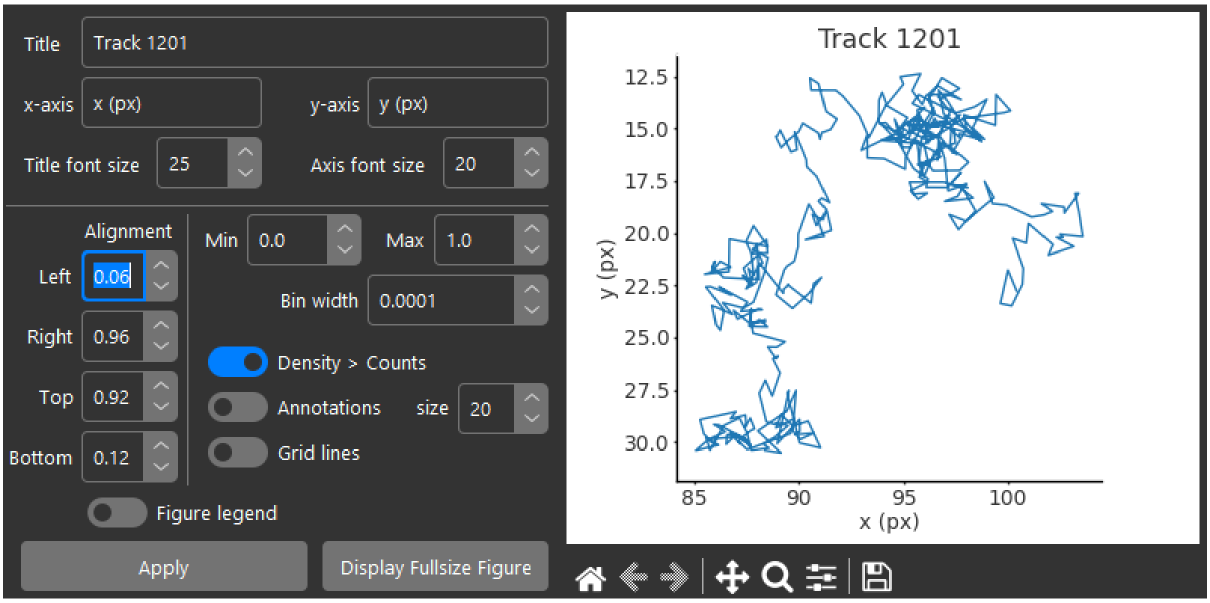


6.2 Tracks

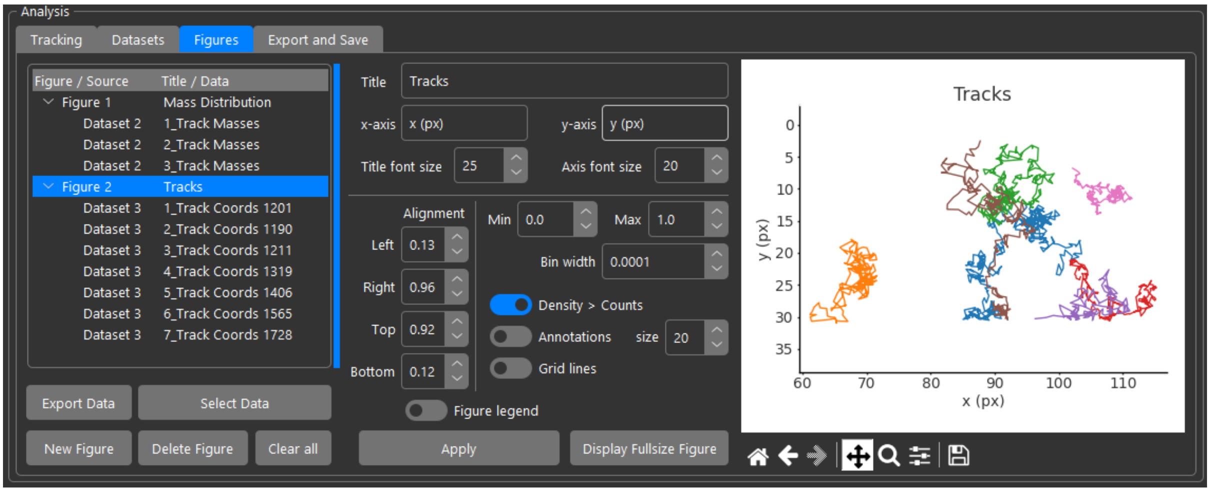
Tracks are plotted from the tracked coordinates of PSFs. They can be plotted individually or all together. When selecting 'Track Coords #NUM' from the drop down in the dataset manager, the current selected track in the Tracking tab will be added. If 'All Tracks' is selected, all tracks in the current filter will be added:



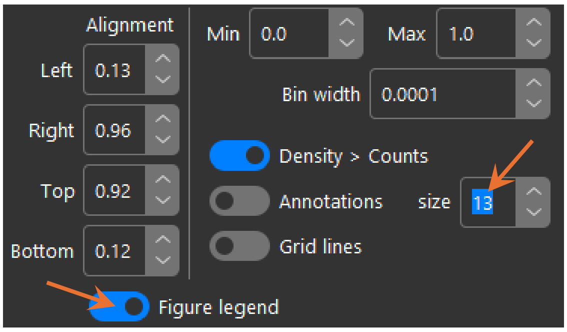
Note that axis labels need to be edited manually:

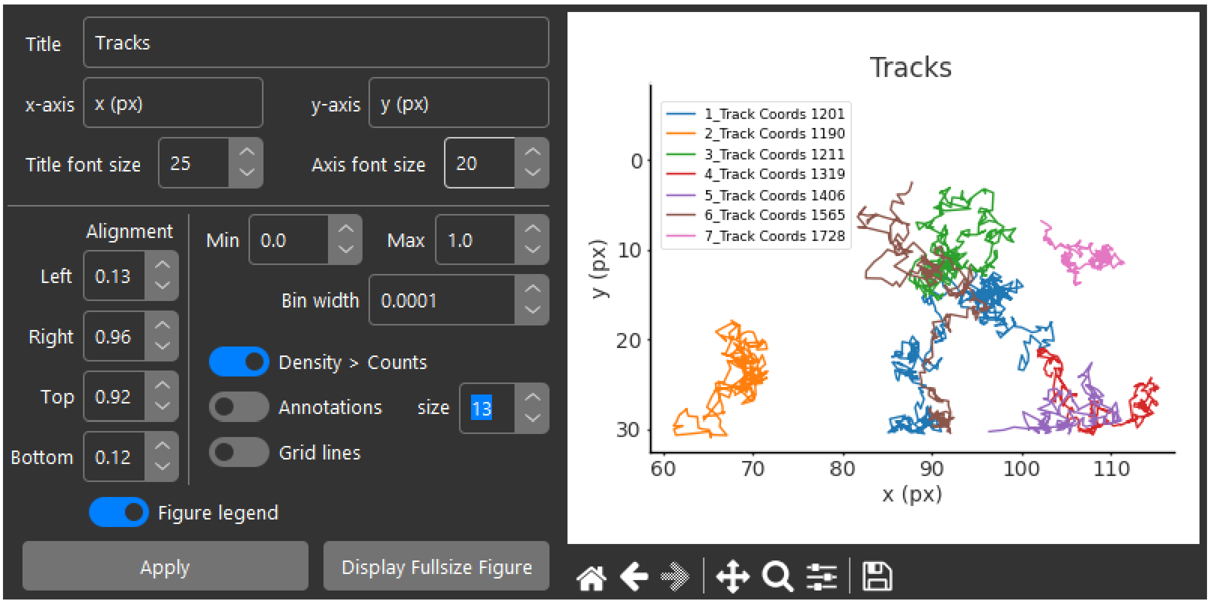


Like histograms, any number of tracks can be plotted on the same axis:

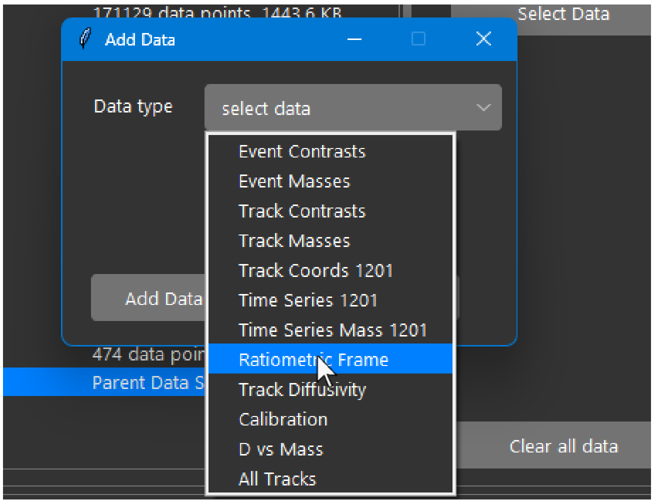


For track plots, the 'Min.', 'Max.', 'Bin width', 'Density > Counts', and 'Annotations' controls do nothing. The annotation size control instead controls the size of the labels in the figure legend:

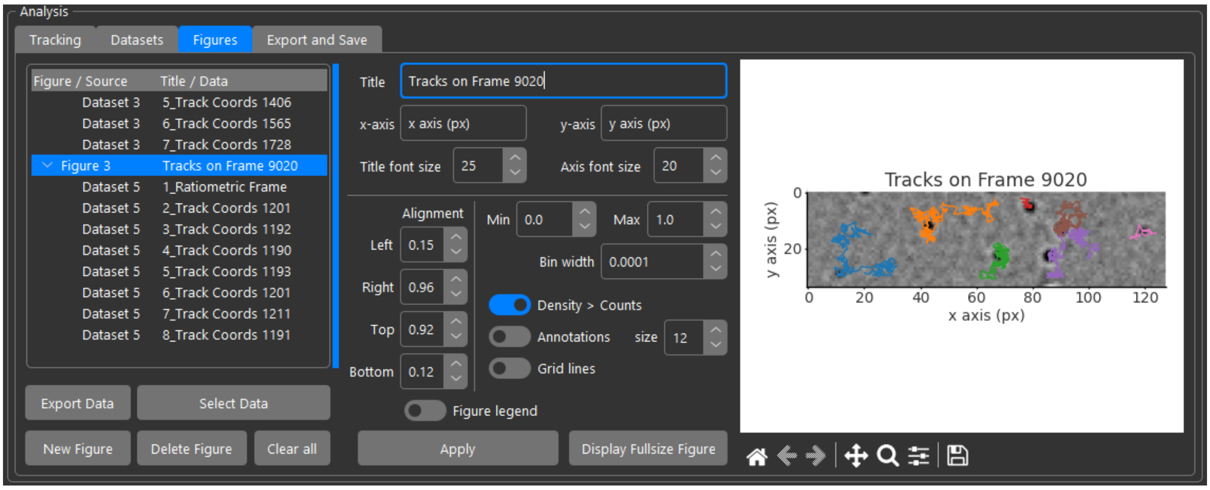




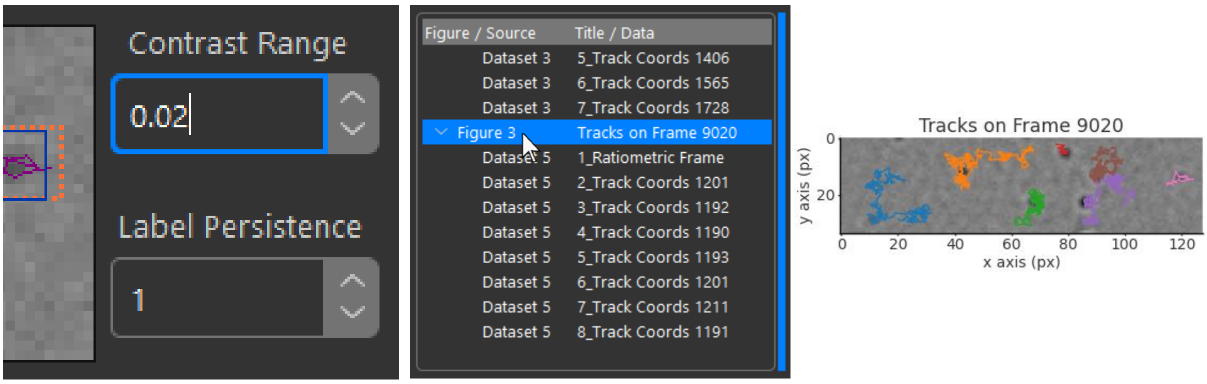
Adding a ratiometric frame to a dataset will add the current ratiometric frame in the ratiometric view:



You can then add the image to the figure with your tracks:

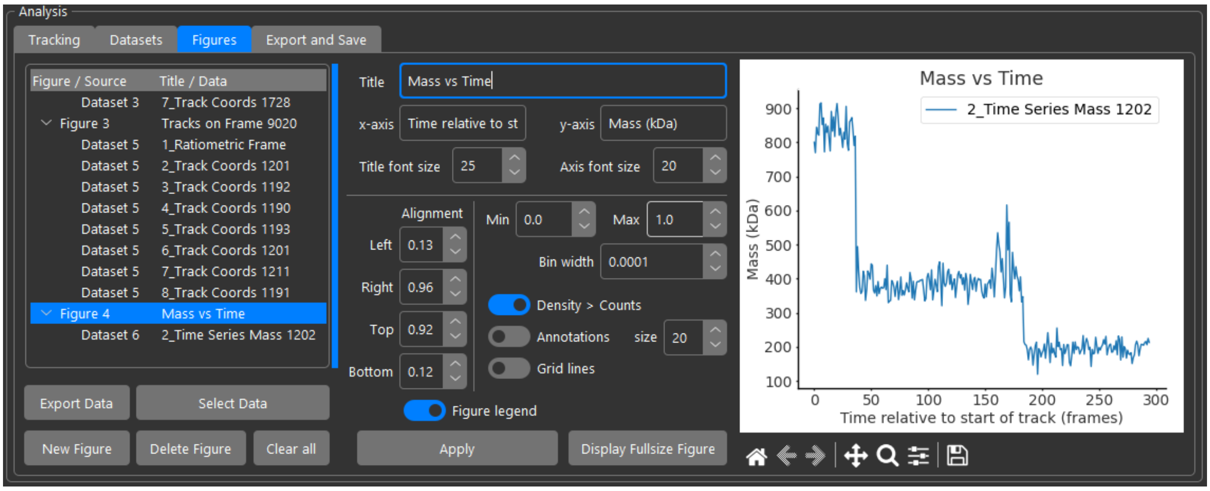


You can adjust the image normalization at any time using the 'Contrast range' control next to the ratiometric view, then reselect the figure in the list box to update the plot:

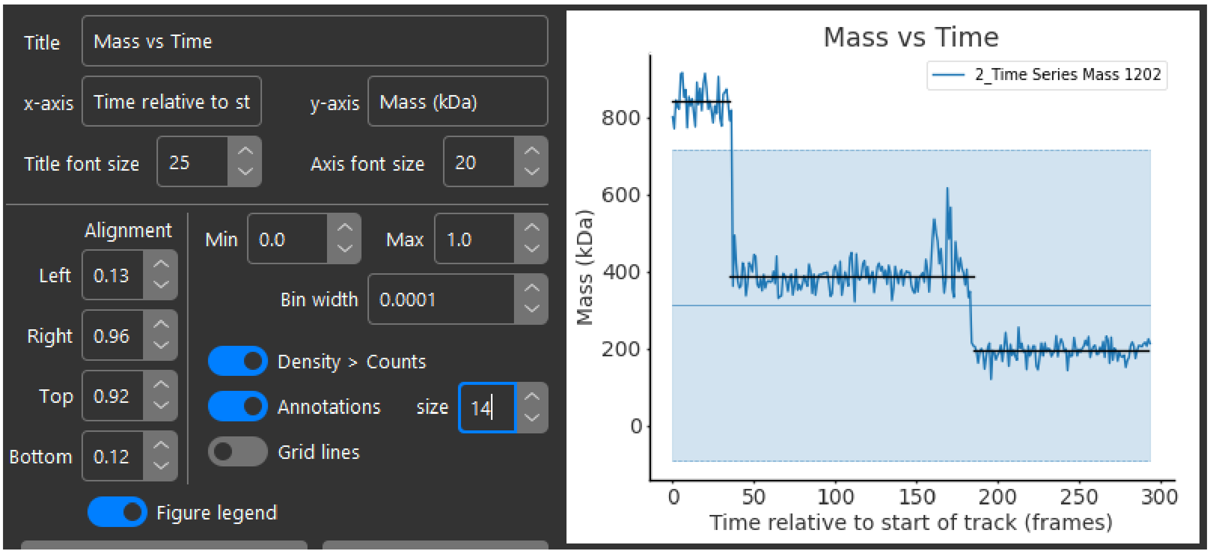


6.3 Time Series

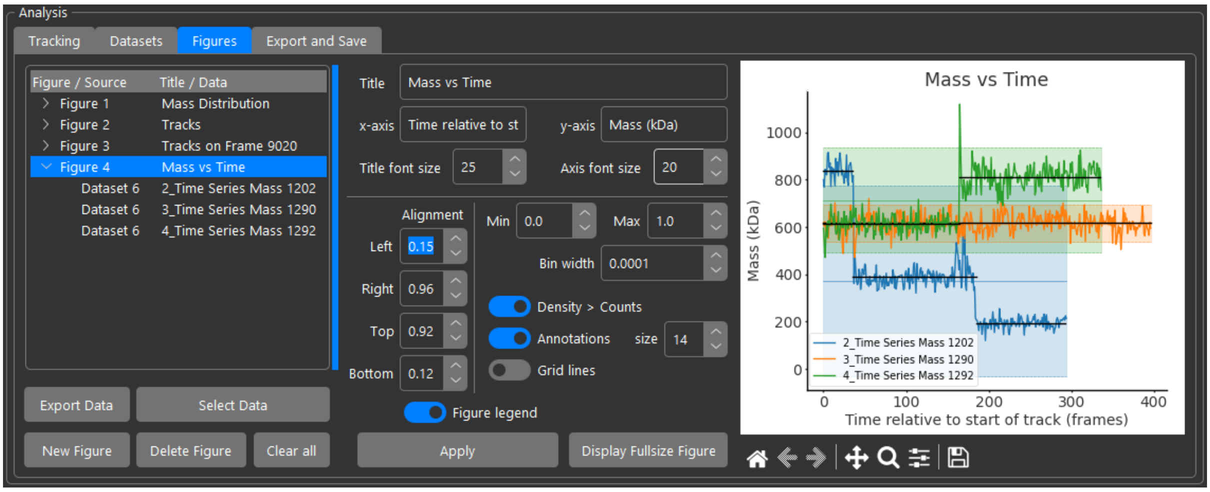
Time series data works in a similar way to tracks, in that the currently selected track's contrast or mass vs time data is added. Once again, multiple selections are possible, however, contrast and mass should be plotted in separate figures as they differ by several orders of magnitude on the y axis:



Turning on annotations will display the standard deviation limit range for mean mass and the fitted plateaus. The 'size' control affects the line width of the plateau fits:

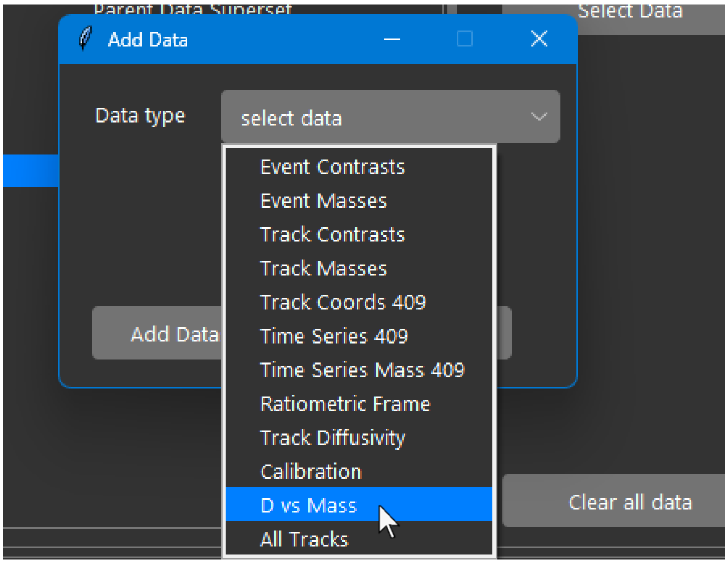


Multiple time series can be plotted on the same axis:

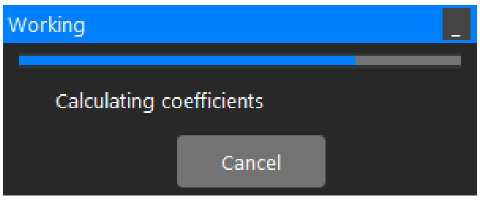


6.4 Diffusivity

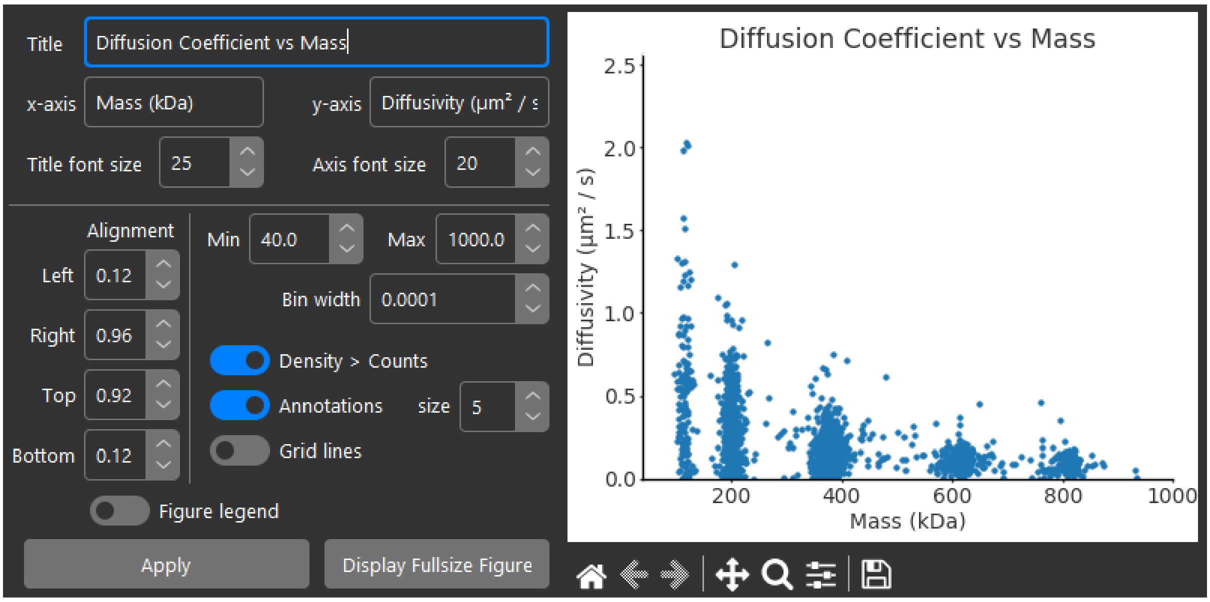
Diffusivity can be plotted as a histogram (see histogram section) or, more interestingly, correlated with mass as a scatter plot. To do so, create a dataset and add 'D vs Mass':



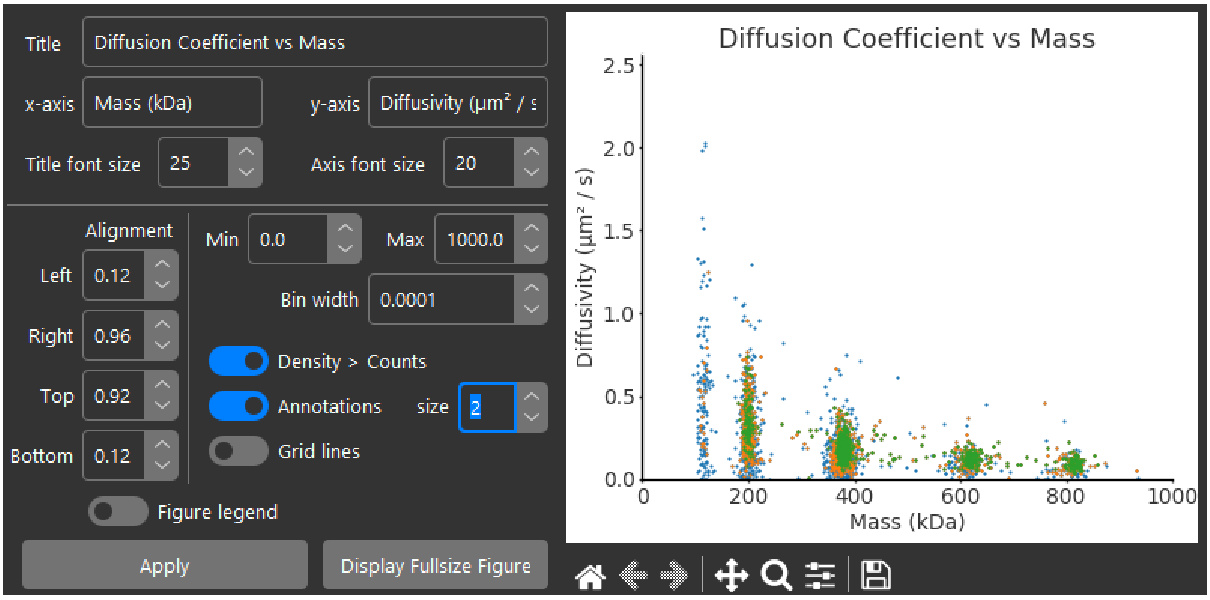
OpenMASS will then calculate the diffusion coefficients (sometimes takes a little time with many long tracks) and a progress bar is displayed:



For D vs Mass plots, Min and Max controls work to set the x range like when plotting histograms:

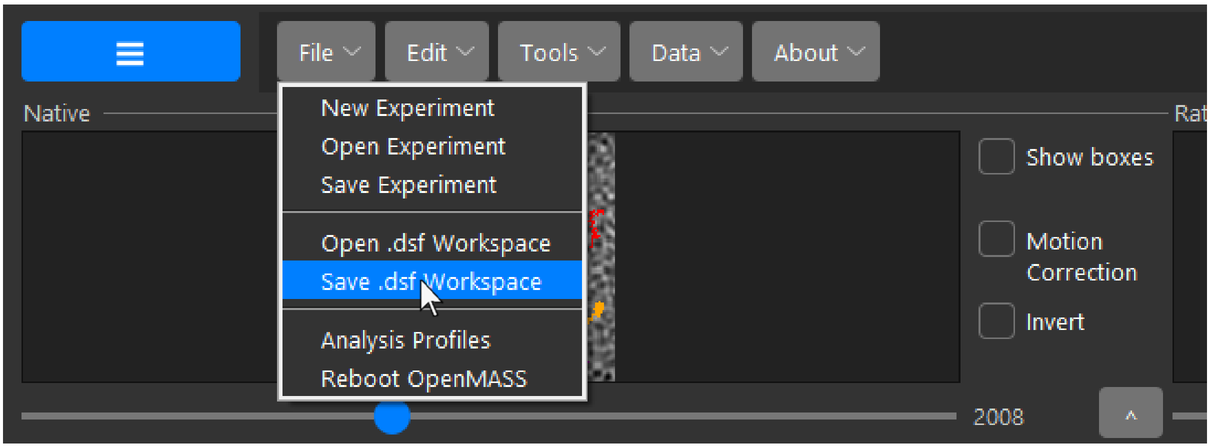


Just like other data types, multiple D vs Mass datasets can be plotted on the same axis. The size control sets the marker size of points in the scatter plot:



6.5 Saving Dataset and Figure Workspaces

The Datasets and Figures you create can be saved and recovered as a workspace, meaning they will be dumped to file 'as is' and can be reopened again after closing and reopening OpenMASS just like individual experiments saved as .lmp files. To save the dataset and figures workspace as a .dsf file, open the menu ribbon and navigate File → Save .dsf Workspace:

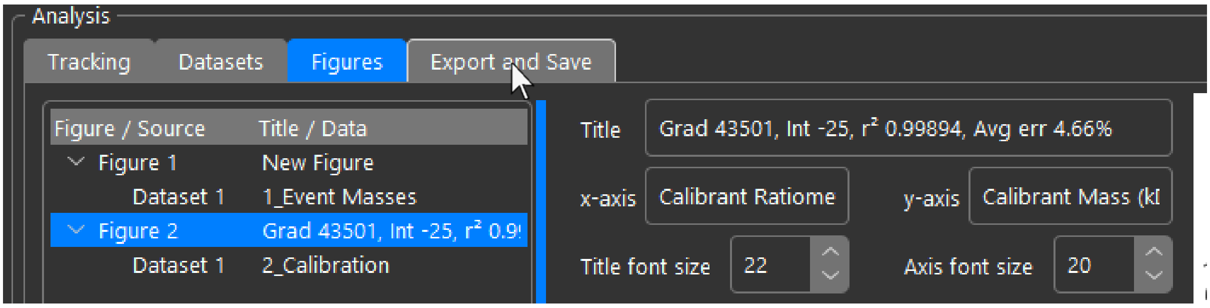


To later reopen the workspace, navigate File → Open .dsf Workspace located just above the command shown above. The workspace will open, and the Analysis panel will automatically switch to the figures tab:

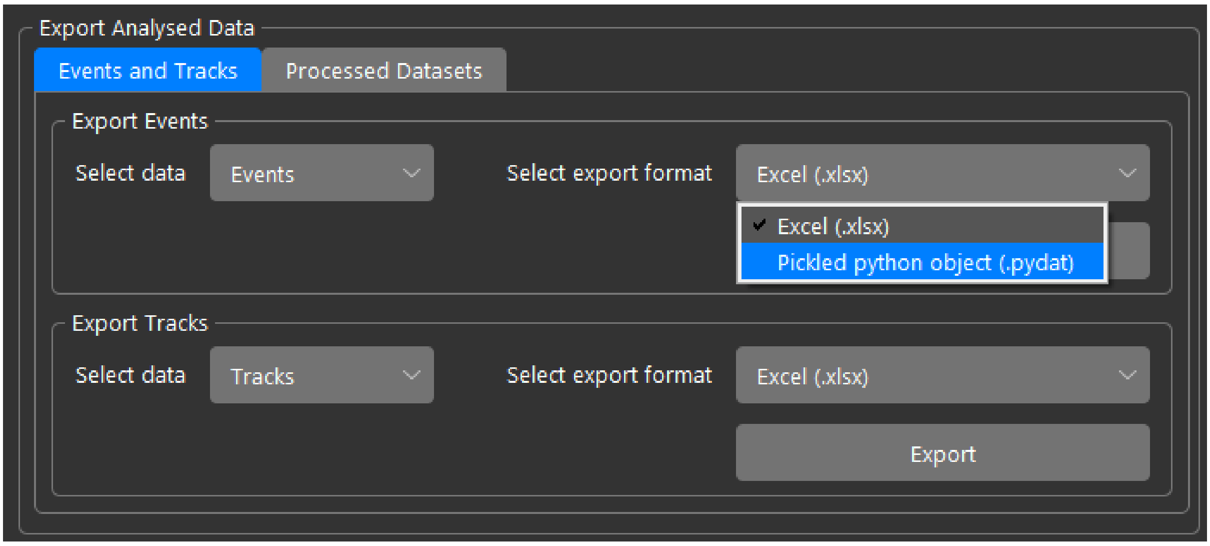
Chapter 7 – Exporting Data and Movies

7.1 Exporting Data

To export data, navigate to the ‘Export and Save’ tab in the Analysis panel:



Exporting Events to Excel is covered in section 1.2, page 21. To export events to a python object, change the export format to ‘.pydat’:



Then click ‘Export’. The following code would open the file and print the first event:

```
import numpy as np
import pickle


path = 'filename.pydat'
with open(path, 'rb') as file:
    events = pickle.load(file)

print(events[0])
```

Events are stored as a nested list where each element is an event and is structured as follows:

```
[
	frame,
	positive convention contrast,
	[
		x coordinate,
		y coordinate,
		sigma x,
		sigma y,
		amplitude,
		baseline,
	],
	contrast trace,
	individual masking state (Boolean or None)
	[
		trace gradient 1,
		trace gradient 2,
		r² 1,
		r² 2,
	],
]
```

Tracks exported to excel have the frames, positions, contrasts, and displacements listed. Since Excel has a maximum number of columns, a new sheet is generated in the workbook for every 1000 tracks. The Exported Excel workbook looks as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Track 1						Track 2						Track 3					
2	Frame	Loc x	Loc y	Contrast	Displacement		Frame	Loc x	Loc y	Contrast	Displacement		Frame	Loc x	Loc y	Contrast	Displacement	
3																		
4	1	104.1469	19.83118	0.008844			1	110.0559	7.854001	0.020193			9	108.9649	30.00993	0.008231		
5	2	102.8515	18.93478	0.008922	1.575273		2	109.3028	7.772749	0.018545	0.757468		10	108.4828	28.64329	0.007778	1.449155	
6	3	102.9325	19.22359	0.00849	0.299955		3	108.9374	7.706851	0.018654	0.371353		11	107.8371	29.12614	0.009221	0.806341	
7	4	101.2429	20.11413	0.009272	1.90997		4	107.9785	7.923799	0.020307	0.983073		12	105.5997	28.63379	0.008755	2.290928	
8	5	100.9944	19.62816	0.007053	0.545818		5	106.8753	7.736463	0.018898	1.119049		13	104.8812	29.83953	0.011116	1.403571	
9	6	102.0933	19.17557	0.009067	1.188435		6	105.7688	8.422661	0.017112	1.30197		14	104.9434	29.14618	0.009845	0.696141	
10	7	101.683	20.53818	0.007853	1.423046		7	106.5492	7.673039	0.017338	1.08206		15	102.843	29.35628	0.009004	2.110929	
11	8	100.8636	21.01912	0.006759	0.950108		8	106.2958	7.498699	0.018189	0.307525		16	103.2531	28.32276	0.00958	1.111937	
12	9	100.002	21.54211	0.008768	1.00787		9	106.3203	8.21689	0.016816	0.718609		17	104.4137	28.16423	0.009679	1.171396	
13	10	100.7397	21.86867	0.009102	0.80672		10	105.6442	8.710404	0.013538	0.837072		18	103.1723	28.95472	0.009363	1.471724	
14	12	99.3075	23.36343	0.002618	2.070133								19	101.9371	28.70935	0.008881	1.259359	
15	13	98.89963	23.81131	0.002189	0.605765								20	100.9477	29.24325	0.008936	1.124297	
16	14	99.20402	23.01838	0.002395	0.849347								21	99.34695	30.09657	0.009904	1.813961	
17																		
18																		
19																		

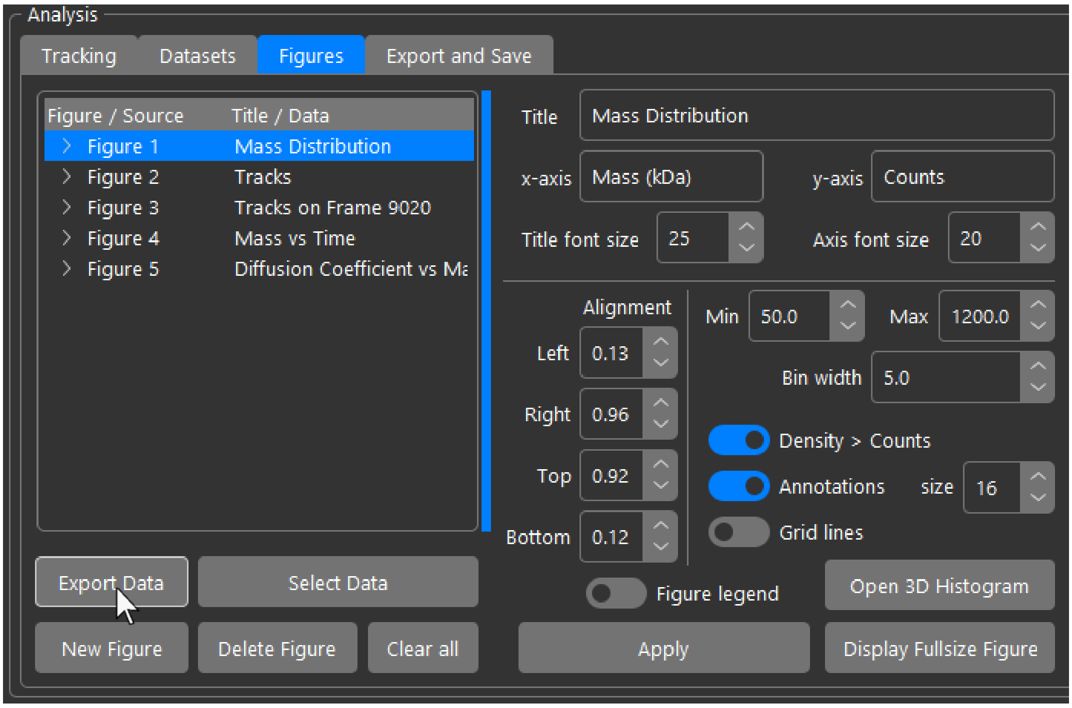
If tracks are exported to python objects (.pydat), the code you write to load them must include the following Track class:

```
class Track:
    def __init__(self):
        self.events = []
        self.frames = []
        self.coords = []
        self.contrasts = []
        self.displacements = []
        self.type = None
        self.ended = False
        self.plateaus = []
        self.plateau_contrasts = []

    def get_distance(self, other):
        return np.sqrt((self.coords[-1][0] - other[0])**2 + (self.coords[-1][1] -
other[1])**2)

    def same_type(self, other):
        if self.type == other:
            return True
        else:
            return False
```

the loaded object will be a list of these track objects.

7.2 Exporting Figure Datasets

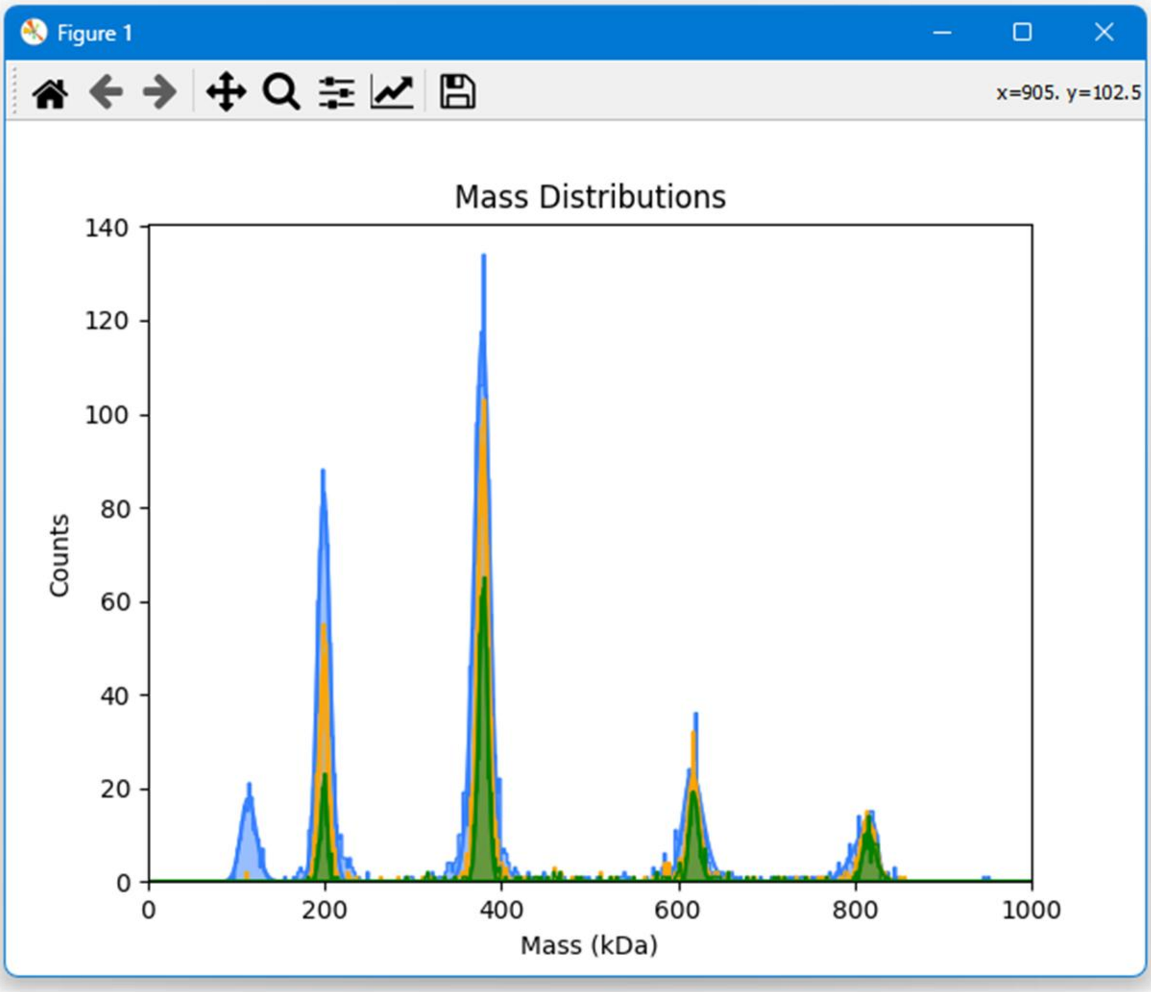
Data plotted in figures can be exported as a python object directly from the Figures tab. Click 'Export Data' under the figure list box:



Since datasets saved from figures are also stored in objects and may contain tracks too, you need to include the Track, Dataset, and Datasubset classes in your code. The code example below is designed to load and plot data exported from a histogram figure:

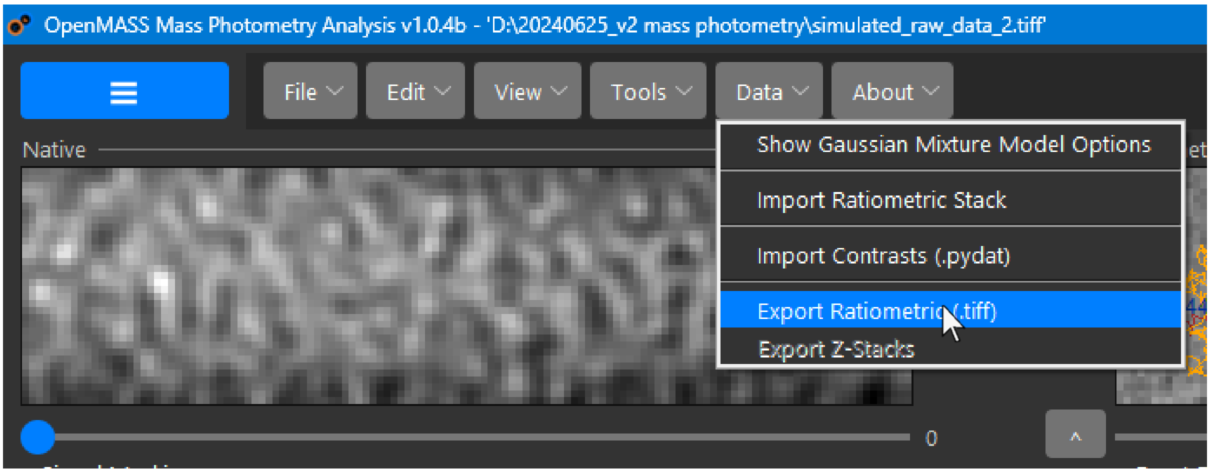
```
import numpy as np
import matplotlib.pyplot as plt
import pickle

class Track:
    def __init__(self):
        self.events = []
        self.frames = []
        self.coords = []
        self.contrasts = []
        self.displacements = []
        self.type = None
        self.ended = False
        self.plateaus = []
        self.plateau_contrasts = []

    def get_distance(self, other):
        return np.sqrt((self.coords[-1][0] - other[0])**2 + (self.coords[-1][1] -
other[1])**2)

    def same_type(self, other):
        if self.type == other:
            return True
        else:
            return False

class Dataset:
    def __init__(self):
        self.subsets = []
        self.ID = None
        self.name = ''
        self.filename = ''
        self.info = 'Parent Data Superset'

class DataSubset:
    def __init__(self):
        self.data = None
        self.fits = []
        self.x_axis = None
        self.fit_vals = None
        self.ID = None
        self.parent = None
        self.data_parent = None
        self.name = ''
        self.type = ''
        self.info = ''
        self.metadata = ''
        self.calib = [0, 1]

path = 'example_data.pydat'
with open(path, 'rb') as file:
    datasets: list = pickle.load(file)

colours = ['#337fff', 'orange', 'green', 'red']
bins = 1000
for idx, dataset in enumerate(datasets):
    distribution, fits, axis = dataset.data, dataset.fits, dataset.x_axis
    col = colours[idx % len(colours)]
    plt.hist(distribution, color=col, bins=bins, density=False, alpha=0.5)
    plt.hist(distribution, color=col, bins=bins, density=False, histtype='step')
    scale_factor = (np.max(distribution) - np.min(distribution)) / bins *
len(distribution)
    for fit in fits:
        plt.plot(axis, fit*scale_factor, color=col, alpha=0.75)
plt.xlim(0, 1000)
plt.title('Mass Distributions')
plt.xlabel('Mass (kDa)')
plt.ylabel('Counts')
plt.show()
```

Running the program plots the histogram data exported from the figure:

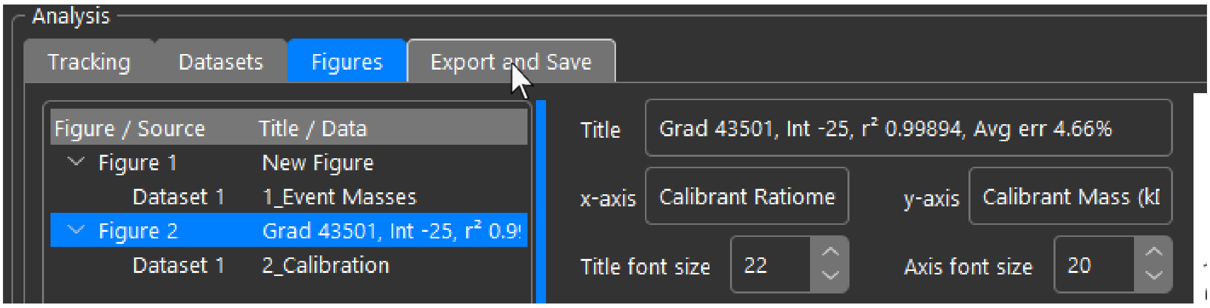


7.3 Exporting Movies

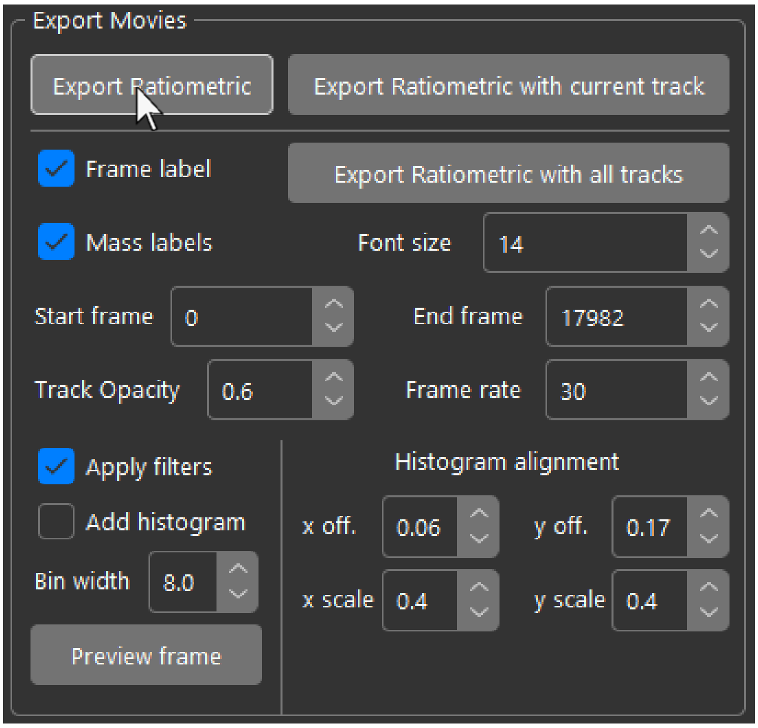
To export the processed ratiometric image stack as a lossless .tiff file that can be used for further analysis, open the menu bar and navigate 'Data' → 'Export Ratiometric (.tiff)':



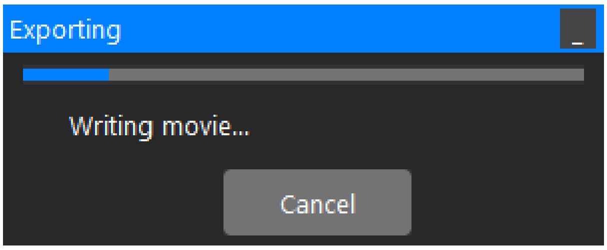
To export movies as mp4 videos for other purposes such as presentations, navigate to the 'Export and Save' tab:



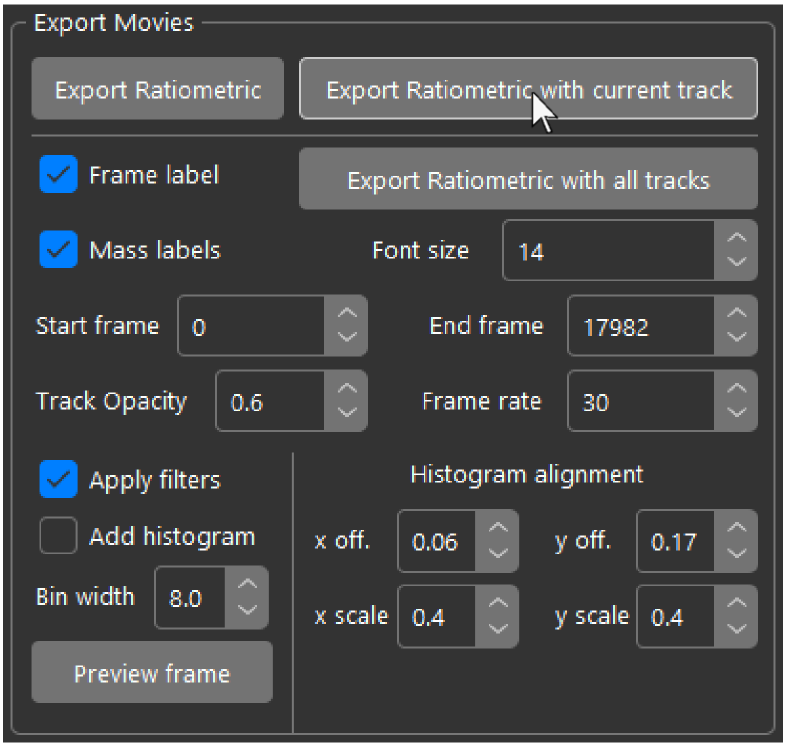
To save the ratiometric stack as an MP4 video, click 'Export Ratiometric':



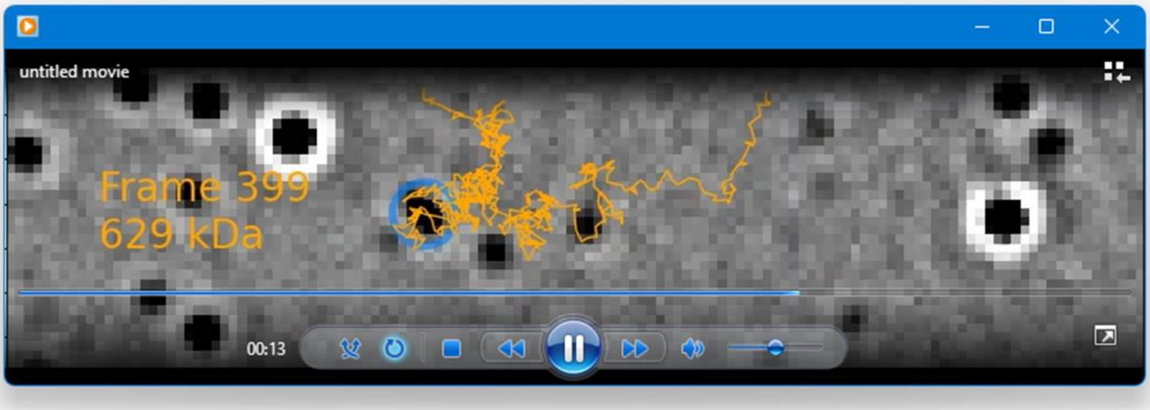
OpenMASS will export the stack as a high-bitrate MP4 file. A progress bar will be displayed while the movie is being exported:



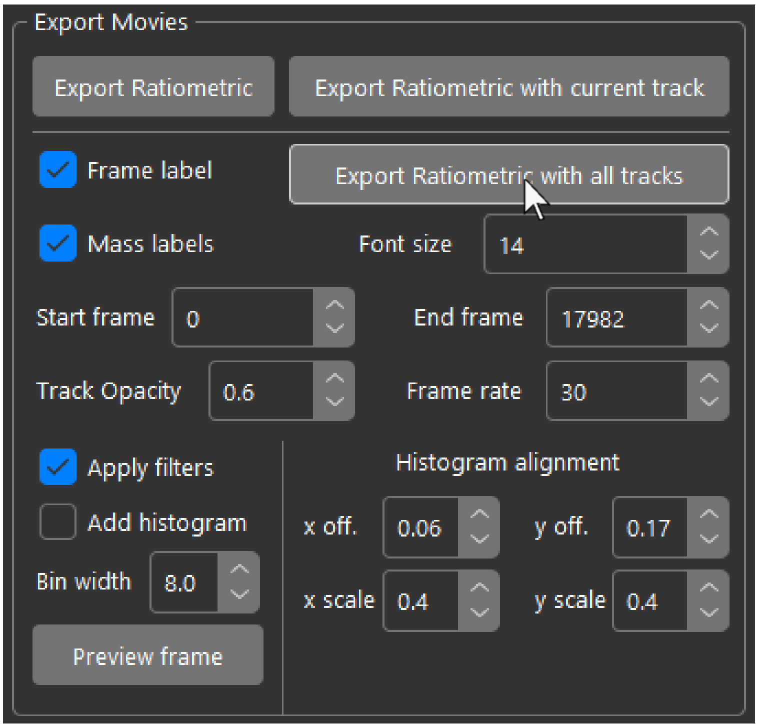
For tracking data, you can also export the ratiometric stack as an MP4 video with tracks superimposed. To export the frames of the movie with the currently selected track, click 'Export Ratiometric with current track':



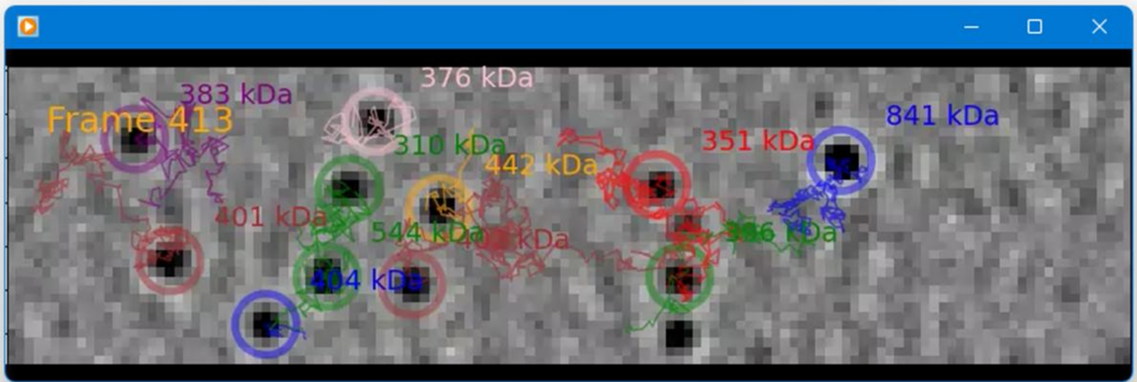
Only the frames with that particular track will be rendered. The current frame and mass of the track will be labelled in the video:



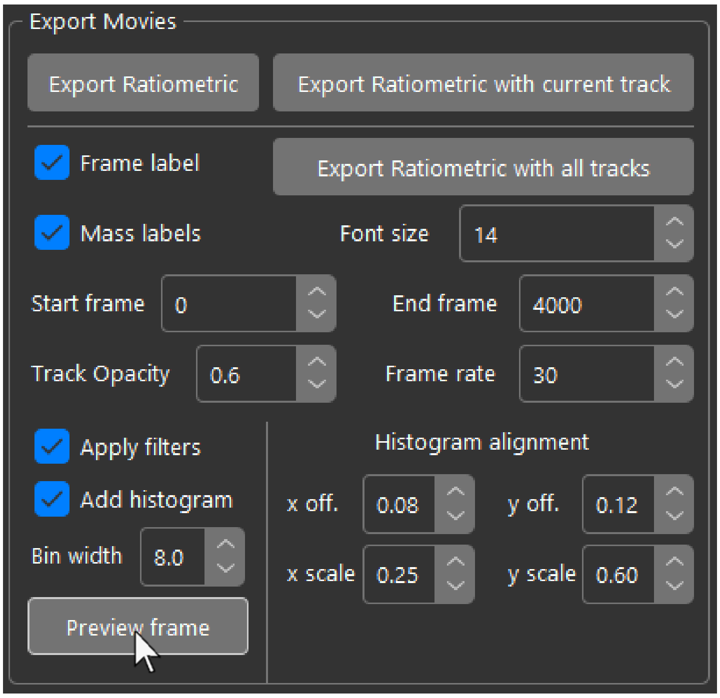
Alternatively, you can render a defined range of frames with all tracks superimposed and labelled in the movie. Set the start and end frame for the range you wish to export. You can also adjust the font size of the mass labels and the opacity of the track overlay. If the 'Apply filters' checkbox is checked, only tracks in the current filter according to minimum track length, standard deviation limit, and minimum displacement will be rendered on top of the ratiometric stack. When ready click 'Export Ratiometric with all tracks:

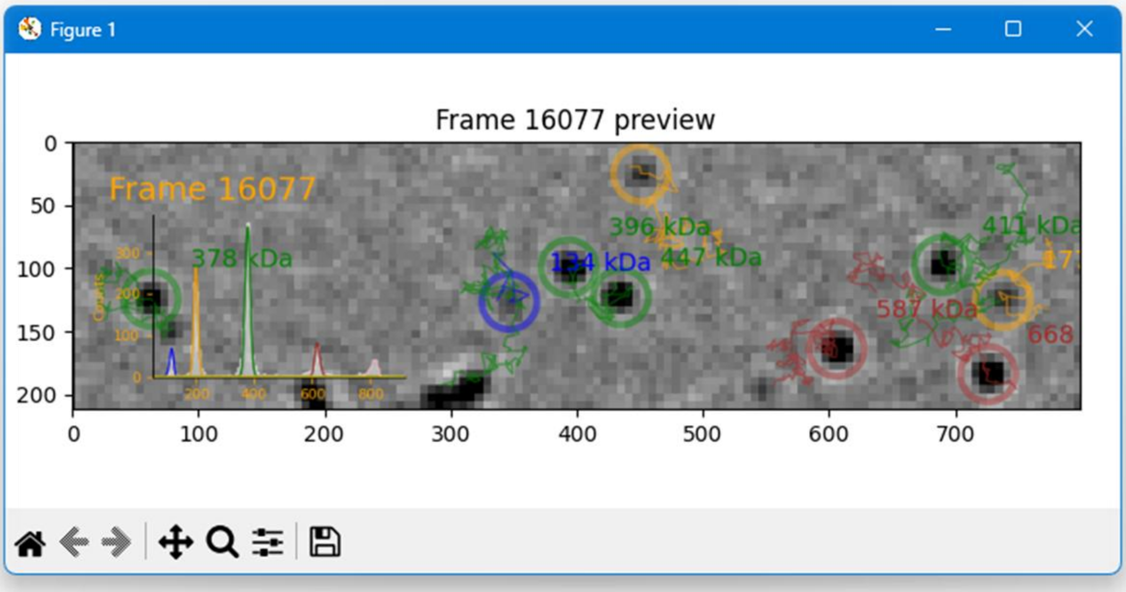


The export process may take several minutes for long movies with many tracks.



With fitted peaks, you can also export the movie with a histogram superimposed. Fitted peaks will be colour coded and tracked molecules in the overlay will be coloured according to the peak they belong to. You can preview what the current frame of the ratiometric view will look like when rendered by clicking 'Preview Frame':





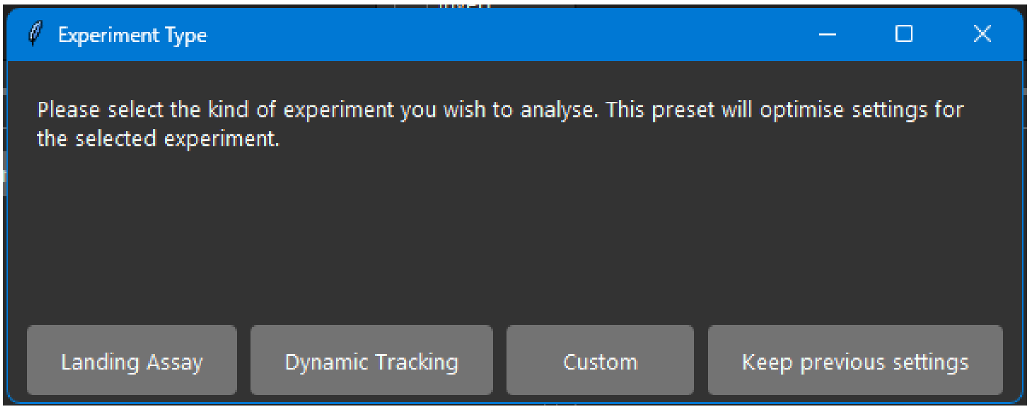
Chapter 8 – Analysis Settings Profiles

One of the most important considerations when batch analysing a lot of data is to use consistent settings for reproducible and comparable results and keeping track of the analysis settings used if they are non-standard. This chapter briefly covers analysis settings profiles which are an easy way to maintain consistency.

The first important thing to note, is that when experiments are saved as '.lmp' files as shown in section 1.2, page 14, the entire workspace apart from datasets and figures, is saved into a file including all analysis settings and the mass calibration if one was loaded. When the workspace is reopened in a different session all settings will be reloaded as if you never closed the session.

8.1 Presets

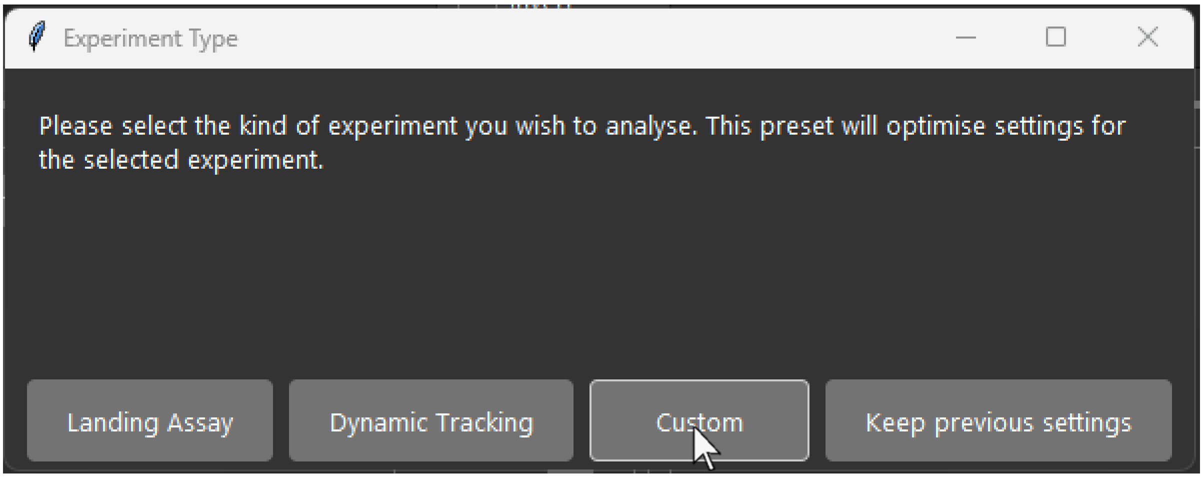
Whenever OpenMASS starts, and when you click 'File' → 'New Experiment', the experiment selection window will appear, prompting you to choose between landing assays and tracking experiments:



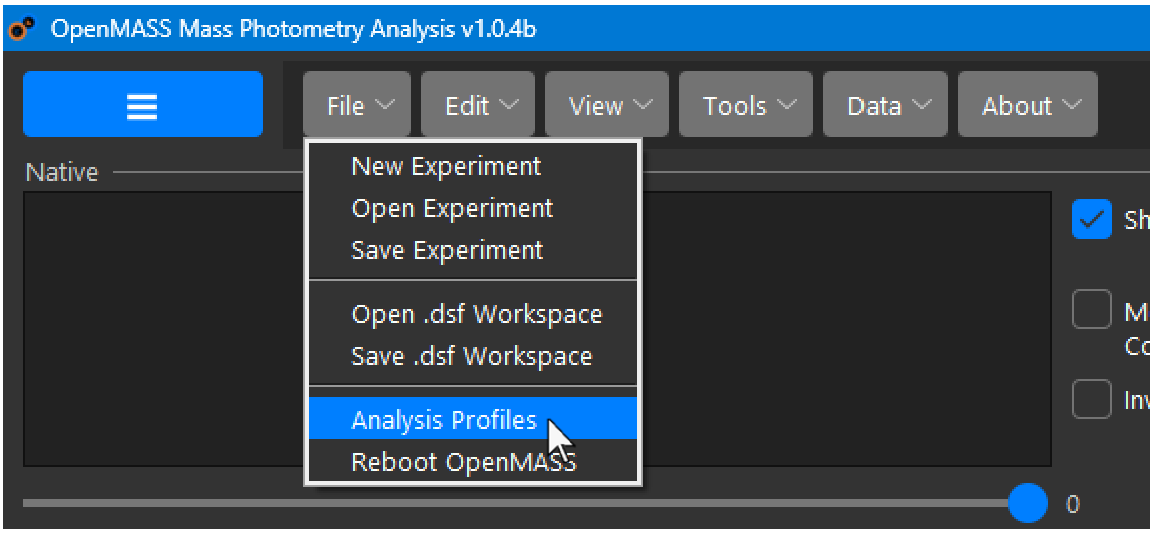
These settings are optimised for quick general use by non-expert users. Note that if you have loaded a profile and want to keep analysing more files with the same settings after clicking 'File' → 'New Experiment', you can just click 'Keep previous settings' when this window appears or just close it. For a faster experience, clicking 'Load Data' will also immediately clear the interface (exactly the same as clicking New Experiment under the hood) and maintain previous settings.

8.2 Custom Profiles

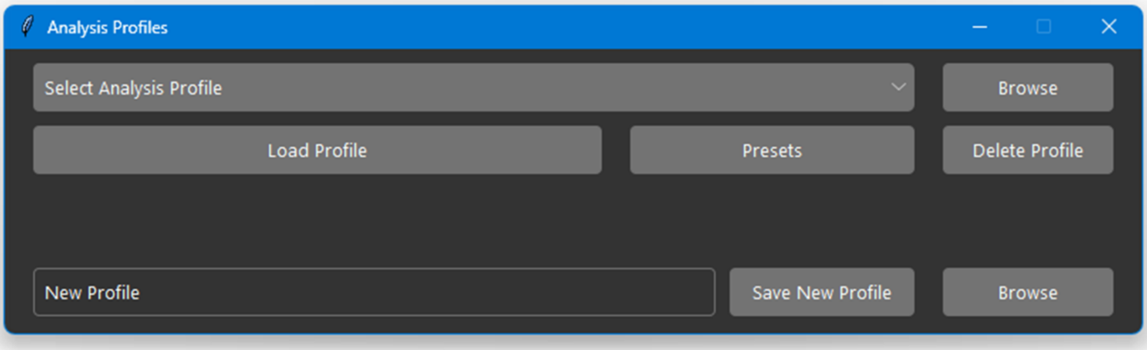
If you know what you're doing and have improved upon the settings for your particular use case and want to be able to preserve them and quickly load them again in the future for consistency, open the analysis profiles window. This can be achieved in one of two ways. When OpenMASS starts, or after clicking 'File' → 'New Experiment', the experiment selection window will appear. Click 'Custom':



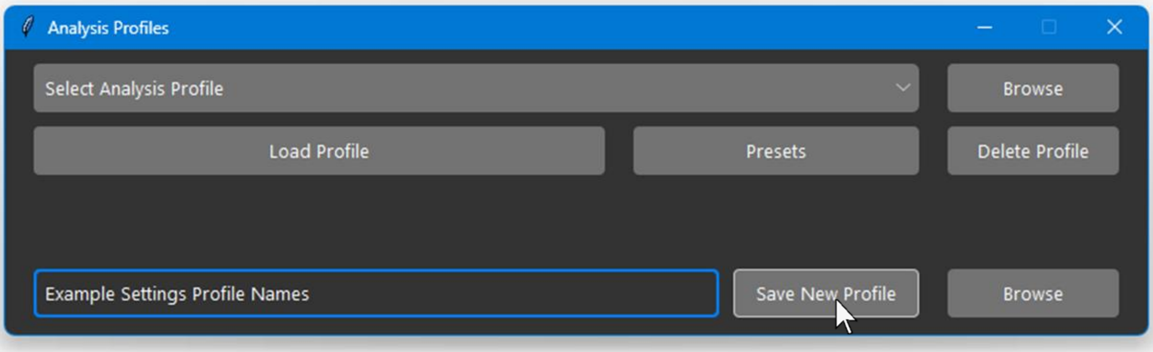
The second way is to navigate 'File' → 'Analysis Profiles':



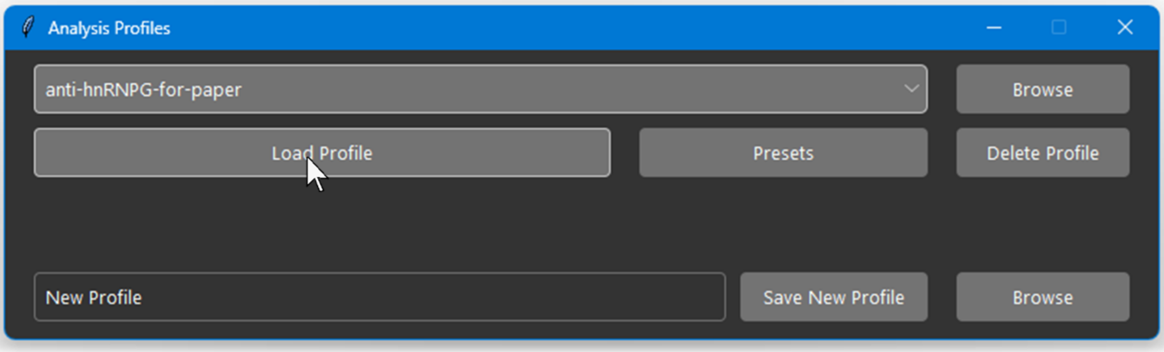
The analysis profiles window will then appear:



To save the settings you created as a profile, type a name for the profile in the entry bar at the bottom and click 'Save New Profile':



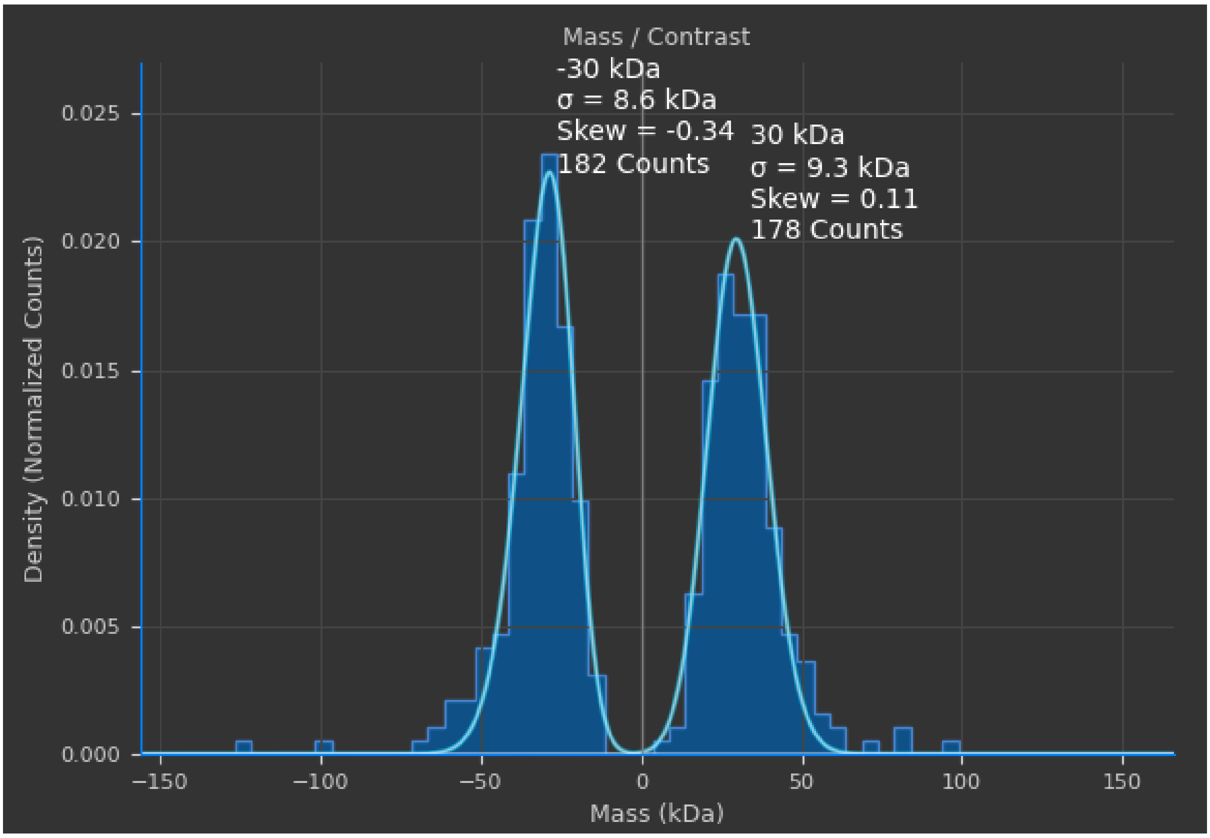
To reload an analysis profile in the future, simply select it in the dropdown menu at the top and click 'Load Profile':



Chapter 9 – Miscellaneous

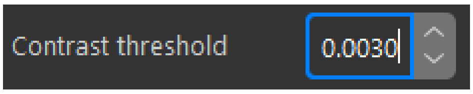
9.1 Calibrating the Detection Threshold

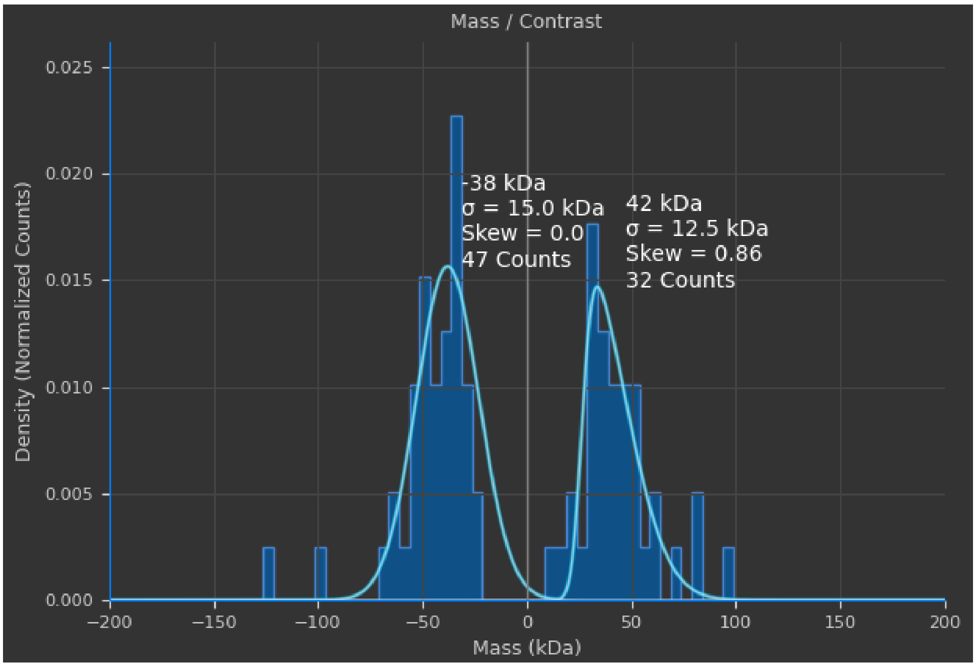
In cases where it is important to get accurate peak integration such as kinetics the contrast threshold for event detection should be calibrated to only allow a minimum number of artifactual events through. To do this, the first step is to acquire a blank acquisition from the buffer sample you intend to use for your experiments. Start a landing assay experiment and load a mass calibration. Load the file and compute the ratiometric stack as described in the quick-start guide in section 1.2. Then analyse the file by detecting contrast events using the default contrast threshold of 0.0025. Fit the distribution to find the number of counts in the blank buffer sample and their masses:



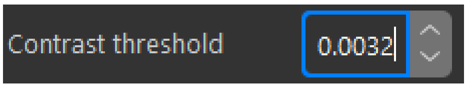
The default threshold is often a little too relaxed and will find events in the noise and fluctuating background. A key indicator here is that the detected contrast is below the detection limit quoted by Refeyn of 40 kDa.

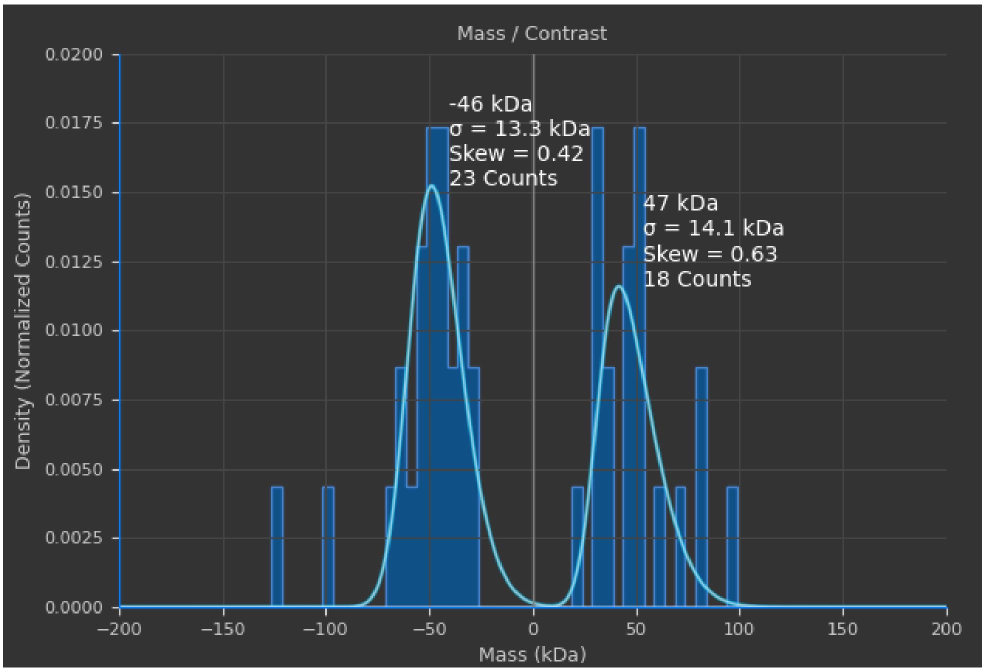
Increase the threshold by 0.0005 to a value of 0.003, then recompute events:





You can repeat this process again if needed, to minimise the number of events detected in the buffer sample. Do not set it too high such that there are no events detected. Leave a small number of events to ensure you aren't cutting off beyond the detection limit:

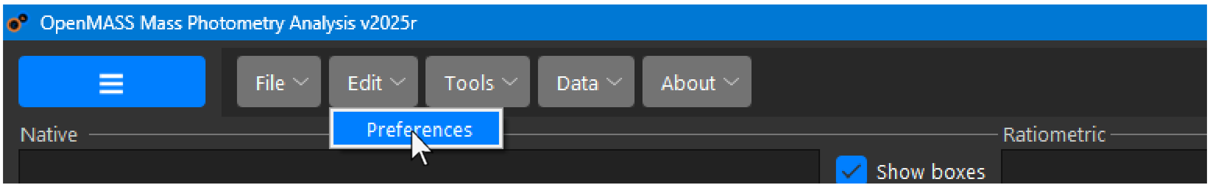




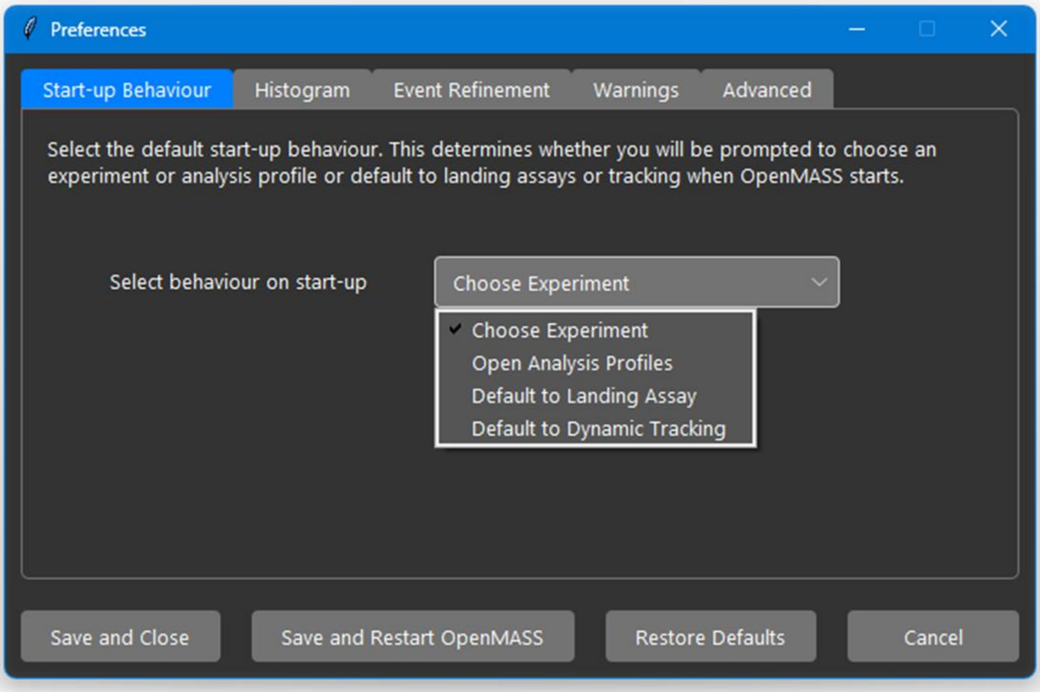
You can then create an analysis profile using the calibrated contrast threshold as described in section 8.2 on page 73.

9.2 Preferences

Some quality-of-life settings are available in the preferences manager. To access preferences, open the menu ribbon and navigate: Edit → Preferences:

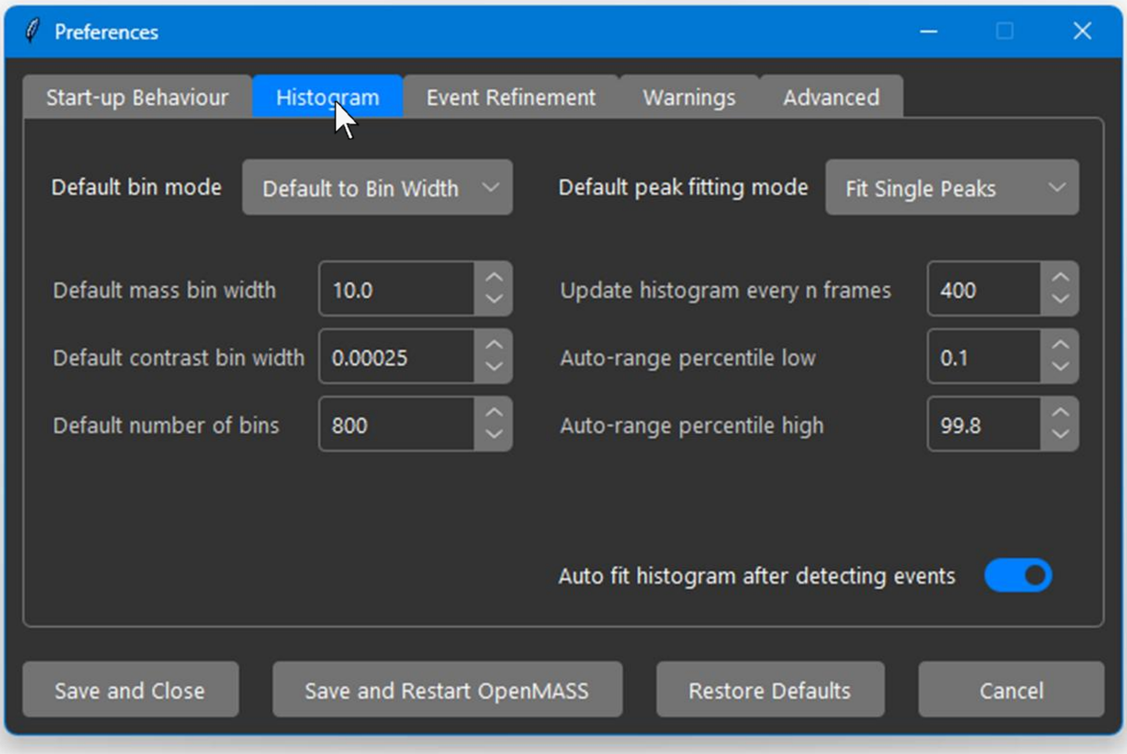


The preferences window will open and allow you to customise the software. The first tab covers default start-up behaviour when you run the software. By default, you will be directed to choose an experiment type. On the other hand, you may wish to go directly to analysis profiles for your custom analysis settings instead of using presets, or you may generally as a user, only do one type of experiment. For example, you may only wish the software to open in landing assay mode since you don't use tracking:



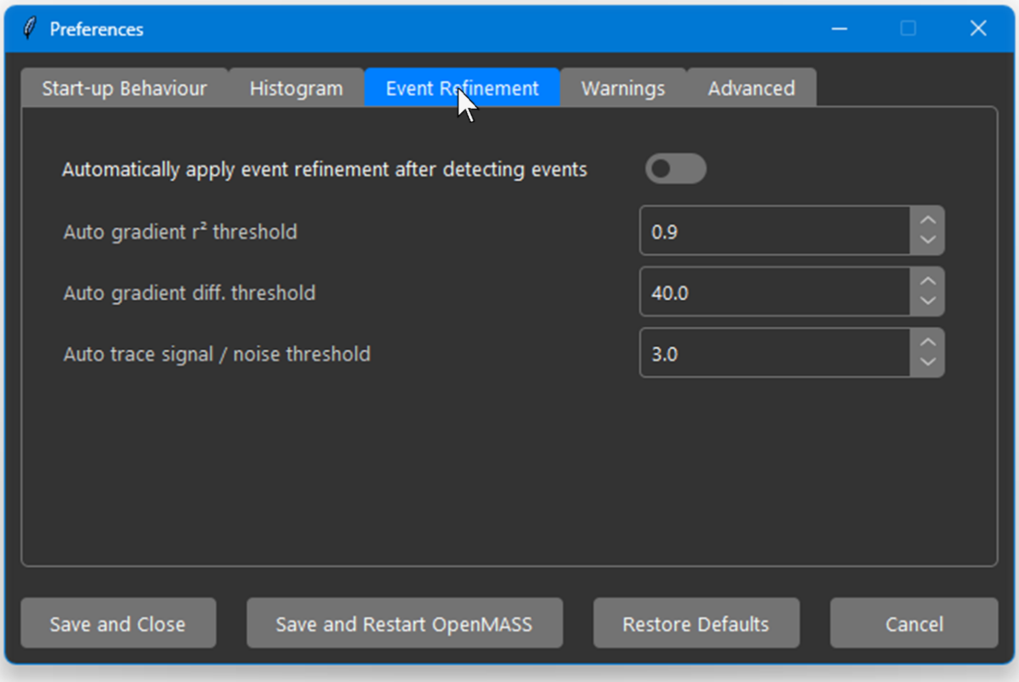
This preference also affects the behaviour of File → New Experiment.

To configure preferences for the histogram, navigate to the histogram tab:

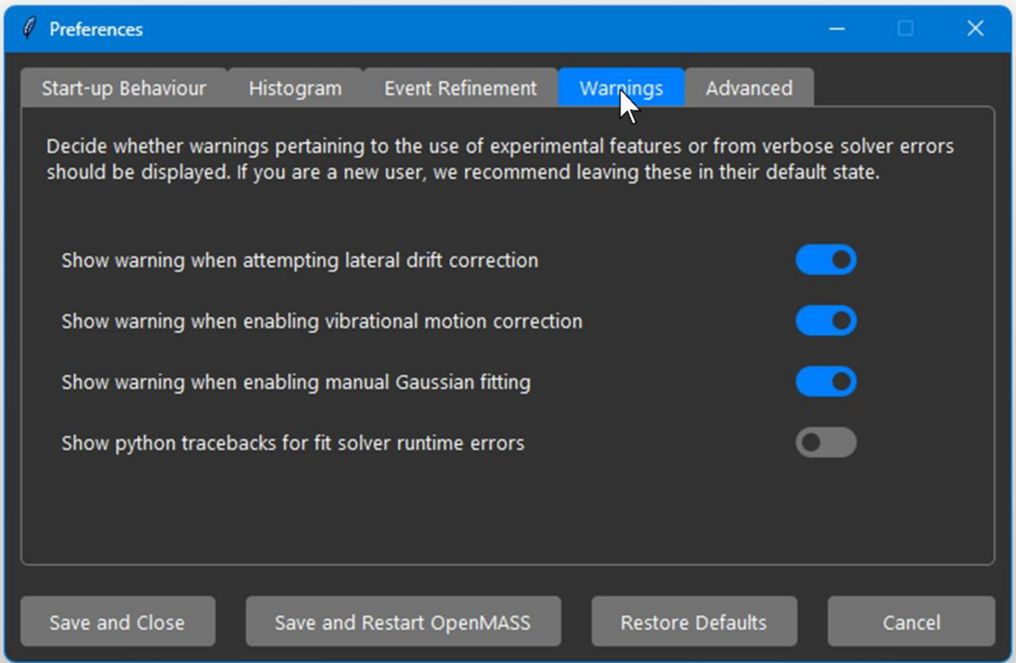


Settings here primarily focus on default bin sizes and count for the histogram when switching between contrast and mass and bin calculation modes. The default peak fitting mode can be set here too. The histogram update frequency determines how often the detected events will be dynamically updated during the event detection process. Percentiles are used to automatically rescale the axis to omit outliers, so the real distributions of your data fill the majority of the histogram. You can also toggle whether you want OpenMASS to automatically fit peaks after detecting events.

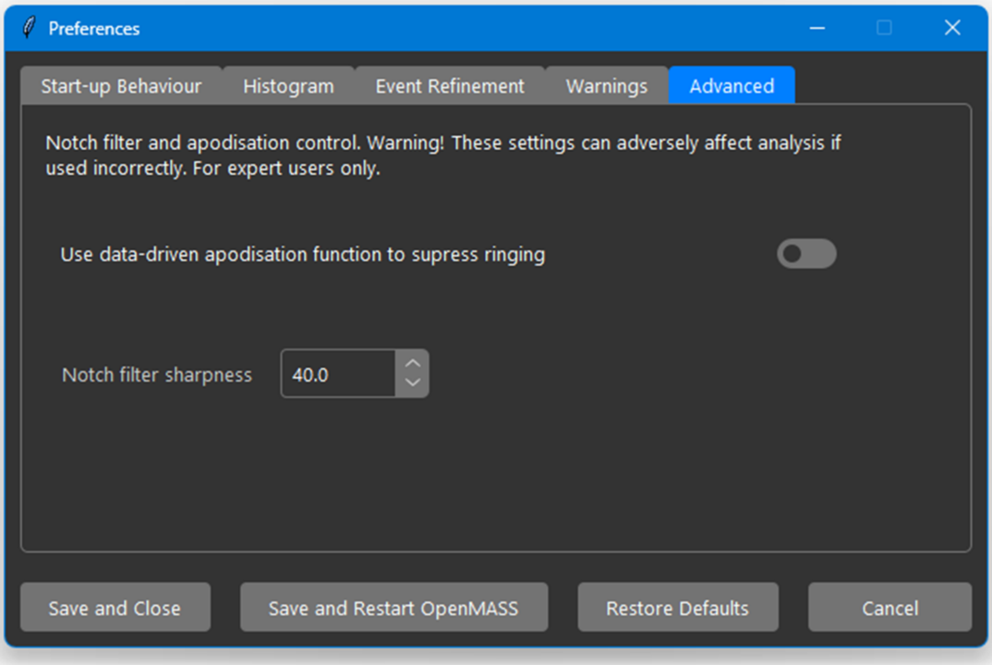
For event refinement preferences, navigate to the Event Refinement tab. Here you can toggle Automatic event refinement and set the thresholds to be used when automatic event refinement is enabled (see section 3.4, page 27 for details):



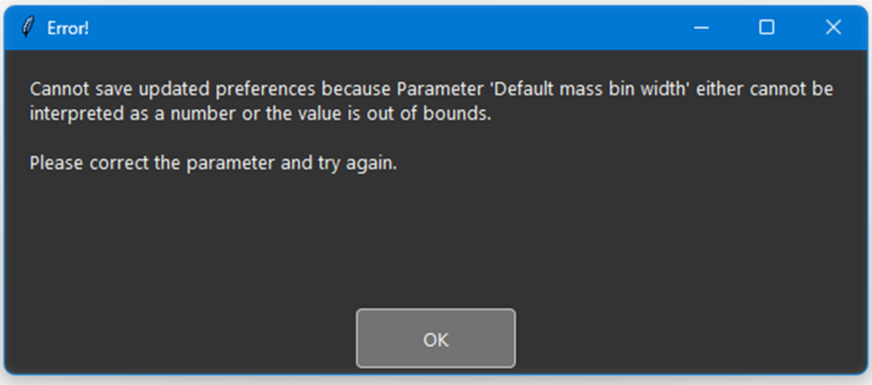
For warnings about experimental features or peak fitting error tracebacks, navigate to the Warnings tab. Here you can toggle whether features will warn you before using them:



In the Advanced tab, you'll find settings pertaining to the notch filter used to correct vibrational motion if motion correction is enabled. This is for advanced users only as you may cause unpredictable behaviour if you don't know what you're doing. Notch filter sharpness controls the width of the notch filter. The higher the sharpness, the narrower the notch width in Fourier space. A narrow filter may not suppress motion as aggressively, but ringing is often longer lived but less intense. A wide filter corrects motion artifacts very aggressively but causes intense but short-lived ringing around events. You can toggle on ringing suppression in this case but be warned that contrast traces will suffer peak broadening and event refinement gradient fitting may no longer work as expected:



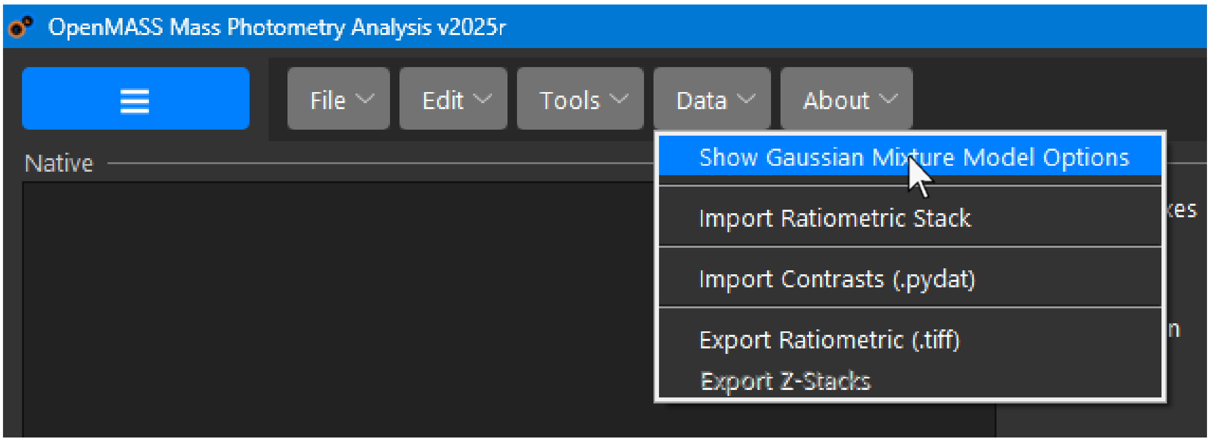
If an invalid quantity is entered into any of the tuneable preferences, the preferences will not be saved, and a warning will be given:



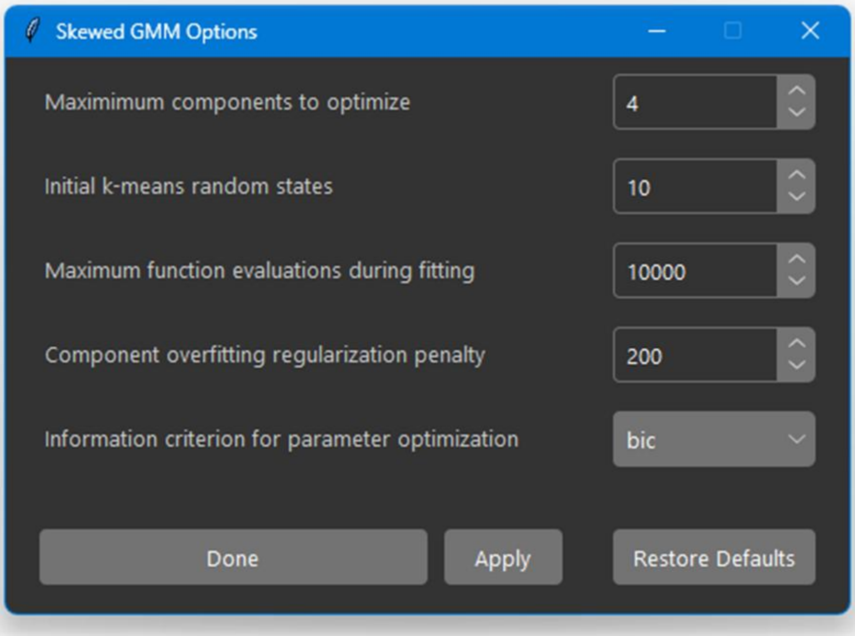
You can reset preferences to defaults by clicking 'Restore Defaults' in the event that your changes break the software, or you are no longer sure if you changed something and wish to revert to the default settings.

9.3 Gaussian Mixture Model Options

To set the default options for the skew-Gaussian mixture model used to fit peaks, open the menu ribbon and navigate Data → Show Gaussian Mixture Model Options:

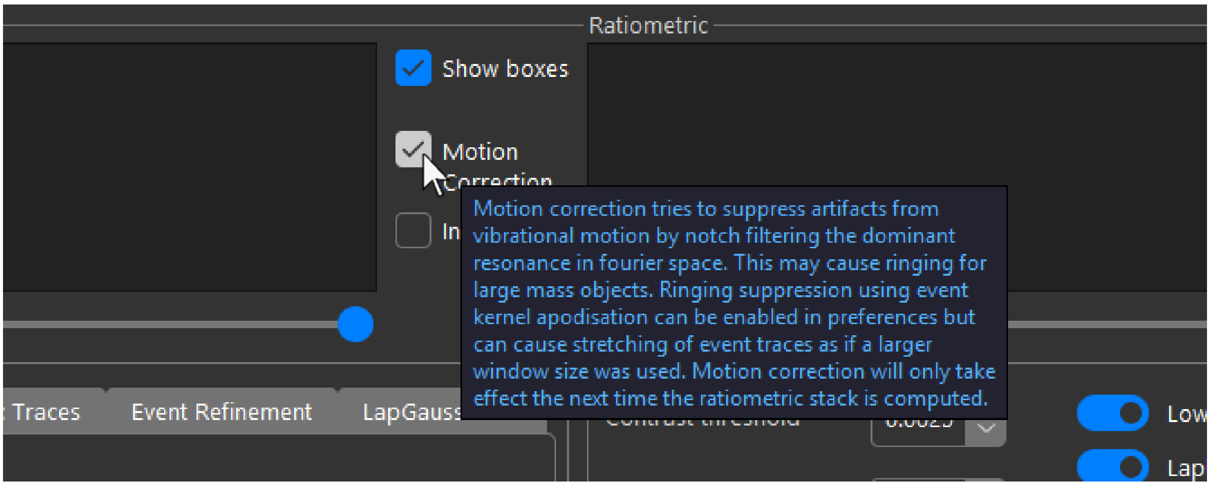


You can then tune the default options including regularisation penalty, number of initial clustering states for K-means initialiser, information criterion to determine components:



9.4 Tooltips

For any important feature or control in OpenMASS, you can hover the mouse over the respective widget for a short description of what it does:



END OF MANUAL